

SVD as a Fast Interpretability Method for Transformers

Min Xue¹ Artur Andrzejak¹

Abstract

Mechanistic interpretability of Transformer models commonly relies on training auxiliary proxy models, such as Sparse Autoencoders or Cross-Layer Transcoders. While effective, these post-hoc approaches introduce approximation bias and incur substantial computational overhead. We propose an alternative, training-free interpretability framework that directly exploits the Singular Value Decomposition (SVD) of weight matrices in Transformer MLP sublayers. By operating natively on model parameters, our method improves scalability while preserving fidelity to the original weights. We show that the projection matrices of MLP sublayers admit a natural decomposition into orthogonal, interpretable rank-1 subspaces, which we term **Detector-Effector Units (DEUs)**. Within each unit, a singular vector functions as a detector of input patterns and modulates a coupled effector vector that encodes output semantics. Building on this structure, we introduce **Subspace Contribution Analysis (SCA)**, a diagnostic method that quantifies the direct causal contribution of individual native subspaces to model predictions. Experiments across the GPT-2 family demonstrate that our framework, **Native Network Anatomy (NaNA)**, identifies dominant functional pathways with orders-of-magnitude efficiency gains over training-based interpretability baselines, while maintaining weight fidelity. Our results suggest that SVD-based analyses provide a scalable and faithful alternative to learned proxy approaches for mechanistic interpretability.

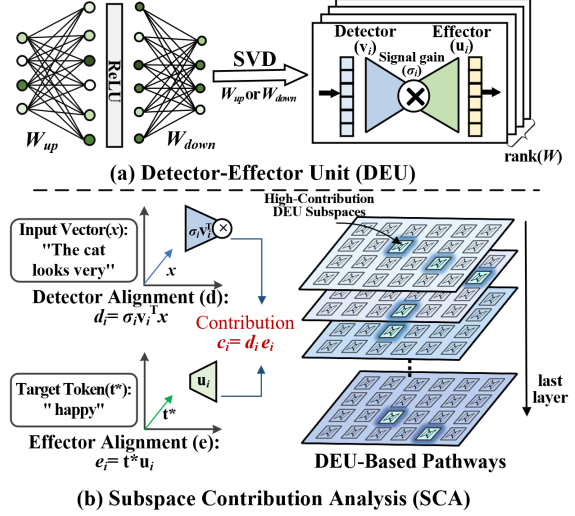


Figure 1. The Native Network Anatomy (NaNA) Framework. (a) **Detector-Effector Unit (DEU)**. The MLP weight matrix is decomposed into interpretable rank-1 subspaces via SVD. Each DEU consists of a detector (v_i), an effector (u_i), and a signal gain (σ). (b) **Subspace Contribution Analysis (SCA)**. This method quantifies the contribution of individual DEU subspaces to a target prediction. By projecting the input onto the detector and the target onto the effector, we obtain a contribution map that reveals sparse, dominant pathways across the layers.

1. Introduction

Large Language Models (LLMs) (Radford et al., 2019; Touvron et al., 2023) built upon the Transformer architecture (Vaswani et al., 2017) have demonstrated remarkable capabilities across a wide range of tasks. However, their internal decision-making mechanisms remain largely opaque. With substantial progress already achieved in analyzing attention heads (Zheng et al., 2024; Zhao et al., 2025; Elhelo & Geva, 2025), the focus has increasingly shifted towards Multi-Layer Perceptron (MLP) components, aiming to disentangle polysemantic neurons (Scherlis et al., 2022).

Recent approaches utilizing Sparse Autoencoders (SAEs) (Cunningham et al., 2023; Gao et al., 2024; Rajamanoharan et al., 2024a) and TransCoders (Dunefsky et al., 2024) have shown promise in extracting interpretable features from MLP activations (Rudin, 2019). However, these dictionary learning methods face two fundamental limitations. First, they are inherently *post-hoc* and *proxy-based*:

¹Institute of Computer Science, Heidelberg University, Germany. Correspondence to: Min Xue <min.xue@uni-heidelberg.de>.

the learned features exist in a reconstructed activation space rather than within the model’s native parameters, inevitably introducing approximation errors (Hase & Bansal, 2020; Chanin et al., 2025). Second, training these massive sparse dictionaries is computationally expensive, often incurring costs comparable to pretraining smaller language models (Templeton et al., 2024). Researchers thus face the dual challenges of high training costs for auxiliary models and the decoupling of interpretation from the model’s pretrained weights.

This motivates a fundamental research question: *Is there a natively interpretable structure within MLP weights, and can it be extracted without relying on auxiliary modules?*

To answer this question, we introduce **Native Network Anatomy** (NaNA), a training-free framework that analyzes Transformer MLPs from a spectral and functional perspective. Building on the observation of Beren & Black (2022) that singular vectors of MLP weights encode semantic structure, we advance beyond a purely representational view to formalize these spectral components as *computational operators* that directly mediate token prediction. Under the NaNA framework, an MLP projection matrix is expressed as a spectral sum of **Detector-Effector Units** (DEUs), defined as rank-1 subspaces obtained via Singular Value Decomposition (SVD). Each DEU functions as a distinct information channel formed by an outer product: a *detector*, given by the right singular vector, which selectively probes latent attributes in the input representation, and an *effector*, given by the corresponding left singular vector, which orients the resulting update in the MLP output. Crucially, this formulation induces an explicit coupling between detector activation and effector influence, enabling direct quantification of each native subspace’s contribution to the residual stream.

We leverage DEUs to propose **Subspace Contribution Analysis** (SCA), a training-free method that identifies sparse, dominant pathways and measures their impact on target token predictions. Moving beyond correlational analysis, we validate the functional necessity of these pathways through targeted intervention and ablation experiments, demonstrating that a small number of top-ranked subspaces determines specific model behaviors. Furthermore, NaNA identifies intrinsic functional pathways, achieving orders-of-magnitude efficiency gains over SAEs and TransCoders. This provides a lightweight yet faithful alternative for mechanistic interpretability.

2. Preliminaries

This section reviews Transformer MLP sublayers and summarizes the spectral interpretation of MLP weight matrices introduced by Beren & Black (2022).

2.1. Transformer Architecture and MLP Sublayers

A Transformer comprises a vocabulary embedding layer followed by L stacked layers, each consisting of a multi-head self-attention module $\text{Attn}^{(l)}(\cdot)$ and a position-wise MLP sublayer. The embedding matrix is $W_{\text{emb}} \in \mathbb{R}^{V \times d}$, where V denotes the vocabulary size and d the model dimension; $\mathbf{t}_j \in \mathbb{R}^{1 \times d}$ denotes its j -th row.

Given an input sequence $X = \{\mathbf{x}_1, \dots, \mathbf{x}_M\}$, let $\mathbf{h}_m^{(l-1)}$ denote the hidden state at position m entering layer l , initialized as $\mathbf{h}_m^{(0)} = \text{emb}(x_m) + \text{pos}(m)$. The attention sublayer produces an intermediate residual state

$$\bar{\mathbf{h}}_m^{(l)} = \mathbf{h}_m^{(l-1)} + \text{Attn}^{(l)}(\gamma(\mathbf{H}^{(l-1)}))_m, \quad (1)$$

where $\gamma(\cdot)$ denotes layer normalization and $\mathbf{H}^{(l-1)} = (\mathbf{h}_1^{(l-1)}, \dots, \mathbf{h}_M^{(l-1)})$ is the sequence of hidden states.

The MLP sublayer applies a two-layer feedforward transformation with residual connection, parameterized by an up-projection matrix $W_{\text{up}}^{(l)} \in \mathbb{R}^{n \times d}$ and a down-projection matrix $W_{\text{down}}^{(l)} \in \mathbb{R}^{d \times n}$, where n is the MLP hidden dimension. With pointwise nonlinearity $\phi(\cdot)$, the MLP output is

$$\mathbf{h}_m^{(l)} = \bar{\mathbf{h}}_m^{(l)} + W_{\text{down}}^{(l)} \phi\left(W_{\text{up}}^{(l)} \gamma(\bar{\mathbf{h}}_m^{(l)})\right). \quad (2)$$

2.2. Spectral Interpretation

For a matrix $W \in \mathbb{R}^{a \times b}$, its *Compact Singular Value Decomposition* (SVD) is defined as $W = U\Sigma V^\top$. Here, $r = \text{rank}(W)$, $U \in \mathbb{R}^{a \times r}$ and $V \in \mathbb{R}^{b \times r}$ are orthogonal matrices, and $\Sigma \in \mathbb{R}^{r \times r}$ is a diagonal matrix containing non-negative singular values $\sigma_1, \dots, \sigma_r$. The i -th column of U , $\mathbf{u}_i \in \mathbb{R}^{a \times 1}$, is a *left* singular vector; analogously, columns $\mathbf{v}_i \in \mathbb{R}^{b \times 1}$ of V are *right* singular vectors. Beren & Black (2022) demonstrated that the singular vectors of MLP weight matrices encode coherent semantic clusters.

Interpreting Left Singular Vectors of W_{down} . Following Beren & Black (2022), for a left singular vector $\mathbf{u}_i$ of W_{down} ($i \in \{1, \dots, d\}$, assuming $d = r$) we obtain a *score vector* $S_i \in \mathbb{R}^V$ by projecting $\mathbf{u}_i$ onto the vocabulary embedding matrix:

$$S_i = W_{\text{emb}} \mathbf{u}_i. \quad (3)$$

The j -th entry of S_i , $s_{ij} = \mathbf{t}_j \mathbf{u}_i$, quantifies the alignment between $\mathbf{u}_i$ and the embedding of the j -th token.

Interpreting Right Singular Vectors of W_{up} . Analogously, any right singular vector $\mathbf{v}_i$ of W_{up} gives rise to a score vector S'_i by projecting the vocabulary embeddings onto $\mathbf{v}_i$:

$$S'_i = W_{\text{emb}} \mathbf{v}_i. \quad (4)$$

Here, the scalar $s'_{ij} = \mathbf{t}_j \mathbf{v}_i$ measures the similarity between the j -th token embedding and the singular vector $\mathbf{v}_i$.

3. Native Network Anatomy Framework

We introduce the *Native Network Anatomy* (NaNA) framework for mechanistically analyzing MLP layers. We first formalize *Detector–Effector Units* (DEUs) as the elementary functional units of linear projections. We then propose *Subspace Contribution Analysis* (SCA) to quantify the predictive role of individual rank-1 components. Finally, we construct *directed DEU-based pathways* and define spectral interventions for causal analysis.

3.1. Detector–Effector Units (DEUs)

Let $W \in \mathbb{R}^{a \times b}$ denote a linear projection (up- or down-projection), with input $x \in \mathbb{R}^{b \times 1}$ and output $y = Wx$. By singular value decomposition, the down-projection admits

$$W_{\text{down}} = \sum_{i=1}^r \sigma_i \mathbf{u}_i \mathbf{v}_i^\top, \quad (5)$$

where each rank-1 term $\sigma_i \mathbf{u}_i \mathbf{v}_i^\top$ defines a *functional subspace*, i.e., a rank-1 operator coupling an input direction to an output direction.

Detector–Effector Units. Applying a single subspace to x yields

$$(\sigma_i \mathbf{u}_i \mathbf{v}_i^\top) x = \sigma_i (\mathbf{v}_i^\top x) \mathbf{u}_i = h_i \mathbf{u}_i, \quad (6)$$

where $h_i = \sigma_i (\mathbf{v}_i^\top x) \in \mathbb{R}$. Thus, each subspace produces an output aligned with $\mathbf{u}_i$, with magnitude controlled by the scalar activation h_i .

We interpret each rank-1 operator as a *Detector–Effector Unit* (DEU): $\mathbf{v}_i$ acts as a *detector* measuring input alignment, $\mathbf{u}_i$ as an *effector* determining output orientation, and σ_i as a *signal gain*.

Detector and Effector Probes. For the i -th DEU, we define the detector probe $f_i(x) = \mathbf{v}_i^\top x$ and effector probe $g_i(y) = \mathbf{u}_i^\top y$. These satisfy

$$g_i(y) = \sigma_i f_i(x), \quad (7)$$

indicating that the detector and effector capture a shared latent attribute before and after projection.

Following Section 2.2, we interpret DEUs by examining singular vectors that interface with the residual stream—specifically, the detector of W_{up} and the effector of W_{down} . Due to SVD sign ambiguity (Kolda & Bader, 2009), we evaluate semantics along both polarities ($\pm \mathbf{u}_i, \pm \mathbf{v}_i$).

MLPs as Superpositions of DEUs. For a full projection matrix W we have:

$$Wx = \sum_{i=1}^r (\sigma_i \mathbf{u}_i \mathbf{v}_i^\top) x = \sum_{i=1}^r h_i \mathbf{u}_i. \quad (8)$$

This formulation reframes the projection W as a linear superposition of orthogonal DEUs, with activations $h_i = \sigma_i (\mathbf{v}_i^\top x)$ modulating each effector direction.

3.2. Subspace Contribution Analysis (SCA)

We now quantify the predictive role of individual DEUs via *Subspace Contribution Analysis* (SCA). Given a down-projection $W_{\text{down}} \in \mathbb{R}^{d \times n}$, input x , and target token embedding $\mathbf{t}^* \in \mathbb{R}^d$, SCA decomposes each DEU’s contribution into detector and effector alignment.

Detector Alignment. We define

$$\mathbf{d} = \Sigma V^\top x \in \mathbb{R}^r, \quad d_i = \sigma_i \mathbf{v}_i^\top x, \quad (9)$$

where d_i captures the input activation of the i -th subspace, incorporating both geometric alignment and gain. Here, $r = \text{rank}(W_{\text{down}})$ which is assumed to be $r = d$ in practice.

Effector Alignment. We define

$$\mathbf{e} = \mathbf{t}^* U \in \mathbb{R}^r, \quad e_i = \mathbf{t}^* \mathbf{u}_i, \quad (10)$$

which measures how strongly each effector aligns with the target embedding.

Subspace Contribution. A subspace contributes meaningfully only when it is both activated and aligned with the target. We therefore define

$$\mathbf{c} = \mathbf{d} \odot \mathbf{e}, \quad c_i = (\sigma_i \mathbf{v}_i^\top x) (\mathbf{t}^* \mathbf{u}_i). \quad (11)$$

When $\mathbf{t}^*$ corresponds to the unembedding vector of the target token, c_i equals the signed contribution of the i -th DEU to the target logit. Let $\mathcal{I}_{\text{top-}k}$ denote the indices of the top- k subspaces with positive contributions; these are referred to as *high-contribution subspaces*.

3.3. Directed DEU-based Pathway

Prior work (Geva et al., 2021; Meng et al., 2022; Paulo et al., 2025) shows that middle-to-late MLP down-projections encode distilled features. We define the *Directed DEU-based Pathway* as the collection of high-contribution subspaces across these layers.

Selective Subspace Projection. We conceptualize the model’s internal computation as a localized information flow within modular subspaces. For each layer l , we decompose

$$W_{\text{down}}^{(l)} = \sum_{i=1}^r \sigma_i^{(l)} \mathbf{u}_i^{(l)} \mathbf{v}_i^{(l)\top}. \quad (12)$$

Let $\mathbf{z}_m^{(l)} \in \mathbb{R}^n$ denote the post-activation MLP state,

$$\mathbf{z}_m^{(l)} = \phi \left(W_{\text{up}}^{(l)} \gamma(\bar{\mathbf{h}}_m^{(l)}) \right), \quad (13)$$

where $\gamma(\cdot)$ denotes normalization and $\phi(\cdot)$ the pointwise nonlinearity.

We define the *subspace filtering operator*

$$\mathcal{P}^{(l)}(\mathbf{z}_m^{(l)}) = \sum_{i \in \mathcal{I}_{\text{top-}k}^{(l)}} \sigma_i^{(l)} \mathbf{u}_i^{(l)} (\mathbf{v}_i^{(l)\top} \mathbf{z}_m^{(l)}), \quad (14)$$

which retains only the high-contribution DEUs.

Full-Model Pathway. To capture the global computational trajectory, we extend the layer-wise operator to a multi-layer pathway formulation. For layers $l \in [l_{\text{st}}, L]$, where l_{st} denotes the starting layer of the pathway and L is the total number of Transformer layers, the pathway-constrained state update is

$$\mathbf{h}_m^{(l)} = \mathbf{h}_m^{(l-1)} + \text{Attn}^{(l)}(\gamma(\mathbf{H}^{(l-1)}))_m + \mathcal{P}^{(l)}(\mathbf{z}_m^{(l)}). \quad (15)$$

Composing these filtered updates traces a directed functional trajectory aligned with the target $\mathbf{t}^*$.

3.4. Subspace Interventions

We assess the role of DEUs using targeted spectral interventions. We consider three complementary intervention types: (i) reconstruction tests sufficiency by isolating a subspace set, (ii) ablation tests necessity by removing it, and (iii) scaling probes directional sensitivity while preserving the remaining spectrum.

Top- k Reconstruction. To test sufficiency, we reconstruct the down-projection using only high-contribution subspaces:

$$\mathbf{W}_{\text{reconst}}^{(l)} = \sum_{i \in \mathcal{I}_{\text{top-}k}^{(l)}} \sigma_i^{(l)} \mathbf{u}_i^{(l)} \mathbf{v}_i^{(l)\top}. \quad (16)$$

Subspace Ablation. To test necessity, we remove these components:

$$\mathbf{W}_{\text{ablated}}^{(l)} = \mathbf{W}_{\text{down}}^{(l)} - \sum_{i \in \mathcal{I}_{\text{top-}k}^{(l)}} \sigma_i^{(l)} \mathbf{u}_i^{(l)} \mathbf{v}_i^{(l)\top}. \quad (17)$$

Performance degradation under ablation quantifies the causal importance of the identified subspaces.

Scaling Directional Sensitivity. To modulate an individual DEU, we apply a *relative perturbation coefficient* α to its rank-1 operator:

$$\mathbf{W}_{\text{interv}}^{(l)} = \mathbf{W}_{\text{down}}^{(l)} + \alpha (\sigma_i^{(l)} \mathbf{u}_i^{(l)} \mathbf{v}_i^{(l)\top}), \quad (18)$$

where $\alpha > 0$ amplifies and $\alpha < 0$ suppresses the subspace.

4. Experiments

In this section, we validate the NaNA framework¹, extending our analysis from case studies to broader systematic generalizations. Our evaluation addresses two core questions: (i) to what extent do high-contribution subspaces determine the semantic outputs of the model, and (ii) do they constitute a reliable, training-free basis for mechanistic interpretability?

¹The code is available at: <https://github.com/minxue29031/NaNA>.

4.1. Experimental Setup

Models and Datasets. We conduct experiments across multiple model scales, including GPT-2, GPT2-Medium, GPT2-Large, and GPT2-XL, using three datasets designed to probe distinct behavioral and functional properties. **Anchor Cases** ($\mathcal{D}_{\text{achr}}$): A set of 20 cases spanning syntax, semantics, code understanding, and factual knowledge (details in Appendix Table 8). **Generalization Dataset** ($\mathcal{D}_{\text{gene}}$): For each model, it comprises 100 diverse prefixes paired with model-specific target tokens, used to quantify the fidelity of behavioral recovery mediated by the identified pathways. **Controlled Variation Sets:** To assess the robustness of subspaces under contextual perturbations, we construct two datasets centered on positive-adjective semantics: (i) *Target Variance Dataset* ($\mathcal{D}_{\text{target}}$), fixing the prefix and varying 100 target adjectives; and (ii) *Prefix Variance Dataset* ($\mathcal{D}_{\text{prefix}}$), varying 100 prefixes while holding the target constant. (Details of the dataset construction are provided in Appendix E.1.)

Baselines. We evaluate NaNA against the following baselines. *SAEs* ($768 \times 24,576$) and *TransCoders* ($768 \times 24,576$) serve as representative training-based methods for decomposing polysemantic features. *Raw MLP neurons* provide a baseline to validate the role of spectral decomposition. *Random DEU subspaces* act as a stochastic control for verifying causal significance. Finally, *Full MLP* and *Full TransCoder* define the upper bound for predictive performance.

Evaluation Metrics. Our evaluation is based on the following four metrics. **Fidelity:** Measures the sufficiency of subspaces or features for recovering predictive performance. We quantify the probability and rank of the original target token by reconstructing the MLP using only the selected components. **Faithfulness:** Evaluates the necessity of subspaces or features by quantifying the resulting degradation in predictive performance. We measure the decline in the target token’s probability and rank after ablating these components from the MLP. **Interpretability:** Assesses semantic consistency. For a given concept across various prefixes and targets, we examine whether the high-contribution subspaces consistently align with the associated semantic properties. **Efficiency:** Reports the computational overhead in terms of total runtime and peak GPU memory consumption.

Intervention Mechanism. We perform causal interventions on MLP projections using PyTorch hooks (Nanda & Bloom, 2022). At each target layer, we intercept the final-token activation and substitute the original output with a projection computed using a modified weight matrix, thereby propagating the intervened signal through the network.

4.2. Case Study on Positive Semantics

In this section, we present a case study on positive semantics (“the cat looks very” $\rightarrow$ “happy”) to dissect how DEU-

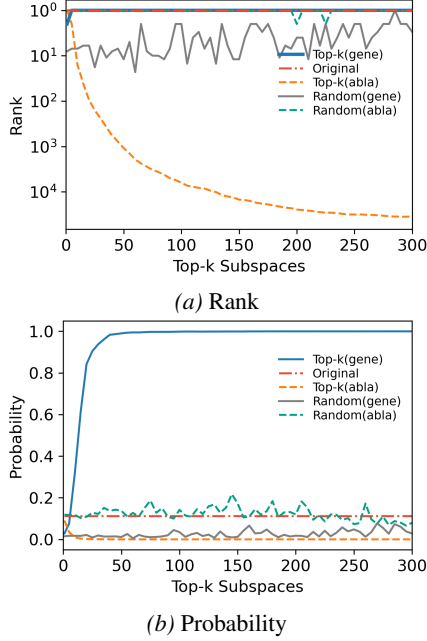


Figure 2. Pathway Analysis via Top- k Subspace Interventions in GPT2-Medium. We evaluate the impact of high-contribution subspaces on the target token’s probability ($\uparrow$) and rank ($\downarrow$) in the positive semantic case (“the cat looks very” $\rightarrow$ “happy”).

based pathways encode positive semantic information across layers. (Other cases are provided in Appendix H.) Unless otherwise stated, top- k subspaces are selected per sample based on SCA scores and then aggregated across samples for frequency analysis. For comparative evaluations we use subspaces that appear most frequently among per-sample top- k sets.

Impact of Pathways on Predictive Performance. We evaluate the identified DEU-based pathways by measuring their impact on target token prediction. To assess fidelity, we reconstruct W_{down} using the top- k high-contribution subspaces identified by the SCA method (Top- k (gene)) and randomly selected DEU subspaces (Random (gene)), with the original weights serving as a baseline. To evaluate faithfulness, we ablate the target pathways by reconstructing W_{down} with the top- k (Top- k (abla)) or random (Random (abla)) DEU subspaces removed. As shown in Figure 2, we intervene on the MLP down-projection matrices across the middle-to-late layers (7–23) of GPT2-Medium, monitoring the resulting fluctuations in target token rank ($\downarrow$) and probability ($\uparrow$). Here, rank denotes the target token’s position in the model’s output distribution.

As illustrated in Figure 2, we observe a distinct phase transition in the model’s behavior. The Top- k (Gene) trajectory reveals that the model’s predictive capability can be almost entirely recovered using only a small fraction of subspaces (e.g., $k < 20$), with the target token’s rank quickly improving and the probability saturating near 1.0. This result suggests that these semantic pathways are intrinsically

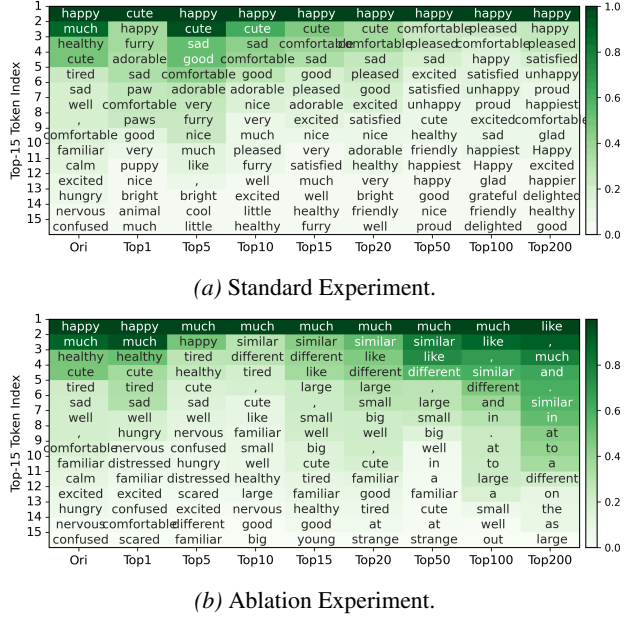


Figure 3. Reconstruction of Middle-to-Late MLP Layers via Pathways within Top- k High-Contribution Subspaces: Standard vs. Ablation. Here, we visualize the top-15 predicted tokens of GPT2-Medium under standard and ablation interventions for the case “The cat looks very” $\rightarrow$ “happy”.

sparse. Conversely, the Top- k (Abla) curves exhibit a sharp performance collapse, where ablating the top- k subspaces triggers an exponential decay in rank. The flat trajectories of the random baselines further confirm that the dominant subspaces identified by SCA are causally necessary for the target prediction. In addition, we extend this analysis to the GPT-2 family, demonstrating that this sparsity pattern remains consistent across parameter scales (see Appendix I.1).

Causal Evidence of Pathway-Driven Predictions. To provide direct causal evidence that SCA-identified pathways drive model predictions, we visualize the evolution of the model’s output distribution across varying pathway widths, defined by the top- k subspaces. Figure 3 illustrates the top-15 predicted tokens for the prompt “The cat looks very” as we reconstruct the middle-to-late MLPs of GPT2-Medium using an increasing number of high-contribution subspaces.

Semantic Recovery (Fidelity). As shown in Figure 3a, reconstructing with only the top-5 subspaces is sufficient to restore the target token “happy”. As more subspaces are included, the top-15 predictions increasingly focus on positive adjectives such as “pleased”, “comfortable”, and “good”. This rapid recovery demonstrates that a small number of high-contribution DEU subspaces play a dominant role in driving the model’s behavior.

Semantic Collapse via Ablation (Faithfulness). Conversely, Figure 3b reveals a dramatic semantic collapse upon targeted ablation. The removal of merely the top 10 subspaces extinguishes the target token “happy” from the top-tier pre-

dictions. As more high-contribution subspaces are systematically ablated, the model loses the ability to generate positive adjectives and instead produces generic terms such as “*much*”, “*similar*”, and “*different*”. This sharp divergence confirms that high-contribution DEUs constitute the primary functional modules for the target prediction. (Detailed results for 20 diverse cases across models are provided in Appendix H.)

Amplifying Target Semantic Contribution ($\alpha > 0$). Increasing the intervention coefficient from 0 to 8 yields a progressive dominance of Subspace 7. Compared to the baseline ($\alpha = 0$), amplifying Subspace 7 forces the model to prioritize tokens such as “happy”, “strong”, and “healthy”. Notably, at high amplification levels, the layer’s output semantics converge almost exclusively toward positive attributes. These results confirm that manipulating Subspace 7 can steer the model’s lexical preferences toward specific semantic concepts.

Figure 1 displays the performance of GPT-2 models (gpt2, gpt2-medium, gpt2-large, gpt2-xl) on the Gene and AbLa datasets. The figure is organized into six subplots, arranged in a 3x2 grid, showing Top-1 Accuracy, Mean Rank, and Mean Target Probability across different Top-k Subspaces.

(a) Top-1 Acc (Gene): Top-1 Accuracy vs. Top-k Subspaces (0 to 50). gpt2-xl (blue) shows the highest accuracy, reaching near 1.0 for k > 20. gpt2 (red) shows the lowest accuracy, reaching near 1.0 for k > 30.

(b) Top-1 Acc (AbLa): Top-1 Accuracy vs. Top-k Subspaces (0 to 40). gpt2-xl (blue) shows the highest accuracy, reaching near 0.8 for k > 10. gpt2 (red) shows the lowest accuracy, reaching near 0.0 for k > 10.

(c) Mean Rank (Gene): Mean Rank vs. Top-k Subspaces (0 to 40). gpt2 (red) shows the highest mean rank, decreasing sharply from ~22 to ~1. gpt2-xl (blue) shows the lowest mean rank, decreasing sharply from ~12 to ~1.

(d) Mean Rank (AbLa): Mean Rank vs. Top-k Subspaces (0 to 100). gpt2-xl (blue) shows the highest mean rank, increasing from ~10⁰ to ~10⁴. gpt2 (red) shows the lowest mean rank, increasing from ~10⁰ to ~10³.

(e) Mean Target Prob (Gene): Mean Target Probability vs. Top-k Subspaces (0 to 100). gpt2-xl (blue) shows the highest mean target probability, reaching near 1.0 for k > 20. gpt2 (red) shows the lowest mean target probability, reaching near 1.0 for k > 60.

(f) Mean Target Prob (AbLa): Mean Target Probability vs. Top-k Subspaces (0 to 50). gpt2-xl (blue) shows the highest mean target probability, reaching near 0.25 for k > 10. gpt2 (red) shows the lowest mean target probability, reaching near 0.0 for k > 10.

4.3. Generalization Evaluation

Standard Experiment (Fidelity): As illustrated in Figures 5a, 5c, and 5e, reconstructing the middle-to-late layers using pathways composed of only the top-20 subspaces yields near-optimal performance. Across all model scales, the mean rank converges toward the optimum, and top-1 accuracy reaches a remarkable 97-99%. Furthermore, the mean probability increases monotonically as additional subspaces are incorporated, confirming that SCA identifies the key subspaces driving the target-directed predictions.

6

Table 1. Layer-wise Interpretation of Subspaces in GPT2-Medium (Layers 14–17). The “Positive” and “Negative” columns denote semantic concepts corresponding to the opposing directions of the singular vector ($\pm \mathbf{u}_i$), accounting for the inherent sign ambiguity in SVD (Kolda & Bader, 2009). Here, parentheses denote subspace interpretability levels (High, Med, or Low).

ID	Positive Direction	Negative Direction	Y/N
L14.SP13	<i>null</i> (null)	vibrant leisure (High)	Y
L14.SP2	<i>null</i> (null)	Punctuation and Symbols (High)	N
L14.SP74	Government Challenges (Med)	<i>null</i> (null)	N
L14.SP3	electricity concepts (Med)	mental states (Med)	Y
L14.SP10	Positive Excellence (High)	Measurement and Time (Med)	Y
L15.SP14	updates and terminations (High)	emotional experiences (High)	Y
L16.SP7	Government Deaths (High)	positive qualities (High)	Y
L16.SP12	Respect and Worthiness (High)	Numbers and Quantities (High)	Y
L16.SP2	Punctuation Marks (High)	<i>null</i> (null)	N
L16.SP3	Observation and Critique (Med)	electric devices (High)	Y
L17.SP14	Cognitive Values (High)	Security incidents (High)	Y

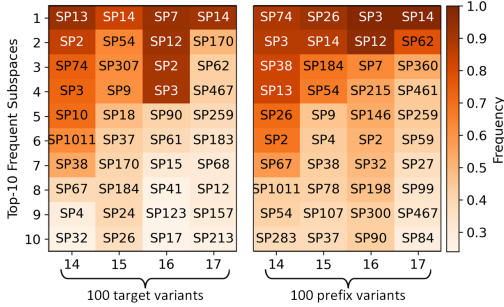
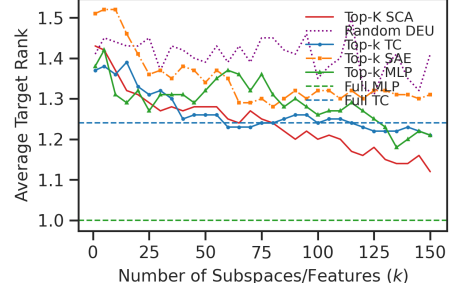


Figure 6. **Top-10 Active Subspaces in GPT2-Medium (Layers 14–17).** This visualizes the activation frequency across target-variance and prefix-variance datasets. Here, color intensity reflects activation frequency, and SP_i denotes subspace i .

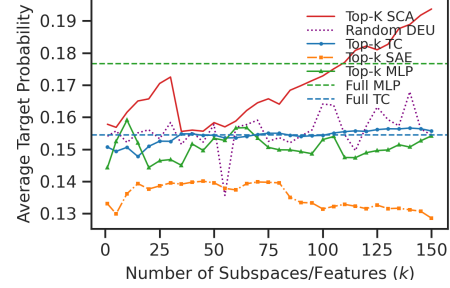
the high-contribution DEU subspaces leads to a catastrophic drop in mean top-1 accuracy and in probability, and an exponential increase in mean rank. This degradation confirms that a marginal fraction of DEU subspaces dominates the model’s predictive behavior, and their removal effectively severs the underlying semantic pathway.

Highly Active Subspaces and Their Semantic Roles. To investigate whether certain subspaces are consistently highly active and serve as interpretable mechanisms for processing specific contextual semantics, we utilize the *Controlled Variation Sets*. These sets consist of a target-variance dataset $\mathcal{D}_{\text{target}}$ and a prefix-variance dataset $\mathcal{D}_{\text{prefix}}$, both focused on positive-adjective semantics. As subspaces derived from $\mathcal{D}_{\text{target}}$ capture intrinsic semantic structures more robustly than the noise-prone $\mathcal{D}_{\text{prefix}}$ setting, we use the former for primary analysis and the latter for validation.

Figure 6 illustrates the top-10 active subspaces in Layers 14–17 of GPT2-Medium, identified by their recurrence frequency among the top-10 contributors across all positive semantic samples. Here, color intensity represents activation frequency, and SP_i denotes subspace i . We find that only a small subset of subspaces are consistently active within each layer. This suggests that a limited number of subspaces act as principal carriers of semantic information, playing a pivotal role in mediating specific semantic features.



(a) Rank.



(b) Probability.

Figure 7. **Fidelity Assessment of Active Components in GPT-2 (Layer 10).** Baselines include original MLP neurons (Top- k MLP), TransCoder features (Top- k TC), SAE features (Top- k SAE), and Random DEU. Full MLP and Full TC denote upper bounds.

We further examine whether these highly active subspaces exhibit direct and interpretable associations with positive semantic attributes. Table 1 details the semantic interpretations of these subspaces under $\mathcal{D}_{\text{target}}$, where $Lm.SP_i$ denotes subspace i in layer m . Our analysis shows that these high-frequency subspaces are typically related to positive semantics. For example, in Layer 14, Subspace 13 encodes the semantic of “*Vibrant Leisure*” and Subspace 10 relates to “*Positive Excellence*”. Similarly, in Layer 16, Subspace 7 encodes “*Positive Qualities*”, Subspace 12 represents “*Respect and Worthiness*”, and Subspace 3 captures “*Observation and Critique*”. These findings suggest that the model’s predictions are mediated by specific, functional modules rather than diffuse representations. Collectively, these subspaces drive the robust generation of positive semantics. (Appendix F.1 provides further layer-wise statistics and interpretations across various models.)

4.4. Comparative Evaluation

Comparison with Neuronal and Dictionary-Based Methods. We utilize the generalization dataset ($\mathcal{D}_{\text{gene}}$) to evaluate the reconstruction fidelity of the top- k active components in recovering the target token’s prediction within GPT-2 (Layer 10). We compare SCA against four representative baselines: original MLP neurons, TransCoder features, SAE features, and random DEU subspaces, where active features are selected based on activation variance. The *Full MLP* and *Full TC* settings serve as performance upper bounds. For

Table 2. Computational Efficiency of Full-Subspace and Feature Interpretation on a Single Projection Matrix in GPT-2 (NVIDIA A100, 80GB). Here, the runtime of SAE and TransCoder includes external module training.

	NaNA	SAE	TransCoder
Runtime	< 2 min	≈ 30 min	≈ 30 min
Training & Tuning	Not needed	Needed	Needed
Training Data	Not needed	60M tokens	60M tokens
GPU Memory	< 1 GB	≈ 16 GB	≈ 16 GB

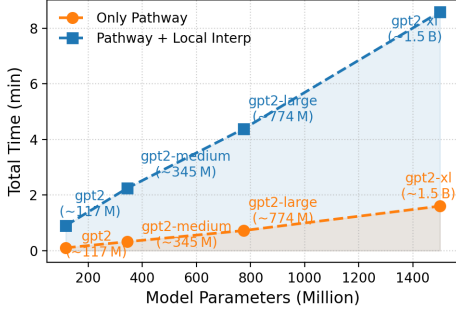


Figure 8. Linear Scaling of the NaNA Framework Across Model Sizes. Here, runtimes are measured on a single A100 (80GB) GPU for pathway extraction and local interpretation using the top-20 subspaces.

SCA, the active subspace is derived from high-frequency components shared across samples.

Figure 7 evaluates reconstruction performance based on Average Target Rank ($\downarrow$) and Average Target Probability ($\uparrow$), demonstrating that SCA consistently outperforms baseline methods on both metrics. Specifically, as illustrated in Figure 7a, while SAE and TransCoder exhibit performance saturation beyond $k > 60$, SCA continues to achieve performance gains.

We attribute this sustained gain to the explicit filtering for *causal bridges*, defined as subspaces that satisfy both input activation (detector alignment) and output relevance (effector alignment). This directed extraction enables the capture of implicit supporters, characterized as subspaces with low activation magnitude yet significant directional influence, which are typically neglected by unsupervised dictionary learning methods.

In terms of stability, SCA exhibits robust and monotonic rank reduction, avoiding the high volatility observed in MLP and Random DEU baselines caused by target-irrelevant semantic interference. Notably, Figure 7b shows a transient probability dip around $k = 30$. We ascribe this to Softmax competition arising from semantic broadening (Holtzman et al., 2020): subspaces at this stage amplify a cluster of related synonyms, temporarily diluting the specific target’s probability mass (as shown in Figure 3).

Computational Efficiency and Resource Overhead. In this section, we evaluate the computational efficiency and scalability of the NaNA framework in comparison to SAEs and Transcoders.

Runtime and GPU Usage on Single-Layer. Table 2 contrasts the computational resource requirements of the NaNA framework against training-based methods for analyzing a single projection matrix in GPT-2. Notably, NaNA is training-free and highly efficient, completing the analysis in under 2 minutes with negligible memory overhead (<1GB) on an NVIDIA A100 (80GB). In sharp contrast, SAEs and TransCoders necessitate training auxiliary modules on approximately 60 million tokens, consuming around 30 minutes and 16GB of memory (Dunefsky et al., 2024). Furthermore, these training-based methods typically require extensive hyperparameter tuning, which substantially amplifies the total computational cost.

On-Demand Pathway Analysis. In training-based paradigms like SAEs and TransCoders, pathway extraction necessitates the computationally intensive pre-computation of global auxiliary modules. Conversely, the NaNA framework facilitates fast, training-free pathway extraction and on-demand analysis by selectively decoding only the top-ranked subspaces. Empirically, pre-computing all modules for GPT2-XL with SAEs or TransCoders takes over 24 hours on a single A100 (80GB) GPU (Dunefsky et al., 2024). By comparison, as shown in Figure 8, our framework completes pathway extraction and local interpretation in only 8.59 minutes using the top-20 subspaces (which, as shown in Figure 5, are sufficient to preserve target predictions). Importantly, NaNA exhibits linear scalability with respect to model size. The runtime for both pathway extraction and local interpretation grows linearly rather than exponentially, ensuring the framework remains computationally efficient as model size increases. This property allows NaNA to overcome the scalability and computational bottlenecks inherent in training-based interpretability methods.

5. Conclusion

In this work, we introduce the Native Network Anatomy (NaNA) framework, which bridges the gap between the linear algebraic structure of weights and the semantic behavior of MLPs. By integrating Detector-Effector Units (DEUs) with Subspace Contribution Analysis (SCA), we demonstrate that model predictions are driven by sparse DEU-based pathways. Crucially, NaNA achieves orders-of-magnitude improvements in efficiency without compromising fidelity, circumventing the approximation bias inherent in proxy-based interpretation methods.

NaNA’s efficacy hinges on SCA’s explicit targeting of “causal bridges”—subspaces that satisfy the dual constraints of input activation (detector alignment) and output relevance (effector alignment). This focus on functional hubs allows the framework to capture “implicit supporters”: low-magnitude components that exert significant directional influence but are typically discarded by unsupervised dictio-

nary learning (Trenton et al., 2023; Dunefsky et al., 2024). Ultimately, NaNA establishes a scalable, high-fidelity foundation for interpreting the internal logic of large-scale models through their native structural geometry.

Impact Statement

This paper introduces a training-free framework for mechanistic interpretability, dissecting MLPs through their native computational structures. We position this work as a foundational step; its broader impact is expected to manifest through downstream applications of pathway analysis, ultimately contributing to the improved understanding, controllability, and reliability of Large Language Models (LLMs).

References

- Ahmad, A., Joshi, A., and Modi, A. Beyond components: Singular vector-based interpretability of transformer circuits. In *The Thirty-ninth Annual Conference on Neural Information Processing Systems*, 2026. URL <https://openreview.net/forum?id=7UbXEQNny7>.
- Arora, S., Li, Y., Liang, Y., Ma, T., and Risteski, A. Linear algebraic structure of word senses, with applications to polysemy. *Transactions of the Association for Computational Linguistics*, 6:483–495, 2018. doi: 10.1162/tacl.a.00034. URL <https://aclanthology.org/Q18-1034/>.
- Belrose, N., Furman, Z., Smith, L., Halawi, D., Ostrovsky, I. V., McKinney, L., Biderman, S., and Steinhart, J. Eliciting latent predictions from transformers with the tuned lens. *ArXiv*, abs/2303.08112, 2023. URL <https://api.semanticscholar.org/CorpusID:257504984>.
- Beren and Black, S. The singular value decompositions of transformer weight matrices are highly interpretable, November 2022. URL <https://www.lesswrong.com/posts/mkbGjzx8D8d8XqKHZA>. AI Alignment Forum.
- Bloom, J., Tigges, C., Duong, A., and Chanin, D. Sae-lens. <https://github.com/decoderresearch/SAELens>, 2024.
- Busmann, B., Leask, P., and Nanda, N. Batchtopk sparse autoencoders, 2024. URL <https://arxiv.org/abs/2412.06410>.
- Chanin, D., Wilken-Smith, J., Dulka, T., Bhatnagar, H., Golechha, S., and Bloom, J. A is for absorption: Studying feature splitting and absorption in sparse autoencoders, 2025. URL <https://arxiv.org/abs/2409.14507>.
- Conmy, A., Mavor-Parker, A., Lynch, A., Heimersheim, S., and Garriga-Alonso, A. Towards automated circuit discovery for mechanistic interpretability. In Oh, A., Naumann, T., Globerson, A., Saenko, K., Hardt, M., and Levine, S. (eds.), *Advances in Neural Information Processing Systems*, volume 36, pp. 16318–16352. Curran Associates, Inc., 2023.
- Cunningham, H., Ewart, A., Riggs, L., Huben, R., and Sharkey, L. Sparse autoencoders find highly interpretable features in language models, 2023. URL <https://arxiv.org/abs/2309.08600>.
- Dumas, C., Wendler, C., Veselovsky, V., Monea, G., and West, R. Separating tongue from thought: Activation patching reveals language-agnostic concept representations in transformers. In Che, W., Nabende, J., Shutova, E., and Pilehvar, M. T. (eds.), *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 31822–31841, Vienna, Austria, July 2025. Association for Computational Linguistics. ISBN 979-8-89176-251-0. doi: 10.18653/v1/2025.acl-long.1536. URL <https://aclanthology.org/2025.acl-long.1536/>.
- Dunefsky, J., Chlenski, P., and Nanda, N. Transcoders find interpretable llm feature circuits. In *Proceedings of the 38th International Conference on Neural Information Processing Systems*, NIPS ’24, Red Hook, NY, USA, 2024. Curran Associates Inc. ISBN 9798331314385.
- Eckart, C. and Young, G. The approximation of one matrix by another of lower rank. *Psychometrika*, 1(3):211–218, 1936. doi: 10.1007/BF02288367.
- Elhage, N., Nanda, N., Olsson, C., Henighan, T., Joseph, N., Mann, B., Askell, A., Bai, Y., Chen, A., Conerly, T., DasSarma, N., Drain, D., Ganguli, D., Hatfield-Dodds, Z., Hernandez, D., Jones, A., Kernion, J., Lovitt, L., Ndousse, K., Amodei, D., Brown, T., Clark, J., Kaplan, J., McCandlish, S., and Olah, C. A mathematical framework for transformer circuits, 2021. URL <https://transformer-circuits.pub/2021/framework/index.html>.
- Elhage, N., Hume, T., Olsson, C., Schiefer, N., Henighan, T., Kravec, S., Hatfield-Dodds, Z., Lasenby, R., Drain, D., Chen, C., Grosse, R., McCandlish, S., Kaplan, J., Amodei, D., Wattenberg, M., and Olah, C. Toy models of superposition, 2022. URL <https://arxiv.org/abs/2209.10652>.
- Elhelo, A. and Geva, M. Inferring functionality of attention heads from their parameters, 2025. URL <https://arxiv.org/abs/2412.11965>.

- Fedus, W., Zoph, B., and Shazeer, N. Switch transformers: scaling to trillion parameter models with simple and efficient sparsity. *J. Mach. Learn. Res.*, 23(1), January 2022. ISSN 1532-4435.
- Gao, L., Biderman, S., Black, S., Golding, L., Hoppe, T., Foster, C., Phang, J., He, H., Thite, A., Nabeshima, N., et al. The pile: An 800gb dataset of diverse text for language modeling. *arXiv preprint arXiv:2101.00027*, 2020.
- Gao, L., la Tour, T. D., Tillman, H., Goh, G., Troll, R., Radford, A., Sutskever, I., Leike, J., and Wu, J. Scaling and evaluating sparse autoencoders, 2024. URL <https://arxiv.org/abs/2406.04093>.
- Gao, L., la Tour, T. D., Tillman, H., Goh, G., Troll, R., Radford, A., Sutskever, I., Leike, J., and Wu, J. Scaling and evaluating sparse autoencoders. In *The Thirteenth International Conference on Learning Representations*, 2025. URL <https://openreview.net/forum?id=tcsZt9ZNKD>.
- Gargiulo, A. A., Crisostomi, D., Bucarelli, M. S., Scardapane, S., Silvestri, F., and Rodolà, E. Task singular vectors: Reducing task interference in model merging, 2025. URL <https://arxiv.org/abs/2412.00081>.
- Geva, M., Schuster, R., Berant, J., and Levy, O. Transformer feed-forward layers are key-value memories. In Moens, M.-F., Huang, X., Specia, L., and Yih, S. W.-t. (eds.), *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, pp. 5484–5495, Online and Punta Cana, Dominican Republic, November 2021. Association for Computational Linguistics. doi: 10.18653/v1/2021.emnlp-main.446. URL <https://aclanthology.org/2021.emnlp-main.446/>.
- Geva, M., Caciularu, A., Wang, K., and Goldberg, Y. Transformer feed-forward layers build predictions by promoting concepts in the vocabulary space. In Goldberg, Y., Kozareva, Z., and Zhang, Y. (eds.), *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pp. 30–45, Abu Dhabi, United Arab Emirates, December 2022. Association for Computational Linguistics. doi: 10.18653/v1/2022.emnlp-main.3. URL <https://aclanthology.org/2022.emnlp-main.3/>.
- Hase, P. and Bansal, M. Evaluating explainable ai: Which algorithmic explanations help users predict model behavior?, 2020. URL <https://arxiv.org/abs/2005.01831>.
- Heimersheim, S. and Nanda, N. How to use and interpret activation patching, 2024. URL <https://arxiv.org/abs/2404.15255>.
- Holtzman, A., Buys, J., Du, L., Forbes, M., and Choi, Y. The curious case of neural text degeneration, 2020. URL <https://arxiv.org/abs/1904.09751>.
- Jiang, A. Q., Sablayrolles, A., Roux, A., Mensch, A., Savary, B., Bamford, C., Chaplot, D. S., de las Casas, D., Hanna, E. B., Bressand, F., Lengyel, G., Bour, G., Lample, G., Lavaud, L. R., Saulnier, L., Lachaux, M.-A., Stock, P., Subramanian, S., Yang, S., Antoniak, S., Scao, T. L., Gervet, T., Lavril, T., Wang, T., Lacroix, T., and Sayed, W. E. Mixtral of experts, 2024. URL <https://arxiv.org/abs/2401.04088>.
- Kolda, T. G. and Bader, B. W. Tensor decompositions and applications. *SIAM review*, 51(3):455–500, 2009.
- Kreutz-Delgado, K., Murray, J. F., Rao, B. D., Engan, K., Lee, T.-W., and Sejnowski, T. J. Dictionary learning algorithms for sparse representation. *Neural computation*, 15(2):349–396, 2003.
- Li, X., Li, S., Song, S., Yang, J., Ma, J., and Yu, J. Pmet: precise model editing in a transformer. In *Proceedings of the Thirty-Eighth AAAI Conference on Artificial Intelligence and Thirty-Sixth Conference on Innovative Applications of Artificial Intelligence and Fourteenth Symposium on Educational Advances in Artificial Intelligence*, AAAI’24/IAAI’24/EAAI’24. AAAI Press, 2024. ISBN 978-1-57735-887-9. doi: 10.1609/aaai.v38i17.29818. URL <https://doi.org/10.1609/aaai.v38i17.29818>.
- Makhzani, A. and Frey, B. k-sparse autoencoders, 2014. URL <https://arxiv.org/abs/1312.5663>.
- Meng, K., Bau, D., Andonian, A., and Belinkov, Y. Locating and editing factual associations in gpt. In *Proceedings of the 36th International Conference on Neural Information Processing Systems*, NIPS ’22, Red Hook, NY, USA, 2022. Curran Associates Inc. ISBN 9781713871088.
- Meng, K., Sharma, A. S., Andonian, A. J., Belinkov, Y., and Bau, D. Mass-editing memory in a transformer. In *The Eleventh International Conference on Learning Representations, ICLR 2023, Kigali, Rwanda, May 1-5, 2023*. OpenReview.net, 2023. URL <https://openreview.net/forum?id=MkbcAHYgyS>.
- Nanda, N. and Bloom, J. Transformerlens. <https://github.com/TransformerLensOrg/TransformerLens>, 2022.
- Nanda, N., Chan, L., Lieberum, T., Smith, J., and Steinhardt, J. Progress measures for grokking via mechanistic interpretability, 2023. URL <https://arxiv.org/abs/2301.05217>.

- Nayak, N. S., Killamsetty, K., Han, L., Bhandwaldar, A., Chanda, P., Xu, K., Silkin, O., Eyceoz, M., Wang, H., Pareja, A., and Srivastava, A. Sculpting subspaces: Constrained full fine-tuning in LLMs for continual learning. In *The Fourteenth International Conference on Learning Representations*, 2026. URL <https://openreview.net/forum?id=vQcyqsGJDw>.
- Park, K., Choe, Y. J., and Veitch, V. The linear representation hypothesis and the geometry of large language models. In *Proceedings of the 41st International Conference on Machine Learning, ICML'24*. JMLR.org, 2024.
- Paulo, G., Shabalin, S., and Belrose, N. Transcoders beat sparse autoencoders for interpretability, 2025. URL <https://arxiv.org/abs/2501.18823>.
- Praggastis, B., Brown, D., Marrero, C. O., Purvine, E., Shapiro, M., and Wang, B. The svd of convolutional weights: A cnn interpretability framework, 2022. URL <https://arxiv.org/abs/2208.06894>.
- Radford, A., Wu, J., Child, R., Luan, D., Amodei, D., and Sutskever, I. Language models are unsupervised multitask learners. 2019. URL <https://api.semanticscholar.org/CorpusID:160025533>.
- Rajamanoharan, S., Conmy, A., Smith, L., Lieberum, T., Varma, V., Kramár, J., Shah, R., and Nanda, N. Improving sparse decomposition of language model activations with gated sparse autoencoders. In *Proceedings of the 38th International Conference on Neural Information Processing Systems, NIPS '24*, Red Hook, NY, USA, 2024a. Curran Associates Inc. ISBN 9798331314385.
- Rajamanoharan, S., Lieberum, T., Sonnerat, N., Conmy, A., Varma, V., Kramár, J., and Nanda, N. Jumping ahead: Improving reconstruction fidelity with jumprelu sparse autoencoders, 2024b. URL <https://arxiv.org/abs/2407.14435>.
- Rudin, C. Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead. *Nature Machine Intelligence*, 1(5):206–215, 2019. ISSN 25225839. doi: 10.1038/s42256-019-0048-x. URL <https://doi.org/10.1038/s42256-019-0048-x>.
- Saxe, A. M., McClelland, J. L., and Ganguli, S. Exact solutions to the nonlinear dynamics of learning in deep linear neural networks. *CoRR*, abs/1312.6120, 2013. URL <https://api.semanticscholar.org/CorpusID:17272965>.
- Scherlis, A., Sachan, K., Jermyn, A. S., Benton, J., and Shlegeris, B. Polysemanticity and capacity in neural networks. *CoRR*, abs/2210.01892, 2022. doi: 10.48550/ARXIV.2210.01892. URL <https://doi.org/10.48550/arXiv.2210.01892>.
- Syed, A., Rager, C., and Conmy, A. Attribution patching outperforms automated circuit discovery, 2023. URL <https://arxiv.org/abs/2310.10348>.
- Tang, Y., Saini, H., Yao, Z., Lin, Z., Liao, Y., Cui, J., Wang, Y., Du, M., and Liu, D. A unified theory of sparse dictionary learning in mechanistic interpretability: Piecewise biconvexity and spurious minima, 2026. URL <https://arxiv.org/abs/2512.05534>.
- Templeton, A., Conerly, T., Marcus, J., Lindsey, J., Bricken, T., Chen, B., Pearce, A., Citro, C., Ameisen, E., Jones, A., Cunningham, H., Turner, N. L., McDougall, C., MacDiarmid, M., Tamkin, A., Durmus, E., Hume, T., Mosconi, F., Freeman, C. D., Summers, T. R., Rees, E., Batson, J., Jermyn, A., Carter, S., Olah, C., and Henighan, T. Scaling monosemanticity: Extracting interpretable features from claude 3 sonnet, May 21 2024. URL <https://transformer-circuits.pub/2024/scaling-monosemanticity/>.
- Touvron, H., Martin, L., Stone, K., Albert, P., Almahairi, A., Babaei, Y., Bashlykov, N., Batra, S., Bhargava, P., Bhosale, S., Bikel, D., Blecher, L., Ferrer, C. C., Chen, M., Cucurull, G., Esiobu, D., Fernandes, J., Fu, J., Fu, W., Fuller, B., Gao, C., Goswami, V., Goyal, N., Hartshorn, A., Hosseini, S., Hou, R., Inan, H., Kardas, M., Kerkez, V., Khabsa, M., Kloumann, I., Korenev, A., Koura, P. S., Lachaux, M.-A., Lavril, T., Lee, J., Liskovich, D., Lu, Y., Mao, Y., Martinet, X., Mihaylov, T., Mishra, P., Molybog, I., Nie, Y., Poulton, A., Reizenstein, J., Rungta, R., Saladi, K., Schelten, A., Silva, R., Smith, E. M., Subramanian, R., Tan, X. E., Tang, B., Taylor, R., Williams, A., Kuan, J. X., Xu, P., Yan, Z., Zarov, I., Zhang, Y., Fan, A., Kambadur, M., Narang, S., Rodriguez, A., Stojnic, R., Edunov, S., and Scialom, T. Llama 2: Open foundation and fine-tuned chat models, 2023. URL <https://arxiv.org/abs/2307.09288>.
- Trenton, B., Adly, T., Joshua, B., Brian, C., Adam, J., Tom, C., Nicholas, L. T., Cem, A., Carson, D., Amanda, A., Robert, L., Yifan, W., Shauna, K., Nicholas, S., Tim, M., Nicholas, J., Alex, T., Karina, N., Brayden, M., Josiah, E. B., Tristan, H., Shan, C., Tom, H., and Chris, O. Towards monosemanticity: Decomposing language models with dictionary learning. 2023. URL <https://transformer-circuits.pub/2023/monosemantic-features>.
- Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, L., and Polosukhin, I. Attention is all you need. In *Proceedings of the 31st International*

Conference on Neural Information Processing Systems, NIPS’17, pp. 6000–6010, Red Hook, NY, USA, 2017. Curran Associates Inc. ISBN 9781510860964.

Wang, B. and Komatsuzaki, A. GPT-J-6B: A 6 Billion Parameter Autoregressive Language Model. <https://github.com/kingoflolz/mesh-transformer-jax>, May 2021.

Wang, K., Variengien, A., Conmy, A., Shlegeris, B., and Steinhardt, J. Interpretability in the wild: a circuit for indirect object identification in gpt-2 small, 2022. URL <https://arxiv.org/abs/2211.00593>.

Wei, J., Wang, X., Schuurmans, D., Bosma, M., Ichter, B., Xia, F., Chi, E. H., Le, Q. V., and Zhou, D. Chain-of-thought prompting elicits reasoning in large language models. In *Proceedings of the 36th International Conference on Neural Information Processing Systems*, NIPS ’22, Red Hook, NY, USA, 2022. Curran Associates Inc. ISBN 9781713871088.

Zhao, D., Liu, X., Feng, X., Wang, H., and Qin, B. Probing and boosting large language models capabilities via attention heads. In Christodoulopoulos, C., Chakraborty, T., Rose, C., and Peng, V. (eds.), *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pp. 28530–28544, Suzhou, China, November 2025. Association for Computational Linguistics. ISBN 979-8-89176-332-6. doi: 10.18653/v1/2025.emnlp-main.1450. URL <https://aclanthology.org/2025.emnlp-main.1450/>.

Zheng, Z., Wang, Y., Huang, Y., Song, S., Yang, M., Tang, B., Xiong, F., and Li, Z. Attention heads of large language models: A survey, 2024. URL <https://arxiv.org/abs/2409.03752>.

APPENDIX

Appendix Contents

A	Related Work	13
B	Discussion and Limitations	14
C	Threats to Validity	14
D	Why Rank-1 SVD Components Define Functional Subspaces	15
E	Experimental Setup: Datasets, Models, and Prompts	16
E.1	Datasets	16
E.2	Models and Prompt	16
F	Layer-wise Analysis of Highly Active DEU Subspaces	17
F.1	Consistently Activated Subspaces	17
F.2	Interpretation of Consistently Activated Subspaces	20
G	Computational Efficiency and Scalability	24
G.1	Computational Cost of Building the Interpretability Datastore	24
G.2	Runtime of Pathway Extraction	25
G.3	Runtime of Pathway Extraction and On-Demand Interpretation	25
H	Twenty Representative Cases with Pathway Extraction Analysis	26
H.1	20 Cases of Diverse Linguistic Phenomena	26
H.2	Evaluation of Pathway Extraction Across Twenty Linguistic Cases: Standard vs. Ablation	27
I	Subspace-Level Analysis of Model Behavior and Lexical Preferences	46
I.1	Case Study: Progressive Effect of Top- k Subspaces.	47
I.2	Layer-wise Distribution of Interpretable Subspaces	48
I.3	Investigating Subspace Contributions to Non-Rank-1 Token Generation	48
I.4	Additional Experiments on Model Lexical Preferences under Subspace Intervention	49
J	Additional Evaluation on Benchmark Datasets and Advanced SAE Variants	49
J.1	Evaluation on the Gender Pronoun Task	49
J.2	Evaluation on the Subject-Verb Agreement Task (Relative Clause)	50
J.3	Evaluation on the Subject-Verb Agreement Task (Simple Structure)	51
K	Properties of Highly Interpretable MLP Subspaces	51
K.1	Layer-wise Count of Highly Interpretable Subspaces	51
K.2	Layer-wise Distribution of Interpretable Subspaces	52
L	Effect of Starting Layer on Pathway Intervention Efficacy	53
M	Illustrative Examples of Interpretable Subspaces	54
M.1	Interpretation of DEU Subspaces Across Models	54
M.2	Semantic Consistency of Top-100 Tokens Captured by DEU Subspaces	59

A. Related Work

Sparse Dictionary Learning. To address the challenge of polysemanticity in neural representations (Elhage et al., 2022), sparse dictionary learning (Tang et al., 2026; Kreutz-Delgado et al., 2003) has emerged as a dominant paradigm for disentangling superposed features (Nanda et al., 2023; Templeton et al., 2024; Elhage et al., 2021). Pioneering work by Cunningham et al. (2023) demonstrated that training sparse autoencoders (SAEs) on model activations can effectively decompose superposition into semantically interpretable feature directions (Arora et al., 2018). Building on this foundation, Trenton et al. (2023) established the scalability of SAEs for extracting interpretable features from large-scale models, enabling systematic circuit-level analyses. Their study further provided a detailed characterization of SAE scaling behavior, along with a taxonomy of learned features across model sizes. Subsequent research has introduced architectural refinements to mitigate specific limitations of standard SAEs. For instance, Rajamanoharan et al. (2024a) addressed feature shrinkage by decoupling direction learning from magnitude estimation, applying L_1 regularization exclusively to a gating mechanism to reduce undesirable suppression effects. In parallel, Gao et al. (2024) proposed k -sparse autoencoders, which impose a hard k -sparsity constraint to explicitly control the sparsity budget (Makhzani & Frey, 2014). Beyond reconstruction-based approaches, Dunefsky et al. (2024) introduced the TransCoder, which learns a direct mapping from MLP inputs to outputs to simulate layer-level functionality (Gao et al., 2024; Paulo et al., 2025). Despite these advancements, existing approaches largely rely on *post-hoc* proxies. These auxiliary models require substantial computational resources for training and

inevitably introduce reconstruction errors, potentially leading to deviations from the original network’s true behavior.

Linear Representations and Mechanistic Interpretability of MLPs. Our work is grounded in the Linear Representation Hypothesis (Park et al., 2024), which posits that semantic concepts are encoded as approximately linear directions in activation space. This linearity assumption provides a principled basis for applying spectral methods to decompose model weights. Within Transformer Feed-Forward Networks (FFNs), Geva et al. (2021) conceptualized each MLP layer as a Key–Value Memory, where the first linear transformation acts as a “key” to detect input patterns, and the second as a “value” to project features into the output space. Building on this view, Geva et al. (2022) further demonstrated that FFNs operate by selectively promoting specific concepts toward the output space. To empirically validate such hypotheses, the field has extensively relied on activation patching and causal tracing to localize and attribute model behaviors (Syed et al., 2023; Wang et al., 2022; Dumas et al., 2025). Notably, Meng et al. (2022) leveraged causal tracing to pinpoint factual knowledge storage, introducing Rank-One Model Editing (ROME) to manipulate down-projection weights (Meng et al., 2023; Li et al., 2024). Similarly, Conmy et al. (2023) proposed automated circuit discovery via recursive patching techniques (Heimersheim & Nanda, 2024). In a separate line of work, Beren & Black (2022) observed that the singular vectors of Transformer weight matrices admit meaningful semantic interpretations (Praggastis et al., 2022). Nevertheless, such spectral analyses have largely remained qualitative or static, lacking a rigorous causal framework for systematically linking individual spectral components to concrete model behaviors. Along this line of work, Ahmad et al. (2026) proposed learning sparse continuous masks over singular directions to identify the key singular vectors that supported a model’s performance on specific tasks. However, this method relies on dataset-specific training optimization and remains limited in enabling targeted, fine-grained attribution and interpretability at the single-sample (prompt–target) level.

While Sparse Autoencoders (Cunningham et al., 2023) and TransCoders (Dunefsky et al., 2024) effectively disentangle polysemantic representations, they act as computationally expensive proxies that approximate, rather than faithfully reflect, the underlying model. Conversely, classical linear analyses (Beren & Black, 2022; Praggastis et al., 2022) avoid proxy training but offer only static interpretations, lacking the causal machinery required to identify functionally relevant pathways. Our work bridges this methodological gap by introducing the **Native Network Anatomy** framework (NaNA). Unlike prior spectral approaches, NaNA extends static weight inspection to dynamic causal attribution. Crucially, unlike learned dictionary methods, it identifies sparse, dominant computational pathways directly from the native spectral decomposition of pre-trained weights, achieving exact functional fidelity without additional training.

B. Discussion and Limitations

While our proposed training-free framework, NaNA, demonstrates significant advantages in both efficiency and fidelity, we acknowledge certain limitations that delineate avenues for future research.

Architectural Scope and Applicability. Our empirical validation is currently focused on the GPT-2 family. The generalizability of NaNA to contemporary LLM architectures with distinct structural paradigms, such as Mixture-of-Experts (MoE) models (Fedus et al., 2022; Jiang et al., 2024), remains an open question. Furthermore, our analysis is strictly confined to MLP layers. Although MLPs constitute the majority of model parameters, attention heads play a pivotal role in information routing and context integration. Extending the spectral decomposition methodology to encompass attention mechanisms represents a critical direction for future work.

Prediction Paradigm and Task Scope. Consistent with prior work on TransCoders and SAEs, our current analysis is primarily grounded in the next-token prediction paradigm. Although this objective underlies the fundamental training of LLMs, interpretations based solely on single-step predictions may not fully capture the models’ macroscopic behavior in complex tasks, such as multi-step reasoning or long-text generation. A key open question is how the local computational pathways identified by NaNA compose to support higher-level capabilities, including Chain-of-Thought reasoning (Wei et al., 2022).

C. Threats to Validity

We acknowledge that the structural heterogeneity between DEU subspaces and baseline feature representations (i.e., SAEs and TransCoders) presents a potential challenge to comparative validity. Baseline methods typically rely on a high-dimensional, overcomplete dictionary, whereas the NaNA framework derives orthogonal DEU subspaces directly from pretrained model weights. This structural disparity implies that neither the intrinsic nature nor the information density per unit is inherently equivalent. Following Dunefsky et al. (2024), we mitigate this concern by imposing a strict sparsity

constraint on the number of active components. This ensures a fair comparison of reconstruction efficiency by focusing on the number of distinct computational units required to recover the target.

A second limitation stems from our reliance on an external LLM for automated subspace interpretation. We recognize that the validity of such assessments is inherently contingent upon the semantic reasoning capabilities and potential biases of the evaluator model. To ensure reproducibility and mitigate variability, we provide the full model configurations and exact prompt templates used in our experiments, facilitating transparent replication and verification of our results.

D. Why Rank-1 SVD Components Define Functional Subspaces

D.1. Theoretical properties distinguishing an SVD-induced basis from an arbitrary orthogonal basis

Following the optimal low-rank approximation (Eckart–Young theorem) (Eckart & Young, 1936), among all rank- k decompositions, the top- k SVD terms minimise the approximation error to the full weight matrix under the Frobenius norm (or any unitarily invariant norm). This means that each additional rank-1 SVD term captures the maximum possible remaining influence on the matrix’s action. For example, a rank-1 component $u_i \sigma_i v_i^\top$ represents the direction in the input space (v_i) that undergoes the maximum possible amplification (σ_i) into the output space (u_i). A random orthogonal basis enjoys no such ordering: its components do not correspond to any notion of functional contribution, and there is no principled way to rank or select among them. In more informal terms, because the SVD isolates the directions of maximum variance, these components represent the most “dominant” pathways through the layer. This is one of the aspects making this basis special.

Further justification comes from the diagonalization property of the SVD. By construction, $W = U\Sigma V^\top$ maps each input direction v_i exclusively onto the output direction u_i , scaled by σ_i , with zero coupling to any other output direction $u_j, j \neq i$. This is not true of an arbitrary orthogonal decomposition, where cross-terms between different basis directions generally remain. As a consequence, each rank-1 SVD term represents a linearly decoupled input–output channel: activating detector v_i produces a response only in effector u_i , not in any other effector. It is in this precise sense, and not merely because the basis vectors happen to be orthogonal as is true of any orthonormal basis, that SVD-induced Detector-Effector units operate with minimal interference between one another, representing the linearly decoupled functional streams the model has learned (Nayak et al., 2026; Gargiulo et al., 2025).

Another indication (yet not a rigid proof) that the SVD basis is privileged comes from studies on the learning dynamics of gradient descent (e.g., Saxe et al. (2013)). This work demonstrates that for deep linear neural networks, gradient descent tends to learn independent input-output mappings sequentially, exactly along the singular vectors of the task’s structural covariance. The network does not learn in an arbitrary orthogonal basis; it explicitly builds its weights by incrementally adding capacity along the principal directions of the data manifold. This provides an existence proof that SVD-aligned structure can emerge from gradient-based learning, rather than being an arbitrary post-hoc decomposition.

D.2. Empirical validation

We provide a series of evidence showing that high-contribution SVD components behave as functional subspaces, rather than merely convenient basis vectors:

Dominance in token prediction. If SVD components were merely a basis artefact, one would expect that predicting a specific token would require a diffuse mixture of many components, and that the identity of the dominant components would vary arbitrarily across prompts. Instead, Sections 4.2 and 4.3 show that a small, consistent set of high-contribution rank-1 terms suffices to dominate target-token prediction, while ablating them causes a marked drop. This selectivity, with specific components for specific predictions, is the empirical signature of functional separability.

Cross-prompt robustness. Section 4.3 (Experiment 2) and Appendix F use Controlled Variation Sets to show that the same subspaces recur as dominant contributors across prompts sharing a common semantic target (e.g., positive-adjective semantics), even when surface form and prefix vary. An arbitrary orthogonal basis would not exhibit this consistency; the fact that SVD components do suggests they track something functionally real in the network’s computation.

E. Experimental Setup: Datasets, Models, and Prompts

E.1. Datasets.

This section details the experimental datasets constructed to evaluate model behavior and pathway generalization. We categorize our data into three primary subsets: **Anchor Cases** for broad linguistic coverage, **Controlled Variation Sets** for isolating semantic variables, and a **Generalization Dataset** for systematic assessment. All datasets are provided in the *Supplementary Material*.

- **Anchor Cases ($\mathcal{D}_{\text{achr}}$).** To facilitate in-depth probing, we selected 20 representative instances spanning diverse functional capabilities, including syntactic dependency, semantic completion, code generation, and factual retrieval. These cases are designed to isolate and observe distinct mechanisms within the model. The complete list is detailed in Table 8.
- **Generalization Dataset ($\mathcal{D}_{\text{gene}}$).** We constructed $\mathcal{D}_{\text{gene}}$ to examine the behavioral generalization of the pathway across diverse contexts in positive-adjective semantics. For each model m in the GPT-2 family, we generated 100 diverse prefixes c_i . To ensure alignment with the model’s native distribution, the target token $t_i^{(m)}$ for each instance is defined as the model’s original top-1 prediction. This yields a model-specific dataset $\mathcal{D}_{\text{gene}}^{(m)} = \{(c_i, t_i^{(m)})\}$ tailored to the predictive behavior of each variant.
- **Controlled Variation Sets.** To assess the robustness of subspaces under contextual perturbations, we construct two complementary datasets focused on positive-adjective semantics:
 - *Target Variance Dataset ($\mathcal{D}_{\text{target}}$).* The target token is varied while the prefix is held fixed. Specifically, we use a fixed prefix $c_{\text{fix}} = \text{“The cat looks very”}$ paired with a set of 100 distinct positive adjectives $T = \{t_1, t_2, \dots, t_{100}\}$ (e.g., “happy”, “sunny”, “glad”, “calm”). Formally, the dataset is defined as $\mathcal{D}_{\text{target}} = \{(c_{\text{fix}}, t_i) \mid t_i \in T\}$.
 - *Prefix Variance Dataset ($\mathcal{D}_{\text{prefix}}$).* The prefix is varied while the target token is held constant. We generate 100 diverse prefixes $C = \{c_1, c_2, \dots, c_{100}\}$ (e.g., “The dancer appears very”, “The baby seems truly”) paired with a fixed target token $t_{\text{fix}} = \text{“happy”}$. Formally, the dataset is defined as $\mathcal{D}_{\text{prefix}} = \{(c_i, t_{\text{fix}}) \mid c_i \in C\}$.

E.2. Models and Prompt.

This section describes the models and prompts used in our experiments. A complete list of evaluated models and their variants is provided in Table 3. Furthermore, the prompt and model employed for the subspace interpretability analysis are illustrated in Figure 9. All experiments were executed on a single **NVIDIA A100 GPU (80GB)**.

Table 3. Summary of the GPT-2 Family, TransCoders, and SAEs Used in Experiments.

Model	Hidden Size	MLP Dim	Layers	Access Link	Reference
GPT-2	768	3072	12	openai-community/gpt2	Radford et al. (2019)
GPT2-Medium	1024	4096	24	openai-community/gpt2-medium	Radford et al. (2019)
GPT2-Large	1280	5120	36	openai-community/gpt2-large	Radford et al. (2019)
GPT2-XL	1600	6400	48	openai-community/gpt2-xl	Radford et al. (2019)
GPT-Neo	2560	10240	32	EleutherAI/gpt-neo-2.7B	Gao et al. (2020)
GPT-J	4096	16384	28	EleutherAI/gpt-j-6b	Wang & Komatsuzaki (2021)
TransCoder	768	24576	-	pchenski/gpt2-transcoders	Dunefsky et al. (2024)
SAE	768	24576	-	tommcmgrath/gpt2-small-mlp-out-saes	Cunningham et al. (2023); Bloom et al. (2024)
JumpReLU SAE	768	24576	-	Codebase	Rajamanoharan et al. (2024b)
Top- k SAE	768	24576	-	Codebase	Gao et al. (2025)
BatchTopK SAE	768	24576	-	Codebase	Bussmann et al. (2024)

²<https://platform.openai.com/docs/models/compare?model=gpt-4o-mini>

SYSTEM PROMPT TEMPLATE

Role: You are an expert in neural network interpretability. Reply strictly in JSON format.

User Instruction: Analyze the semantic consistency of the following direction.

Input Data:

- **Direction ID:** <ID> (<Polarity>)
- **Top Tokens:** [<Token List>]

Analysis Criteria: My goal is to identify directions that represent a specific semantic meanings or grammatical functions (concepts, topics, or distinct grammatical roles).

1. **Concept Label:**

- Summarize the common meaning using 2-4 English words.
- Do NOT use labels like “Subwords” or “Prefixes” – these are structural, not semantic. If no semantic theme exists, label it *null*.

2. **Consistency Score:** Assign strictly based on MEANING, not spelling.

- **High:** Tokens clearly share a specific concept.
- **Medium:** Tokens are thematically related but broad.
- **Low:** Tokens have totally unrelated meanings. If tokens share structure (e.g., all start with ‘#’) but have unrelated meanings, score as *low*.

Output Format:

{ { “label”: “Your concept label” or null, “score”: “High” / “Medium” / “Low” or null } }

Layer	Direction	Top Activating Tokens	Label	Score
L26_SP15	Positive	['compliant', 'qualified', 'certified', 'suitable', 'suited', 'wisely', 'capable', 'preferred', 'licensed', 'ideal', 'recommended', 'approved', 'proficient', 'Certified', 'worthy', 'specialize', 'authorized', '#Works', 'competent', 'compatible']	qualified approval	High
	Negative	['disruptions', 'disturbances', 'outbreaks', 'earthquakes', 'storms', 'glitches', 'crashes', 'collisions', 'riots', 'explosions', 'casualties', 'occurrences', 'spills', 'turbulence', 'floods', 'surges', '#effects', 'scandals', 'leakage', 'wildfires']	natural disasters	High

Figure 9. Prompt Design for Automated Interpretation. For example, we illustrate the interpretation of Subspace 15 in Layer 26 of W_{down} (GPT2-XL). The “Positive” and “Negative” directions correspond to the semantic concepts aligned with the opposing directions of the singular vector ($\pm \mathbf{u}_i$), reflecting the inherent sign ambiguity in SVD (Kolda & Bader, 2009). Here, *gpt-4o-mini*² is used to provide semantic interpretations for the subspaces.

F. Layer-wise Analysis of Highly Active DEU Subspaces

In this section, we examine the layer-wise highly active subspaces in the GPT-2 family. This analysis assesses whether specific subspaces contribute consistently and robustly to target semantic features, and further examines whether these highly active subspaces exhibit direct and interpretable associations with particular semantic scenarios. Because subspaces identified under the target variance setting ($\mathcal{D}_{\text{target}}$) more directly reflect intrinsic semantic structure, whereas the prefix variance setting ($\mathcal{D}_{\text{prefix}}$) is comparatively more susceptible to contextual noise, we use the former for primary analysis and the latter for validation.

F.1. Consistently Activated Subspaces

Figures 10-13 illustrate the ten most active subspaces, defined as those that most frequently appear among the top-10 contributors across both target token variants ($\mathcal{D}_{\text{target}}$) and prefix variants ($\mathcal{D}_{\text{prefix}}$). We observe that only a small subset of subspaces appears with high frequency within each layer, indicating that semantic contributions are highly concentrated rather than uniformly distributed, and that these subspaces function as dominant semantic carriers. This suggests that a limited number of subspaces play a disproportionately central and stable role in mediating specific semantic features.

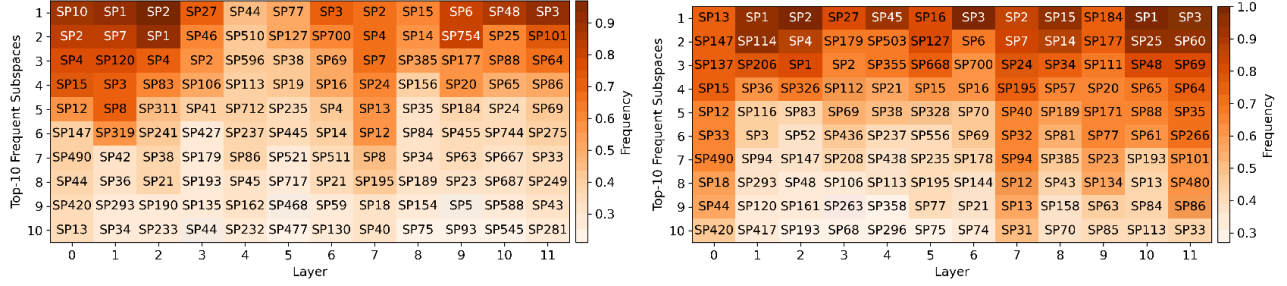


Figure 10. Layer-wise Consistency of Top-10 Subspace Activations Across Target Token Variants ($\mathcal{D}_{\text{target}}$) and Prefix Variants ($\mathcal{D}_{\text{prefix}}$) in GPT-2. The heatmaps illustrate, for each layer, the frequency with which each subspace appears among the top-10 contributors across all variants, thereby identifying subspaces that are consistently activated across diverse target tokens. Here, SP_i denotes subspace i .

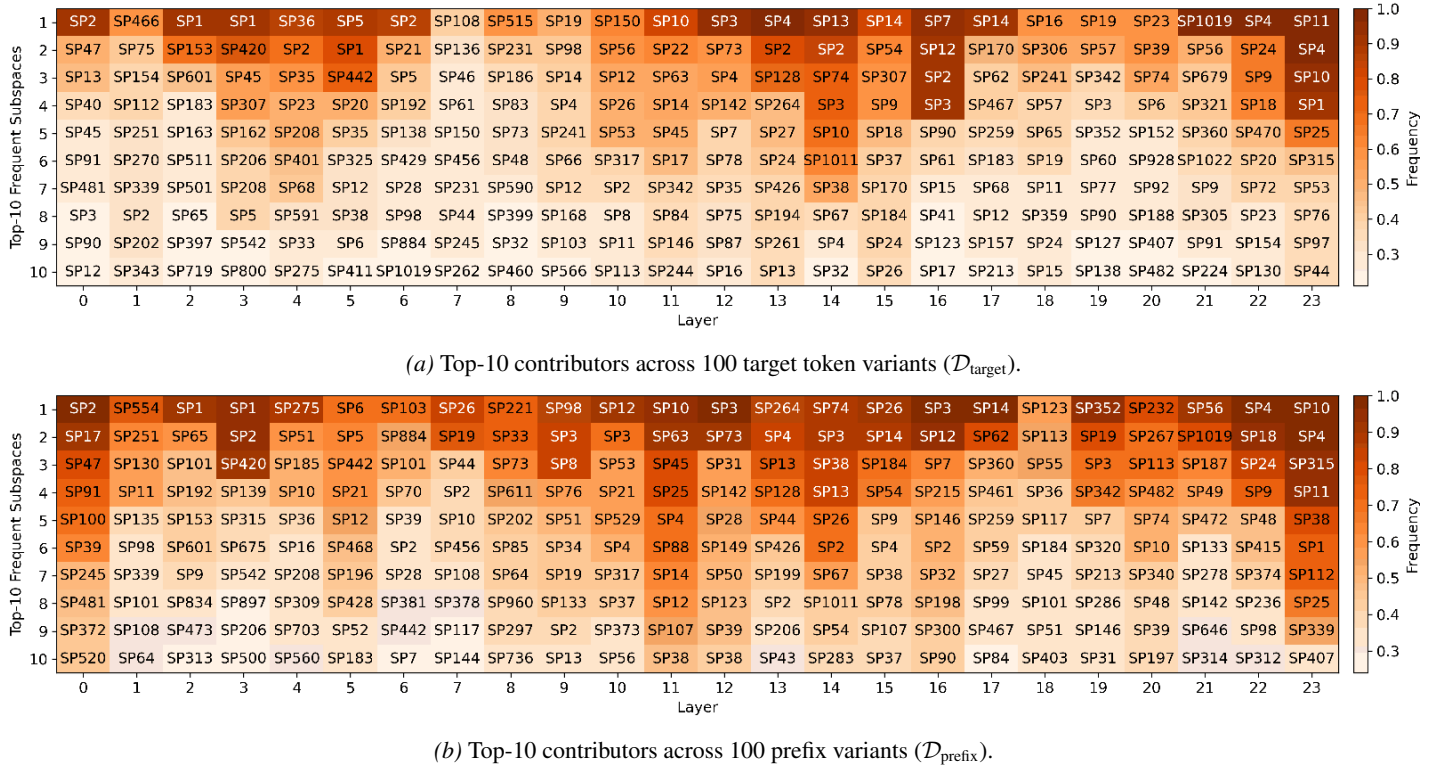
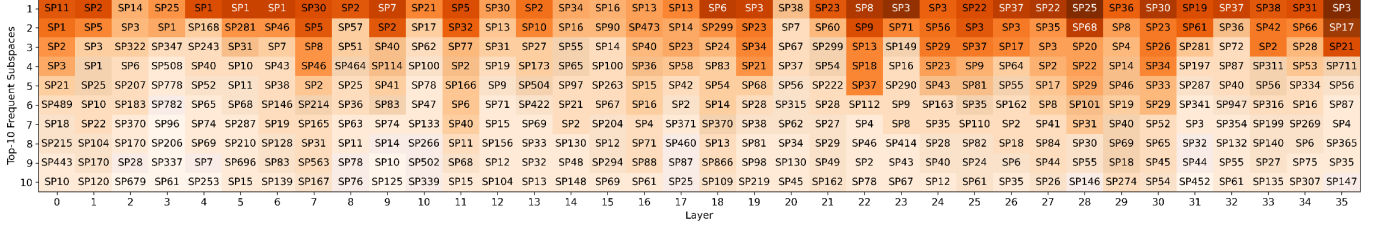
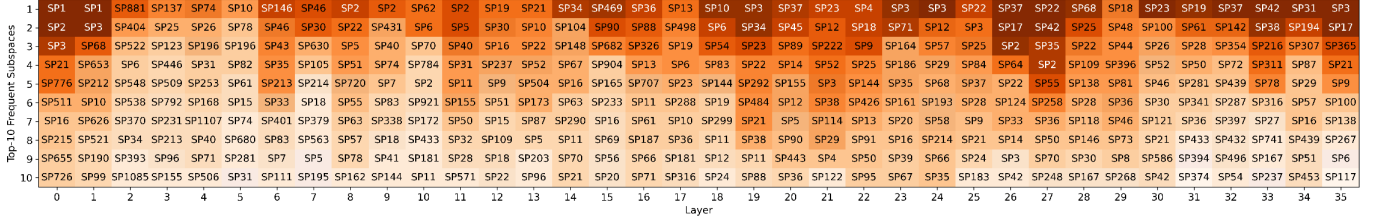


Figure 11. Layer-wise Consistency of Top-10 Subspace Activations Across Target Token Variants ($\mathcal{D}_{\text{target}}$) and Prefix Variants ($\mathcal{D}_{\text{prefix}}$) in GPT2-Medium. The heatmaps illustrate, for each layer, the frequency with which each subspace appears among the top-10 contributors across all variants, thereby identifying subspaces that are consistently activated across diverse target tokens. Here, SP_i denotes subspace i .

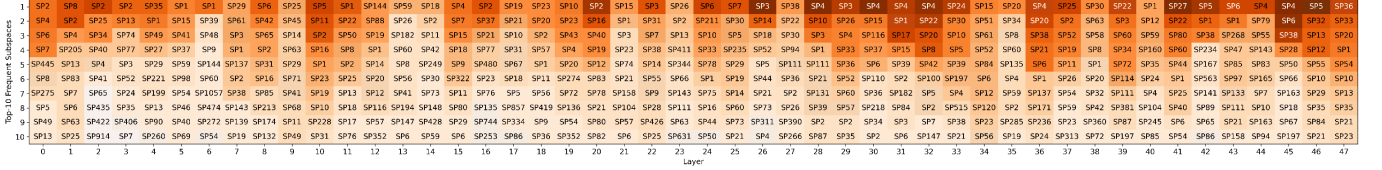


(a) Top-10 contributors across 100 target token variants ($\mathcal{D}_{\text{target}}$).

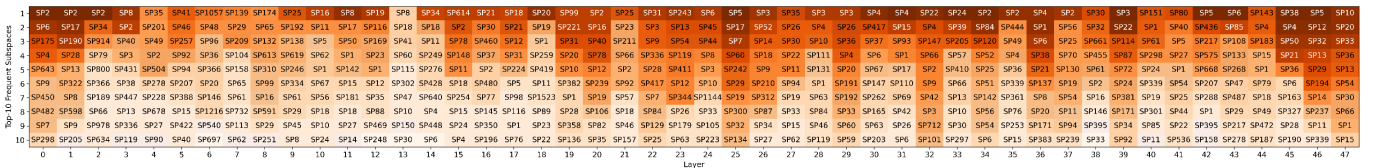


(b) Top-10 contributors across 100 prompt variants ($\mathcal{D}_{\text{prefix}}$).

Figure 12. Layer-wise Consistency of Top-10 Subspace Activations Across Target Token Variants ($\mathcal{D}_{\text{target}}$) and Prefix Variants ($\mathcal{D}_{\text{prefix}}$) in GPT2-Large. The heatmaps illustrate, for each layer, the frequency with which each subspace appears among the top-10 contributors across all variants, thereby identifying subspaces that are consistently activated across diverse target tokens. Here, SP_i denotes subspace i .



(a) Top-10 contributors across 100 target token variants ($\mathcal{D}_{\text{target}}$).



(b) Top-10 contributors across 100 prompt variants ($\mathcal{D}_{\text{prefix}}$).

Figure 13. Layer-wise Consistency of Top-10 Subspace Activations Across Target Token Variants ($\mathcal{D}_{\text{target}}$) and Prefix Variants ($\mathcal{D}_{\text{prefix}}$) in GPT2-XL. The heatmaps illustrate, for each layer, the frequency with which each subspace appears among the top-10 contributors across all variants, thereby identifying subspaces that are consistently activated across diverse target tokens. Here, SP_i denotes subspace i .

F.2. Interpretation of Consistently Activated Subspaces

In this section, we examine whether these highly active subspaces exhibit direct and interpretable associations with specific semantic contexts. Tables 4-7 report the interpretability analysis of these subspaces under $\mathcal{D}_{\text{target}}$, where $Lm.SP_i$ denotes subspace i in layer m . The result indicates that these high-frequency subspaces are typically related to positive semantics. It suggests that the model’s predictions are mediated by specific, functional subspaces rather than by diffuse representations. Multiple subspaces, while semantically distinct, exhibit strong correlations and jointly contribute to robust predictions of positive semantics.

Table 4. **Layer-wise Interpretation of Highly Active Subspaces in GPT-2.** Here, parentheses denote subspace interpretability levels (High, Med, or Low). **Green** highlights denote subspaces associated with positive semantics.

ID	Positive Direction	Negative Direction	ID	Positive Direction	Negative Direction
L0.SP10	Retail and Online Shopping (Med)	Prep. & Conj. (High)	L7.SP2	<i>null</i> (null)	<i>null</i> (null)
L0.SP2	punctuation marks (High)	Retail and Nature (Med)	L7.SP4	Legal and Social Issues (Med)	Photography Terms (Med)
L0.SP4	punctuation marks (High)	<i>null</i> (null)	L7.SP7	Neglect and Awareness (High)	game activities (High)
L0.SP15	<i>null</i> (null)	Content Management (Med)	L7.SP24	Prison-related terms (Med)	mindfulness and diligence (High)
L0.SP12	<i>null</i> (null)	Awakening Determination (Med)	L7.SP13	<i>null</i> (null)	recent events (High)
L1.SP1	function words (High)	<i>null</i> (null)	L7.SP12	death and decline (High)	Review and Insight (High)
L1.SP7	function words (High)	Retail and Delivery (Med)	L8.SP15	advantageous roles (Med)	time periods (Med)
L1.SP120	real estate topics (Med)	<i>null</i> (null)	L8.SP14	value and importance (High)	Deletion actions (High)
L1.SP3	Wrestling concepts (High)	Magnitudes and Rarity (Med)	L8.SP385	Flavoring Concepts (Med)	<i>null</i> (null)
L1.SP8	punctuation&conjunctions(Med)	Dragon Magazine (Med)	L9.SP6	Location references (High)	<i>null</i> (null)
L1.SP319	Menopause Related Terms (Med)	<i>null</i> (null)	L9.SP754	<i>null</i> (null)	<i>null</i> (null)
L2.SP2	<i>null</i> (null)	Utility Organizations (High)	L9.SP177	<i>null</i> (null)	political themes (Med)
L2.SP1	common conjunctions (High)	<i>null</i> (null)	L9.SP20	<i>null</i> (null)	<i>null</i> (null)
L2.SP4	<i>null</i> (null)	Travel and Logistics (Med)	L10.SP48	day references (Med)	oratory terms (Med)
L2.SP83	<i>null</i> (null)	Media and Learning (Med)	L10.SP25	<i>null</i> (null)	<i>null</i> (null)
L3.SP27	<i>null</i> (null)	social dynamics (Med)	L10.SP88	communication expressions (Med)	Skiing Activities (Med)
L3.SP46	Criminality and Exclusion (High)	advertising concepts (High)	L11.SP3	Retail and Confirmation (Med)	punctuation usage (High)
L3.SP2	Chance and Opportunity (Med)	<i>null</i> (null)	L11.SP101	system concepts (Med)	Nationalities and Locations (High)
L3.SP106	<i>null</i> (null)	Food items (Med)	L11.SP64	ability concepts (High)	Locations and Places (Med)
L6.SP3	User Interface Elements (Med)	<i>null</i> (null)	L11.SP86	advice and guidance (Med)	Emotional expressions (Med)
L6.SP700	<i>null</i> (null)	Public Figures (Med)			

Table 5. **Layer-wise Interpretation of Highly Active Subspaces in GPT2-Medium.** Here, parentheses denote subspace interpretability levels (High, Med, or Low). **Green** highlights denote subspaces associated with positive semantics.

ID	Positive Direction	Negative Direction	ID	Positive Direction	Negative Direction
L0.SP2	<i>null</i> (null)	punctuation usage (High)	L14.SP13	<i>null</i> (null)	vibrant leisure (High)
L1.SP466	Rewards and Characters (Med)	Detention and Control (Med)	L14.SP2	<i>null</i> (null)	Punctuation and Symbols (High)
L2.SP1	Common conjunctions (High)	<i>null</i> (null)	L14.SP74	Government Challenges (Med)	<i>null</i> (null)
L2.SP153	Cultural References (Med)	<i>null</i> (null)	L14.SP3	electricity concepts (Med)	mental states (Med)
L3.SP1	<i>null</i> (null)	punctuation usage (Med)	L14.SP10	Positive Excellence (High)	Measurement and Time (Med)
L3.SP420	Government and Authority (Med)	<i>null</i> (null)	L14.SP101	struggle and challenges (Med)	<i>null</i> (null)
L3.SP45	<i>null</i> (null)	Information Retrieval (Med)	L14.SP38	termination concepts (Med)	indifference and desire (Med)
L3.SP307	Management Concepts (Med)	Political Terms (Med)	L15.SP14	updates and terminations (High)	emotional experiences (High)
L4.SP36	virtual balance (Med)	Social Events (Med)	L15.SP54	Geographical Terms (Med)	Social Media Roles (Med)
L4.SP2	<i>null</i> (null)	punctuation&conjunctions (Med)	L15.SP307	Weddings and Events (Med)	<i>null</i> (null)
L4.SP35	Nebraska and related terms (Med)	Critique and Analysis (High)	L15.SP9	similarity concepts (High)	travel actions (High)
L5.SP5	<i>null</i> (null)	Energy Industries (High)	L16.SP7	Government Deaths (High)	positive qualities (High)
L5.SP1	articles and conjunctions (High)	<i>null</i> (null)	L16.SP12	Respect and Worthiness (High)	Numbers and Quantities (High)
L5.SP442	Media and Communication (Med)	<i>null</i> (null)	L16.SP2	Punctuation Marks (High)	<i>null</i> (null)
L6.SP2	<i>null</i> (null)	Political Issues (Med)	L16.SP3	Observation and Critique (Med)	electric devices (High)
L8.SP515	Addiction and Faces (Med)	Conflict and Grievance (High)	L17.SP14	Cognitive Values (High)	Security incidents (High)
L10.SP150	<i>null</i> (null)	<i>null</i> (null)	L18.SP16	one and variations (High)	second person pronouns (High)
L10.SP56	<i>null</i> (null)	Business Management (Med)	L18.SP306	Data Management (Med)	<i>null</i> (null)
L10.SP12	Scientific terminology (Med)	<i>null</i> (null)	L19.SP19	<i>null</i> (null)	<i>null</i> (null)
L11.SP10	Names and Places (Med)	Quantity and Rarity (Med)	L19.SP57	Face Tracking (Med)	Advertising Concepts (Med)
L11.SP22	trustworthiness attributes (High)	<i>null</i> (null)	L20.SP23	Outsider references (Med)	action or activity (High)
L11.SP63	Food Quality (Med)	Financial Terms (Med)	L20.SP39	<i>null</i> (null)	Creation and Time (Med)
L11.SP14	Education and Travel (Med)	negatives and errors (Med)	L20.SP74	work and charity (Med)	growth and vitality (Med)
L12.SP3	<i>null</i> (null)	<i>null</i> (null)	L21.SP1019	<i>null</i> (null)	Marketing Concepts (Med)
L12.SP73	<i>null</i> (null)	Positive Traits (High)	L22.SP4	<i>null</i> (null)	articles and conjunctions (High)
L12.SP4	<i>null</i> (null)	people and organizations (High)	L22.SP24	<i>null</i> (null)	people and workers (High)
L13.SP4	game items (Med)	warning concepts (Med)	L22.SP9	Nationalities and Places (Med)	Prep. & Conj. (High)
L13.SP2	<i>null</i> (null)	Punctuation and Symbols (Med)	L22.SP18	comparative adjectives (High)	action verbs (Med)
L13.SP128	education assessment (Med)	<i>null</i> (null)	L23.SP11	the definite article (High)	Gaming References (Med)
			L23.SP4	<i>null</i> (null)	punctuation&conjunctions (Med)
			L23.SP10	<i>null</i> (null)	of and related (High)
			L23.SP1	<i>null</i> (null)	punctuation&conjunctions (Med)
			L23.SP25	Creation and Discussion (Med)	Veterinary Medicine (High)

Table 6. **Layer-wise Interpretation of Highly Active Subspaces in GPT2-Large.** Here, parentheses denote subspace interpretability levels (High, Med, or Low). **Green** highlights denote subspaces associated with positive semantics.

ID	Positive Direction	Negative Direction	ID	Positive Direction	Negative Direction
L0.SP11	<i>null</i> (null)	Voting and Associations (Med)	L23.SP3	Fictional Characters (High)	challenges and limitations (High)
L0.SP1	Media and Gaming (Med)	<i>null</i> (null)	L23.SP71	emotional decline (High)	impact assessment (Med)
L0.SP2	Financial Analysis (High)	<i>null</i> (null)	L24.SP3	Names of People (High)	action and consequences (High)
L1.SP2	Government Support (Med)	material properties (High)	L24.SP56	Celebration and Excellence (High)	Statements and Predictions (High)
L1.SP5	<i>null</i> (null)	Music and Culture (Med)	L24.SP29	management concepts (High)	vivid sensations (High)
L3.SP25	Celebration Events (Med)	Developmental Concepts (Med)	L25.SP22	investigation actions (High)	positive attributes (High)
L4.SP1	<i>null</i> (null)	social issues (Med)	L25.SP3	Names of People (High)	discussion engagement (High)
L5.SP1	<i>null</i> (null)	people and actions (Med)	L25.SP37	modification terms (Med)	bravery and acclaim (High)
L5.SP281	Courage and Visibility (Med)	<i>null</i> (null)	L26.SP37	negative traits (High)	historical interpretations (High)
L6.SP1	people and actions (Med)	<i>null</i> (null)	L26.SP3	Names and Titles (High)	<i>null</i> (null)
L6.SP46	Health and Technology (Med)	Tools and Equipment (Med)	L26.SP17	<i>null</i> (null)	Positive Emotions (High)
L7.SP30	Cells and Structures (Med)	<i>null</i> (null)	L27.SP22	uncertainty concepts (High)	<i>null</i> (null)
L7.SP5	overcoming challenges (High)	Screen elements (Med)	L27.SP35	protection and safety (High)	Adventure Themes (Med)
L7.SP8	visual representation (High)	Raids and Settlements (High)	L27.SP3	<i>null</i> (null)	thematic references (Med)
L7.SP46	official documentation (Med)	<i>null</i> (null)	L28.SP25	Positive Attributes (High)	nature elements (Med)
L8.SP2	<i>null</i> (null)	general expressions (Med)	L28.SP68	war and conflict (High)	carelessness and rights (Med)
L9.SP7	<i>null</i> (null)	Booking and Recording (Med)	L29.SP36	rejection themes (Med)	Between Points (Med)
L9.SP2	conjunctions and discourse (Med)	<i>null</i> (null)	L30.SP30	extreme quality (High)	<i>null</i> (null)
L10.SP21	sports competition (Med)	<i>null</i> (null)	L30.SP23	<i>null</i> (null)	data representation (Med)
L11.SP5	<i>null</i> (null)	real estate activities (High)	L30.SP26	stability and safety (Med)	<i>null</i> (null)
L11.SP32	competition winners (High)	development abilities (Med)	L30.SP34	Quality Descriptors (High)	<i>null</i> (null)
L12.SP30	natural disasters (High)	Fashion and Style (Med)	L31.SP19	Punctuation marks (High)	Punctuation and Symbols (High)
L13.SP2	Unleashed Outbreak (Med)	<i>null</i> (null)	L31.SP61	<i>null</i> (null)	<i>null</i> (null)
L13.SP10	surrounding contexts (High)	<i>null</i> (null)	L32.SP37	Right Direction (Med)	Gaming Entities (High)
L17.SP13	Apocalyptic Events (High)	collaboration roles (High)	L33.SP38	<i>null</i> (null)	Financial Transactions (High)
L18.SP6	uniqueness and perfection (Med)	International Organizations (High)	L33.SP42	Danger and States (Med)	public technology (Med)
L18.SP299	Licensing&Engagements (Med)	<i>null</i> (null)	L33.SP2	<i>null</i> (null)	Prep. & Conj. (High)
L19.SP3	Measurement Terms (Med)	Famous Individuals (High)	L34.SP31	Time references (High)	Ordinal Numbers (High)
L19.SP23	Financial Disqualification (High)	creativity and inspiration (High)	L34.SP66	power and energy (High)	Nautical Terms (Med)
L19.SP34	monitoring techniques (Med)	Preferred Choices (High)	L35.SP3	causal conjunctions (Med)	<i>null</i> (null)
L19.SP21	Positive Feedback (High)	underground structures (High)	L35.SP17	<i>null</i> (null)	Years Mentioned (High)
L21.SP23	chaos management (Med)	Friendship and Kindness (High)	L35.SP21	degree comparison (High)	Geographical Entities (High)
L22.SP8	abandonment themes (Med)	inspiration and insight (High)			
L22.SP9	Event References (High)	human behaviors (High)			
L22.SP13	Advocacy and Roles (High)	quantitative measures (High)			
L22.SP18	struggles and recovery (High)	Geometric Shapes (High)			
L22.SP37	change and history (High)	emotional distress (High)			

Table 7. **Layer-wise Interpretation of Highly Active Subspaces in GPT2-XL.** Here, parentheses denote subspace interpretability levels (High, Med, or Low). **Green** highlights denote subspaces associated with positive semantics.

ID	Positive Direction	Negative Direction	ID	Positive Direction	Negative Direction
L0.SP2	Behavioral Goals (Med)	<i>null</i> (null)	L30.SP4	cautionary observations (High)	<i>null</i> (null)
L0.SP4	Financial Instruments (High)	<i>null</i> (null)	L30.SP15	social interactions (High)	Administrative Terms (Med)
L0.SP6	Financial Terms (Med)	Friendship Concepts (Med)	L30.SP116	Media and Culture (Med)	<i>null</i> (null)
L0.SP7	Travel and Hospitality (Med)	environmental concepts (Med)	L30.SP37	resource allocation (Med)	daily activities (High)
L1.SP8	<i>null</i> (null)	Organizations&Agencies (High)	L31.SP4	<i>null</i> (null)	emotional responses (High)
L1.SP2	Media and Promotion (Med)	<i>null</i> (null)	L31.SP1	<i>null</i> (null)	Punctuation Marks (High)
L1.SP4	Automotive Advertising (Med)	Liquid Enhancement (High)	L31.SP17	time duration (High)	positive attributes (High)
L2.SP2	<i>null</i> (null)	<i>null</i> (null)	L31.SP15	Economic Hardship (High)	Reports and References (Med)
L2.SP25	<i>null</i> (null)	Communication Actions (Med)	L32.SP4	<i>null</i> (null)	briefing documents (High)
L2.SP34	optimism themes (Med)	<i>null</i> (null)	L32.SP22	failure and success (Med)	Visual Media (High)
L3.SP2	<i>null</i> (null)	negative experiences (High)	L32.SP20	excellence and power (High)	negation terms (High)
L3.SP13	technology usage (Med)	Publishing Concepts (Med)	L32.SP8	Locations and Events (Med)	Negative Evaluation (High)
L4.SP35	Plants and Growth (High)	<i>null</i> (null)	L33.SP24	<i>null</i> (null)	personal growth (High)
L4.SP1	articles and conjunctions (High)	<i>null</i> (null)	L33.SP30	rejection behaviors (High)	Validation Process (High)
L4.SP49	Social Activities (Med)	food substances (Med)	L33.SP10	<i>null</i> (null)	extreme qualities (High)
L5.SP1	function words (High)	<i>null</i> (null)	L33.SP5	<i>null</i> (null)	greatness and limits (Med)
L5.SP15	<i>null</i> (null)	romantic themes (High)	L34.SP15	documents and media (Med)	improvement actions (High)
L6.SP1	function words (High)	<i>null</i> (null)	L35.SP20	<i>null</i> (null)	numerical references (High)
L8.SP6	transportation services (High)	Action and Conditions (Med)	L36.SP4	<i>null</i> (null)	<i>null</i> (null)
L8.SP42	Computing Processes (High)	Relief and Assurance (Med)	L36.SP20	action and evaluation (Med)	high quality (Med)
L8.SP65	Energy Resources (Med)	Technology Terms (Med)	L36.SP38	negation terms (High)	Health Risks (High)
L10.SP5	<i>null</i> (null)	Rental Properties (High)	L36.SP21	Personal Pronouns (High)	possessive pronouns (High)
L10.SP11	events and activities (Med)	color attributes (High)	L36.SP6	Possessive Pronouns (High)	Fashion and Activities (Med)
L10.SP2	<i>null</i> (null)	Prep. & Conj. (High)	L37.SP25	<i>null</i> (null)	Travel and Aviation (High)
L10.SP16	<i>null</i> (null)	Environmental Issues (High)	L37.SP2	business topics (Med)	<i>null</i> (null)
L10.SP1	large scale (High)	<i>null</i> (null)	L37.SP52	Informative Communication (Med)	Measurement Terms (Med)
L11.SP1	<i>null</i> (null)	large scale (High)	L37.SP19	<i>null</i> (null)	Numerical References (High)
L11.SP22	Buyers and Sellers (High)	disruption and change (High)	L37.SP11	<i>null</i> (null)	<i>null</i> (null)
L11.SP50	artistic expression (Med)	accumulation and tracking (Med)	L38.SP30	Emotional expressions (Med)	returning actions (Med)
L12.SP144	Awakening Concepts (Med)	Urban Areas (High)	L39.SP22	<i>null</i> (null)	comparative qualities (Med)
L12.SP88	news reporting (High)	Trade and Supply (High)	L39.SP3	<i>null</i> (null)	Political Power (High)
L15.SP4	justification and meaning (Med)	Urban Institutions (High)	L39.SP60	<i>null</i> (null)	Financial Transactions (High)
L15.SP7	<i>null</i> (null)	conjunctions&functions (High)	L39.SP34	office locations (Med)	Emotional expressions (Med)
L15.SP15	Creative expression (Med)	financial regulations (High)	L39.SP72	Length Measurement (High)	<i>null</i> (null)
L16.SP2	<i>null</i> (null)	<i>null</i> (null)	L40.SP1	large scale (High)	<i>null</i> (null)
L16.SP37	chaos and greatness (Med)	Professional Roles (Med)	L41.SP27	<i>null</i> (null)	<i>null</i> (null)
L17.SP19	gathering activities (High)	Legal and Safety Terms (High)	L41.SP22	Corporate Terms (Med)	positive attributes (High)
L18.SP23	success and persistence (Med)	emotional concepts (High)	L41.SP80	<i>null</i> (null)	Command Functions (Med)
L18.SP20	individuality concepts (Med)	Media&Communication (Med)	L41.SP60	Direction and Action (High)	Motherhood Concepts (Med)
L19.SP10	indifference and rigidity (Med)	Financial Institutions (High)	L42.SP5	Space Exploration (Med)	<i>null</i> (null)
L19.SP23	requirements and options (High)	adventure themes (Med)	L42.SP1	large scale concepts (High)	<i>null</i> (null)
L19.SP43	navigation concepts (Med)	Deception Concepts (High)	L42.SP38	have and had forms (High)	Financial Terms (Med)
L20.SP2	Conj. & Punct. (Med)	<i>null</i> (null)	L43.SP6	<i>null</i> (null)	finance and work (Med)
L20.SP16	love and inspiration (High)	dependency concepts (High)	L43.SP1	large scale concepts (High)	<i>null</i> (null)
L20.SP40	Health and Technology (Med)	pleasant and weather (High)	L44.SP4	<i>null</i> (null)	Advertising Terms (Med)
L20.SP19	large quantities (High)	professionalism themes (Med)	L45.SP4	Mathematical Symbols (Med)	<i>null</i> (null)
L21.SP15	Legal & Finance (High)	Aesthetic Descriptions (High)	L45.SP6	<i>null</i> (null)	Prep. & Conj. (High)
L22.SP3	advertising concepts (Med)	<i>null</i> (null)	L45.SP38	Cultural References (Med)	Logical Operators (High)
L24.SP6	governmental authority (High)	<i>null</i> (null)	L45.SP28	Punctuation and Symbols (High)	Quotations and Speech (High)
L24.SP211	scheduling concepts (Med)	warning and action (Med)	L46.SP5	<i>null</i> (null)	<i>null</i> (null)
L24.SP10	collaboration and growth (High)	errors and failures (High)	L46.SP32	time references (High)	openness and closure (High)
L24.SP33	Behavioral Analysis (High)	financial support (High)	L46.SP13	<i>null</i> (null)	Numerical References (High)
L25.SP7	innovation and impact (High)	indoor activities (Med)	L46.SP12	Influencing actions (High)	possessive pronouns (High)
L26.SP3	Names of People (High)	emotional struggle (High)	L47.SP36	<i>null</i> (null)	Dates and Months (High)
L26.SP14	emotional experiences (High)	representation concepts (High)	L47.SP33	games and players (High)	web documents (High)
L27.SP38	education contribution (High)	hidden truths (High)	L47.SP20	question words (High)	Reports and Publications (High)
L28.SP4	confusion themes (High)	<i>null</i> (null)	L47.SP1	Fantasy Characters (High)	<i>null</i> (null)
L28.SP10	Completion status (High)	Social Groups (High)	L47.SP54	legal rules (High)	positive emotions (High)
L28.SP3	<i>null</i> (null)	Corporate Entities (Med)			
L29.SP3	negative emotions (High)	<i>null</i> (null)			
L29.SP26	Unpredictable roles (Med)	disappearance&removal (High)			
L29.SP4	misunderstanding (High)	Names and Places (Med)			
L29.SP33	Reports and Analysis (High)	Trust and Relationships (High)			

G. Computational Efficiency and Scalability

In this section, we describe the operational flexibility of our framework, which supports two distinct modes:

- **Global Pre-Computation:** This mode constructs a comprehensive interpretability datastore for every DEU subspace across all layers. Consequently, during pathway extraction, the interpretations of high-contribution subspaces can be directly queried, enabling efficient and reusable access.
- **On-Demand Interpretation:** This model first extracts the pathway and subsequently generates interpretations solely for the required top- k high-contribution subspaces. This procedure minimizes upfront computational overhead and enables efficient targeted analysis.

G.1. Computational Cost of Building the Interpretability Datastore

To assess the computational feasibility of constructing the interpretability datastore, we evaluated resource consumption across four GPT-2 variants. The analysis quantifies the computational overhead per layer as a function of matrix size, defined as $hidden_size \times mlp_dim$. To ensure consistent performance metrics, all benchmarks were executed on a single **NVIDIA A100 GPU (80GB)**.

As illustrated in Figure 14, the system demonstrates efficient scaling behavior regarding both runtime and memory usage. The total runtime, which aggregates the W_{up} and W_{down} interpretation stages, exhibits a strictly linear correlation with matrix size. Notably, even for the largest variant, GPT2-XL (characterized by 20.5M parameters per layer), the process remains highly efficient and requires approximately **7 minutes per layer**. Similarly, peak GPU memory consumption scales linearly, ranging from less than 1 GB for GPT-2 to approximately 6.5 GB for GPT2-XL, a range that falls well within the capacity of standard consumer and enterprise hardware.

The linear scaling trends and low absolute resource requirements confirm that building the interpretability datastore is computationally feasible. The process is highly scalable, allowing for the efficient interpretation of larger language model layers without requiring prohibitive hardware resources.

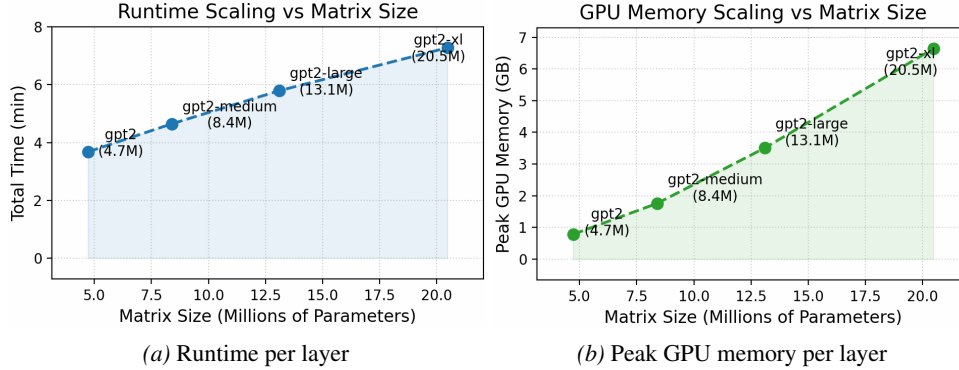


Figure 14. **Computational Cost of Constructing an Interpretability Datastore per Layer for Models of Varying Scale on a Single NVIDIA A100 GPU (80GB).** Total runtime (in minutes) is reported as the sum of the W_{up} and W_{down} interpretation stages. Matrix size is measured in millions of parameters, defined as $hidden_size \times mlp_dim$.

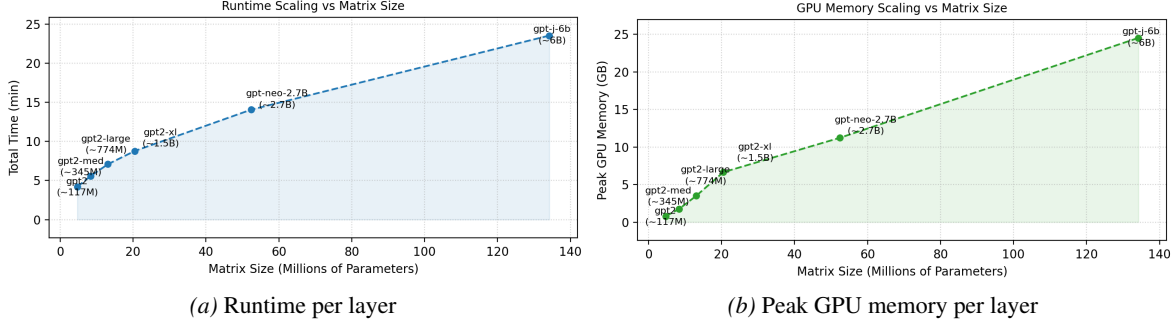


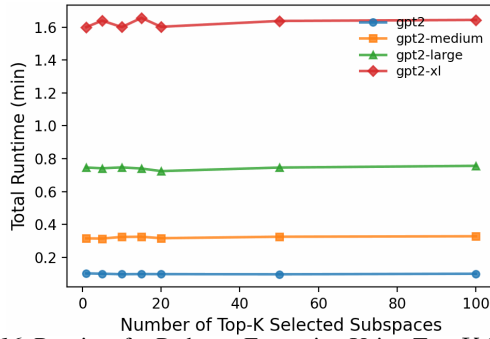
Figure 15. **Computational Cost of Constructing an Interpretability Datastore per Layer for Models of Varying Scale on a Single NVIDIA A100 GPU (40GB).** Total runtime (in minutes) is reported as the sum of the W_{up} and W_{down} interpretation stages. Matrix size is measured in millions of parameters, defined as $\text{hidden_size} \times \text{mlp_dim}$.

G.2. Runtime of Pathway Extraction

We further evaluate the computational cost of pathway extraction under varying sparsity levels, considering top- k high-contribution subspaces with $k \in [1, 100]$ across different model variants.

Figure 16 shows that the total runtime remains effectively constant whether pathway extraction is restricted to a single subspace or extended to 100 subspaces. This indicates that increasing the granularity of subspace analysis incurs no additional computational overhead. Moreover, pathway extraction is consistently efficient across model scales, requiring approximately 0.32 minutes for GPT2-Medium and 0.75 minutes for GPT2-Large. Even for GPT2-XL, the maximum recorded runtime is merely 1.64 minutes.

Overall, pathway extraction is completed within two minutes across all configurations, underscoring the method’s efficiency and applicability in real-world settings.



Top-K Subspaces	GPT-2	GPT2-Medium	GPT2-Large	GPT2-XL
1	0.10	0.31	0.75	1.60
5	0.10	0.31	0.74	1.64
10	0.10	0.32	0.75	1.60
15	0.10	0.32	0.74	1.66
20	0.10	0.32	0.72	1.60
50	0.10	0.32	0.75	1.64
100	0.10	0.33	0.76	1.64

Figure 16. Runtime for Pathway Extraction Using Top- K Subspaces Across the GPT-2 Family on a Single NVIDIA A100 (80GB).

G.3. Runtime of Pathway Extraction and On-Demand Interpretation

Unlike SAEs and TransCoders that generally rely on extensive precomputation of global features, our framework enables flexible, on-demand analysis. This method enables direct interpretation of the top-ranked subspaces within the identified pathways, while avoiding the computational overhead associated with processing irrelevant model components.

Figure 17 reports the runtime of the complete analysis pipeline, which encompasses both pathway extraction and local interpretation across all layers. The results exhibit an approximately linear relationship between runtime and the number of top- k subspaces, with the overall pipeline remaining highly efficient. For the largest model, GPT2-XL, the pipeline requires 6.62 minutes for a single subspace ($k = 1$) and 17.56 minutes for a comprehensive analysis of 100 subspaces ($k = 100$). For smaller models such as GPT-2, the entire pipeline is completed in approximately 3 minutes even at $k = 100$. These findings indicate that the framework can efficiently extract mechanistic pathways and yield interpretable insights within a short timeframe, highlighting its suitability for practical deployment in large-scale model analyses.

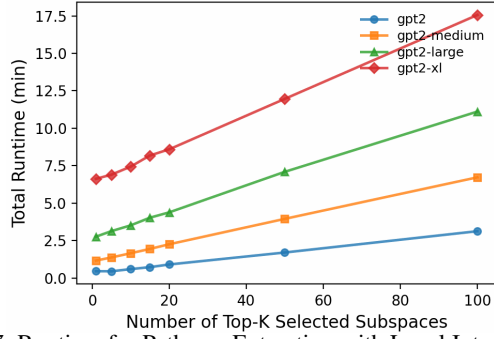


Figure 17. Runtime for Pathway Extraction with Local Interpretation Using Top- K Subspaces Across the GPT-2 Family on a Single NVIDIA A100 (80GB).

H. Twenty Representative Cases with Pathway Extraction Analysis

In this section, we present twenty representative cases covering a broad spectrum of linguistic phenomena. Each case illustrates distinct patterns in model behavior, enabling a detailed evaluation of pathway extraction and interpretability across syntactic, semantic, and pragmatic contexts.

H.1. 20 Cases of Diverse Linguistic Phenomena

Within the NaNA framework, SCA serves as a directed pathway extraction method, distinguished by its explicit filtering of *causal bridges*. We define such bridges as subspaces that simultaneously exhibit input activation (detector alignment) and target alignment (effector alignment). To validate the extracted pathways, we evaluate whether they can recover or steer generation toward the specified targets. Notably, for the 20 anchor cases, certain target tokens do not correspond to the model’s original top-1 predictions. Since our method is explicitly designed for directed pathway extraction toward arbitrary targets, this setting is entirely valid.

Table 8. Twenty Representative Cases Covering Diverse Linguistic Phenomena.

#	Input Example	Target Token	Pathway Type	Linguistic Phenomenon
1	The cat looks very	happy	Syntax–Semantics	Adjective completion
2	Code snippet: for i in range(5): print	(	Code Syntax	Parenthesis completion
3	The square root of 64 is	8	Factual / Math	Arithmetic fact retrieval
4	The dogs in the park	are	Syntax	Subject–verb agreement
5	The capital of Germany is	Berlin	Factual Knowledge	Geographic knowledge retrieval
6	Water boils at 100 degrees	Celsius	Factual Knowledge	Scientific fact retrieval
7	Code snippet: for char in ‘hello’: print	char	Code Syntax	Loop variable completion
8	John and Mary went to the	church	Semantics	Plausible location inference
9	She didn’t see the sign, so she	didn	Morphology	Negation contraction
10	She quickly ran to the	door	Semantics	Goal argument completion
11	The sun rises in the	east	Factual Knowledge	Common-sense astronomy
12	The food was not bad, it was	good	Semantics / Pragmatics	Scalar implicature
13	The boy who cried wolf finally	got	Semantics	Pragmatic inference
14	The committee decided to postpone the	vote	Semantics	Event nominal completion
15	During the thunderstorm, the lights	of	Discourse	Causal continuation
16	The movie was thrilling but also	sad	Semantics	Contrastive adjective completion
18	The recipe calls for two cups of	sugar	Factual / Procedural	Ingredient selection
16	To solve the equation, one must first	simplify	Procedural Knowledge	Step ordering
19	Alice gave Bob the book because she	wanted	Discourse / Coreference	Implicit intention inference
20	If I had studied harder, I	would	Syntax / Semantics	Counterfactual construction

H.2. Evaluation of Pathway Extraction Across Twenty Linguistic Cases: Standard vs. Ablation

Case 1: Prompt “The cat looks very” with Target Token “happy”.

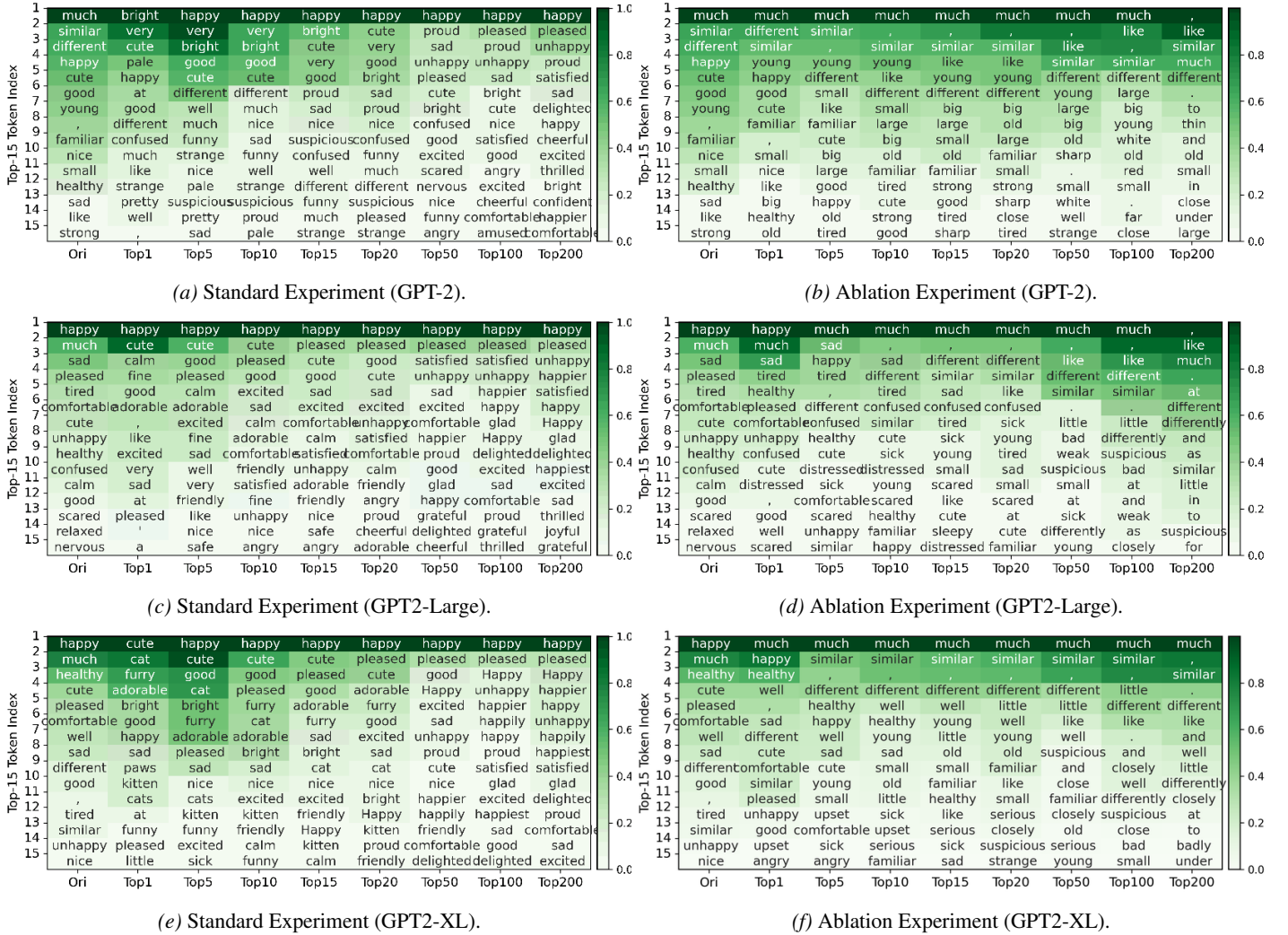


Figure 18. Reconstruction of Middle-to-Late MLP Layers via Pathways within Top- k High-Contribution Subspaces: Standard vs. Ablation Studies. This figure illustrates a case study using the prompt “The cat looks very” with the target token “happy”. We visualize the top-15 predicted tokens across GPT-2 family under standard and ablation settings.

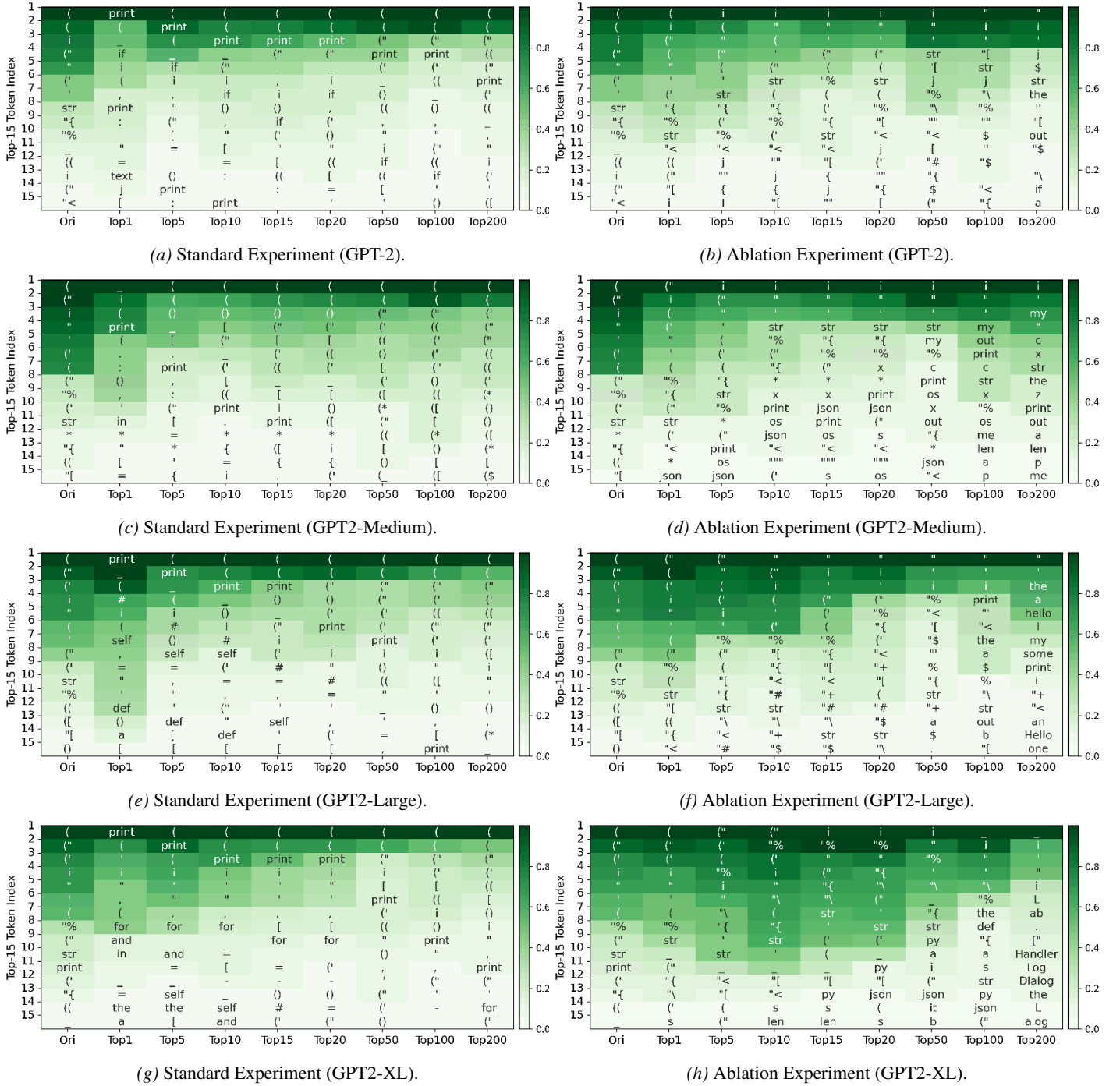
Case 2: Prompt “Code snippet: for i in range(5): print” with Target Token “(”.

Figure 19. Reconstruction of Middle-to-Late MLP Layers via Pathways within Top- k High-Contribution Subspaces: Standard vs. Ablation Studies. This figure illustrates a case study using the prompt “Code snippet: for i in range(5): print” with the target token “(”. We visualize the top-15 predicted tokens across GPT-2 variants under standard and ablation settings.

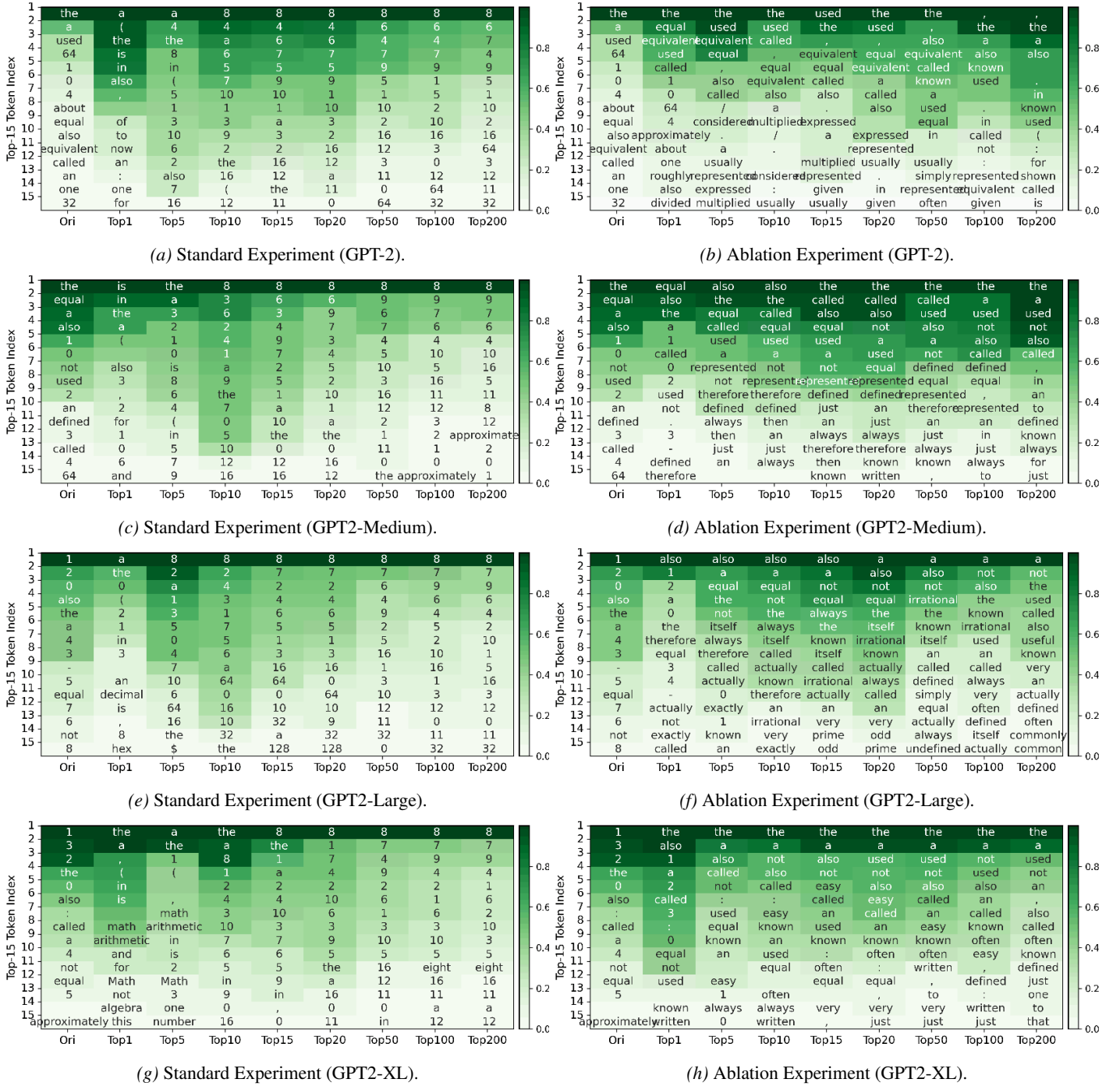
Case 3: Prompt “The square root of 64 is” with Target Token “8”.

Figure 20. Reconstruction of Middle-to-Late MLP Layers via Pathways within Top- k High-Contribution Subspaces: Standard vs. Ablation Studies. This figure illustrates a case study using the prompt “The square root of 64 is” with the target token “8”. We visualize the top-15 predicted tokens across GPT-2 variants under standard and ablation settings.

Case 4: Prompt “The dogs in the park” with Target Token “are”.

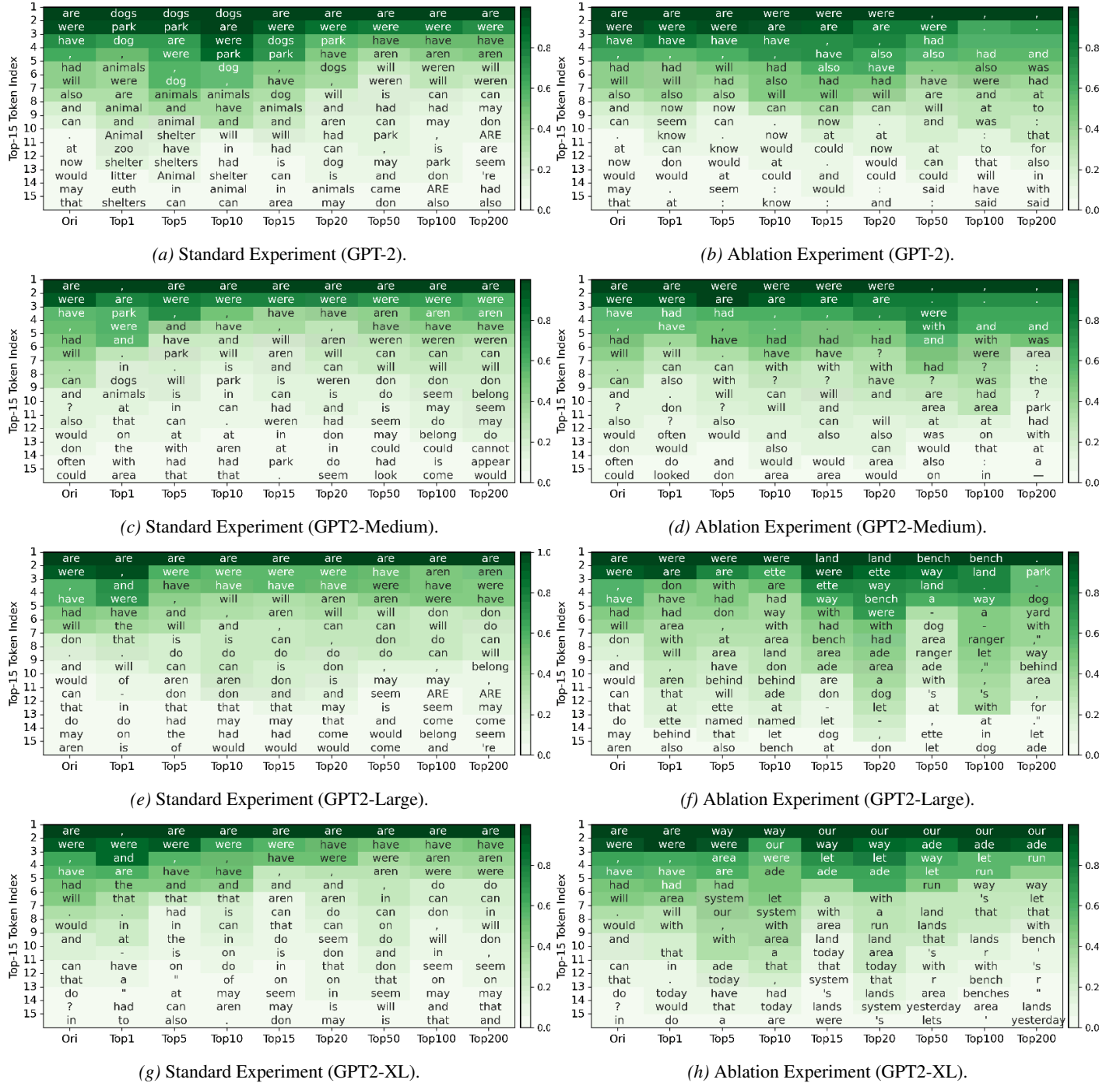


Figure 21. **Reconstruction of Middle-to-Late MLP Layers via Pathways within Top- k High-Contribution Subspaces: Standard vs. Ablation Studies.** This figure illustrates a case study using the prompt “The dogs in the park” with the target token “are”. We visualize the top-15 predicted tokens across GPT-2 variants under standard and ablation settings.

Case 5: Prompt “The capital of Germany is” with Target Token “Berlin”.

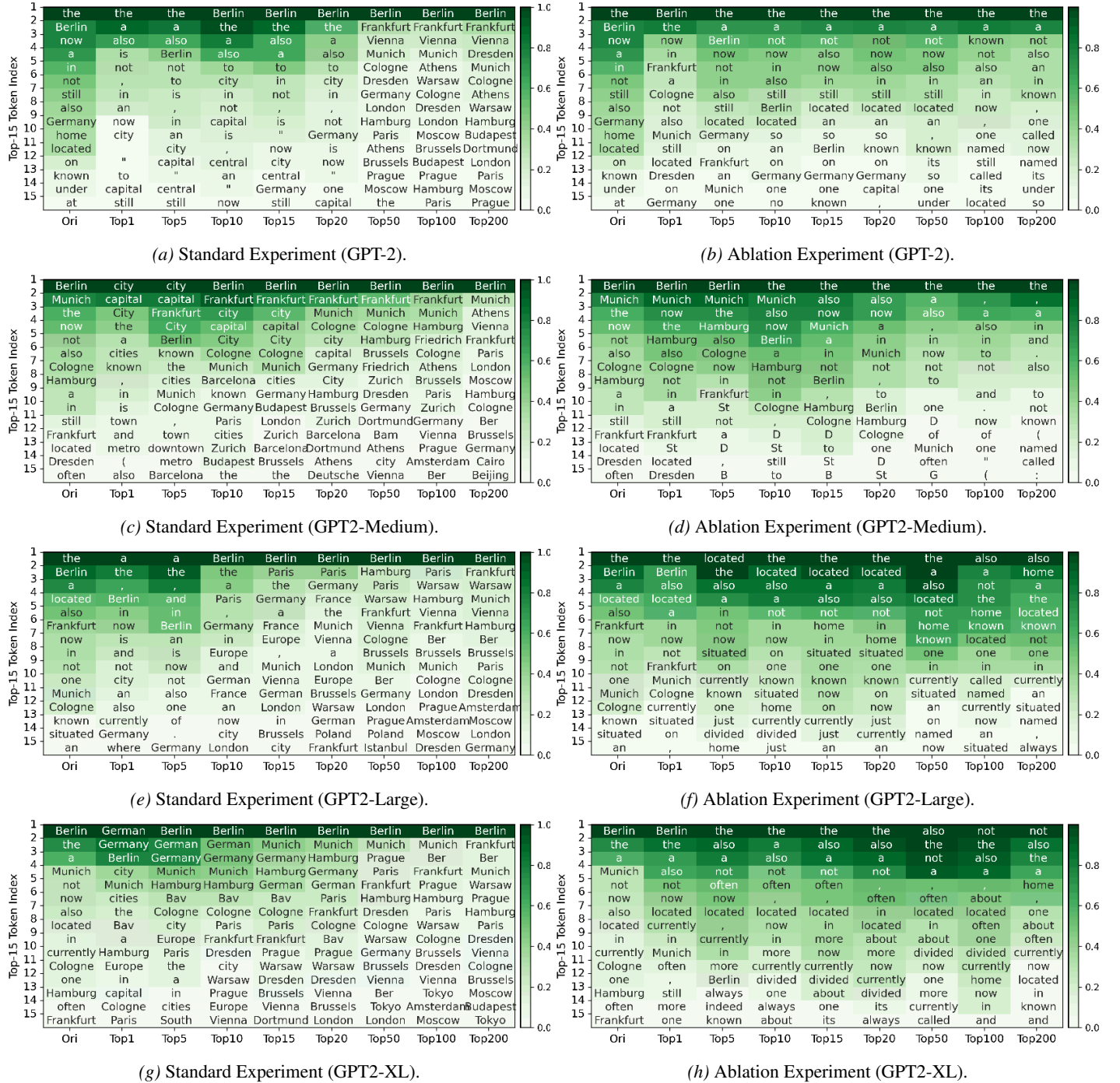


Figure 22. **Reconstruction of Middle-to-Late MLP Layers via Pathways within Top- k High-Contribution Subspaces: Standard vs. Ablation Studies.** This figure illustrates a case study using the prompt “The capital of Germany is” with the target token “Berlin”. We visualize the top-15 predicted tokens across GPT-2 variants under standard and ablation settings.

Case 6: Prompt “Water boils at 100 degrees” with Target Token “Celsius”.

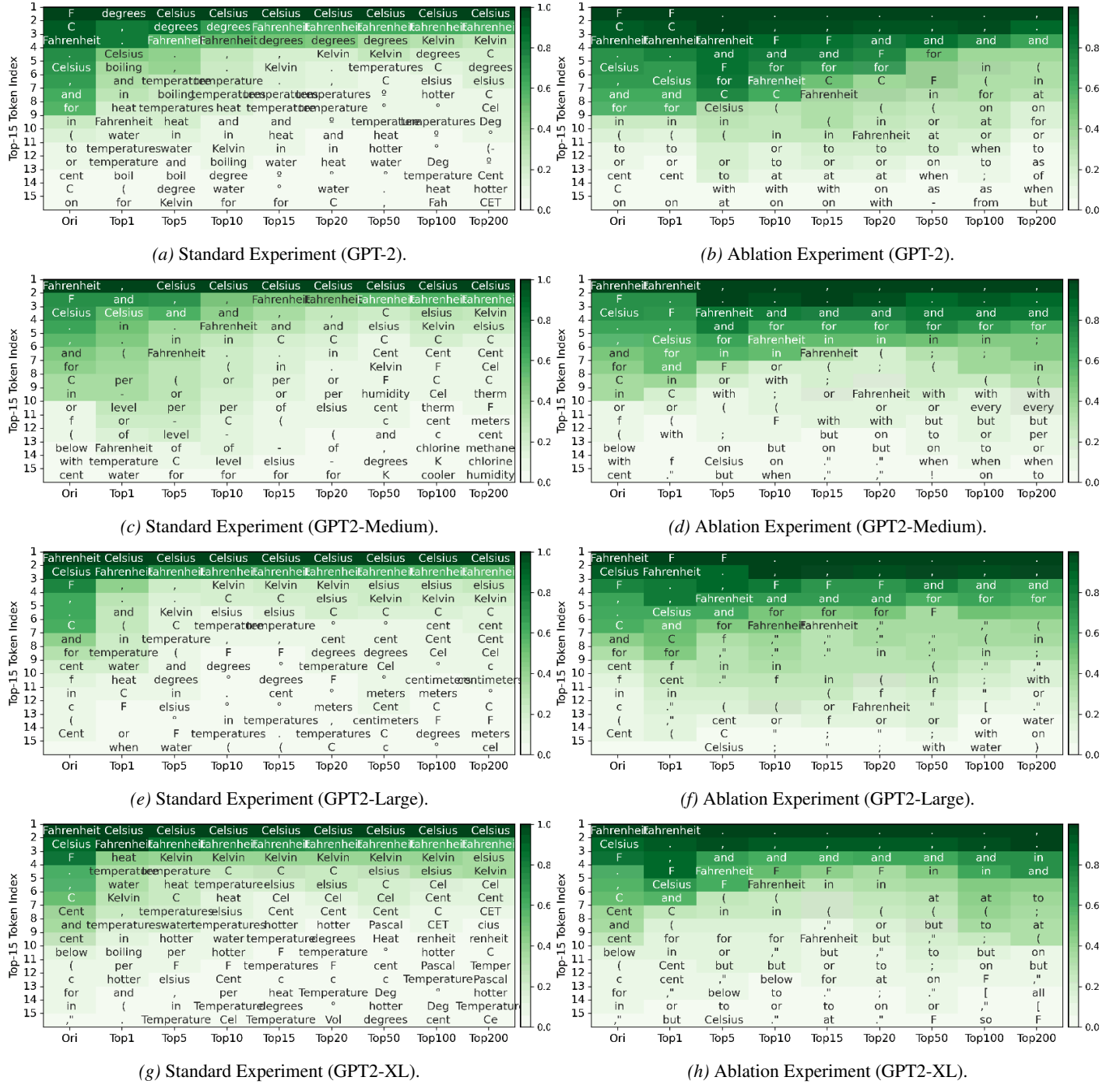


Figure 23. **Reconstruction of Middle-to-Late MLP Layers via Pathways within Top- k High-Contribution Subspaces: Standard vs. Ablation Studies.** This figure illustrates a case study using the prompt “Water boils at 100 degrees” with the target token “Celsius”. We visualize the top-15 predicted tokens across GPT-2 variants under standard and ablation settings.

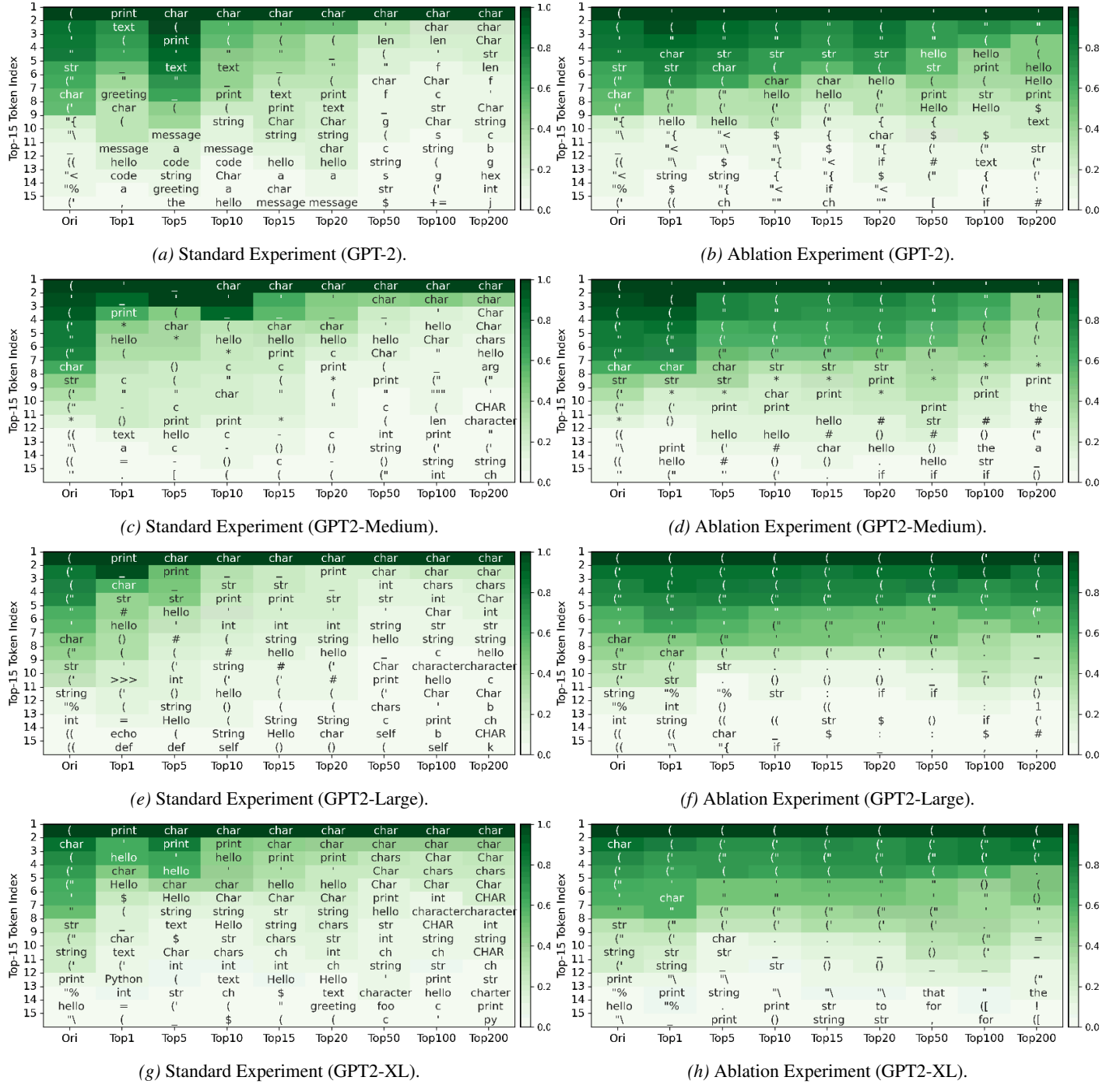
Case 7: Prompt “Code snippet: for char in ‘hello’: print” with Target Token “char”.

Figure 24. Reconstruction of Middle-to-Late MLP Layers via Pathways within Top- k High-Contribution Subspaces: Standard vs. Ablation Studies. This figure illustrates a case study using the prompt “Code snippet: for char in ‘hello’: print” with the target token “char”. We visualize the top-15 predicted tokens across GPT-2 variants under standard and ablation settings.

Case 8: Prompt “John and Mary went to the” with Target Token “church”.

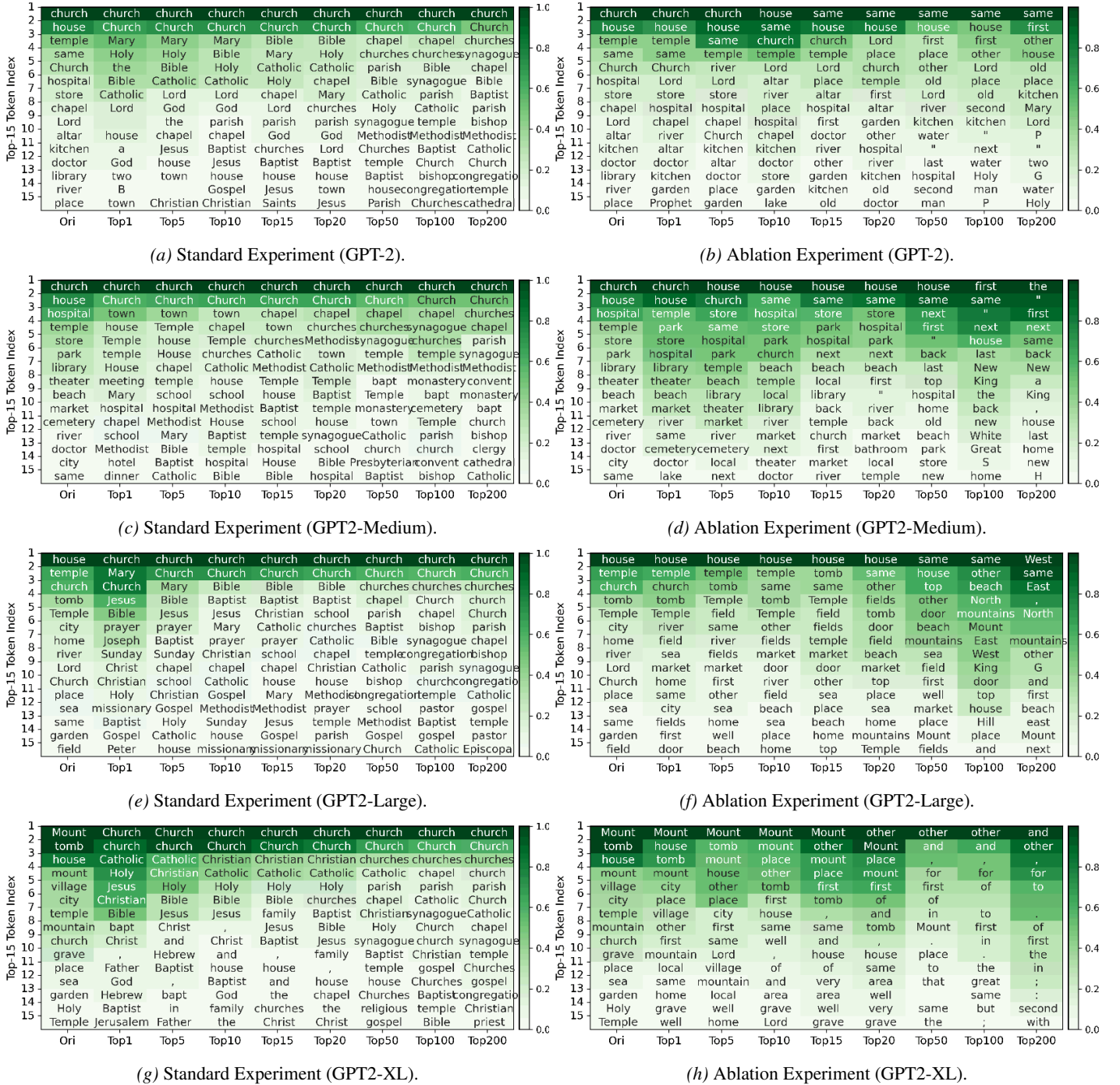


Figure 25. **Reconstruction of Middle-to-Late MLP Layers via Pathways within Top- k High-Contribution Subspaces: Standard vs. Ablation Studies.** This figure illustrates a case study using the prompt “John and Mary went to the” with the target token “church”. We visualize the top-15 predicted tokens across GPT-2 variants under standard and ablation settings.

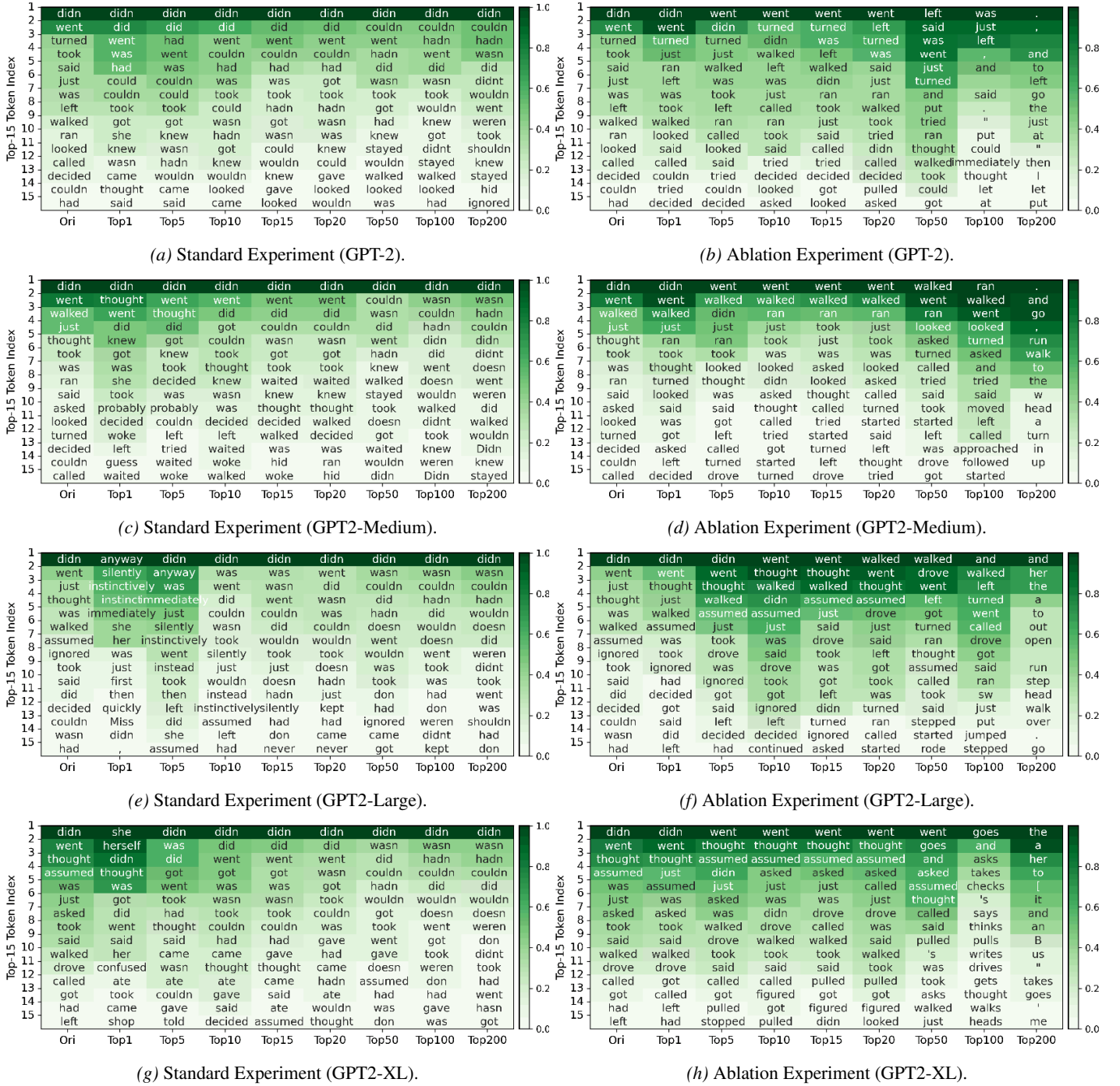
Case 9: Prompt “She didn’t see the sign, so she” with Target Token “didn”.

Figure 26. Reconstruction of Middle-to-Late MLP Layers via Pathways within Top- k High-Contribution Subspaces: Standard vs. Ablation Studies. This figure illustrates a case study using the prompt “She didn’t see the sign, so she” with the target token “didn”. We visualize the top-15 predicted tokens across GPT-2 variants under standard and ablation settings.

Case 10: Prompt “She quickly ran to the” with Target Token “ door”.

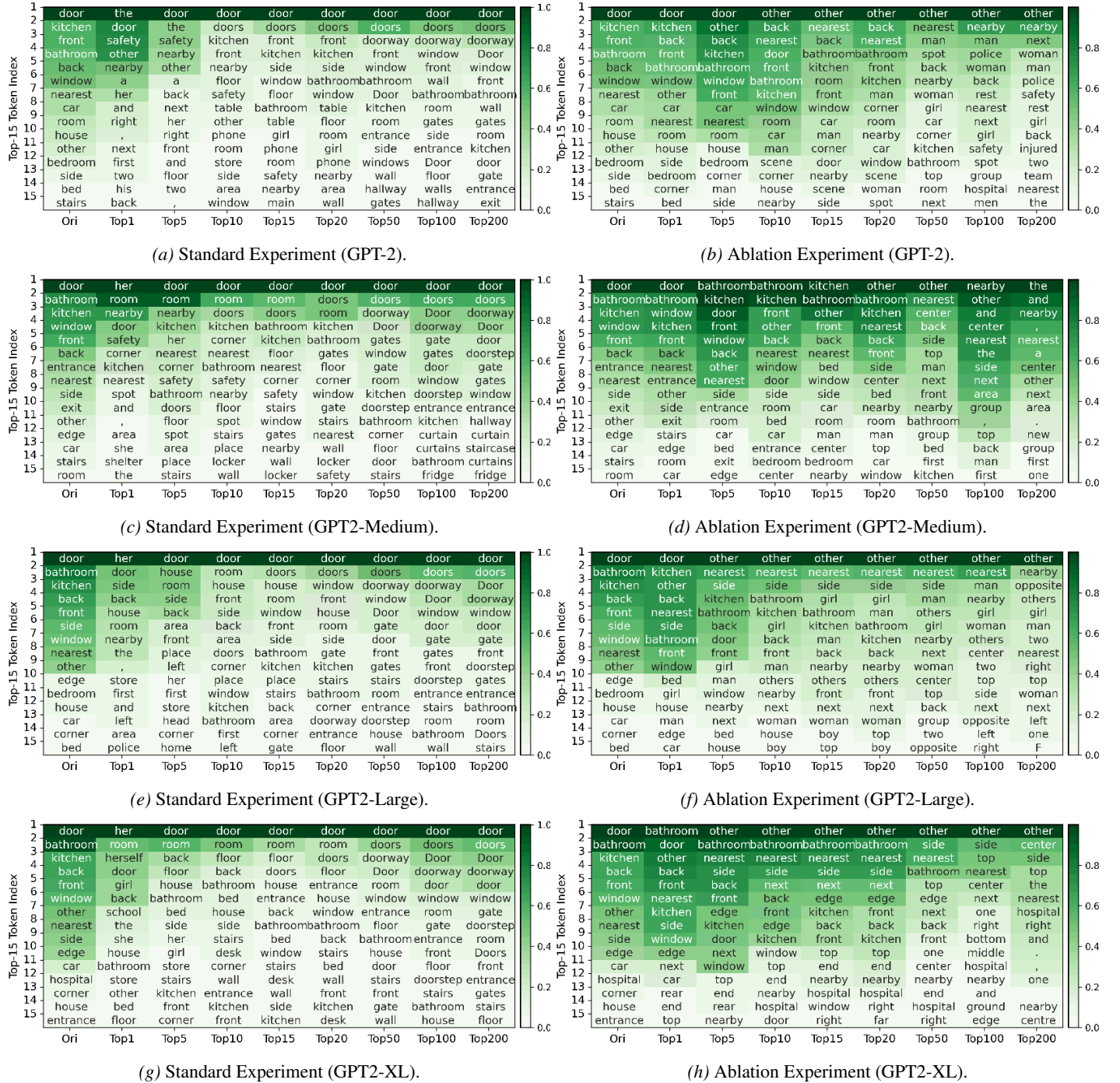


Figure 27. **Reconstruction of Middle-to-Late MLP Layers via Pathways within Top- k High-Contribution Subspaces: Standard vs. Ablation Studies.** This figure illustrates a case study using the prompt “She quickly ran to the” with the target token “ door”. We visualize the top-15 predicted tokens across GPT-2 variants under standard and ablation settings.

Case 11: Prompt “The sun rises in the” with Target Token “east”.

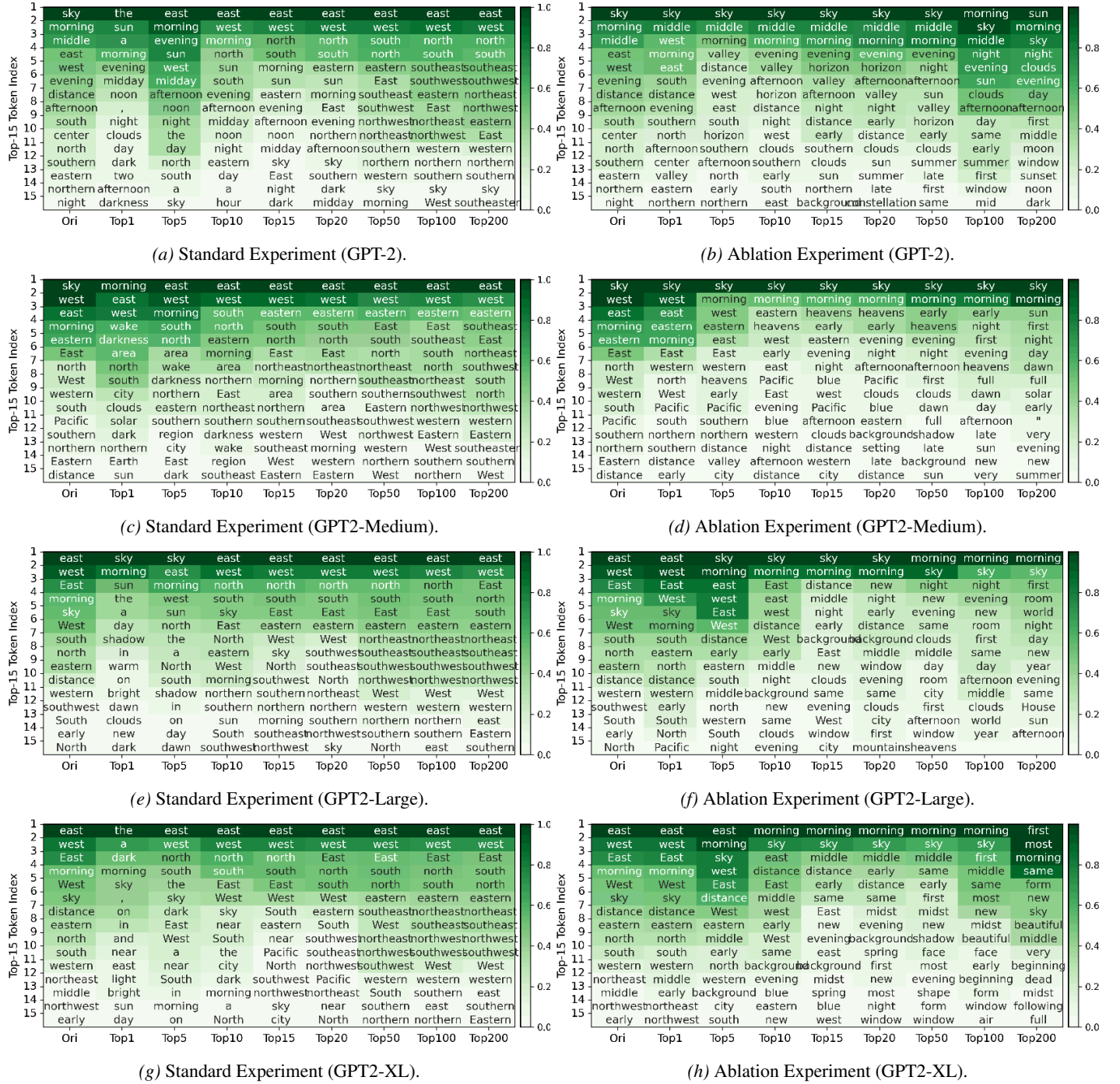


Figure 28. Reconstruction of Middle-to-Late MLP Layers via Pathways within Top- k High-Contribution Subspaces: Standard vs. Ablation Studies. This figure illustrates a case study using the prompt “The sun rises in the” with the target token “east”. We visualize the top-15 predicted tokens across GPT-2 variants under standard and ablation settings.

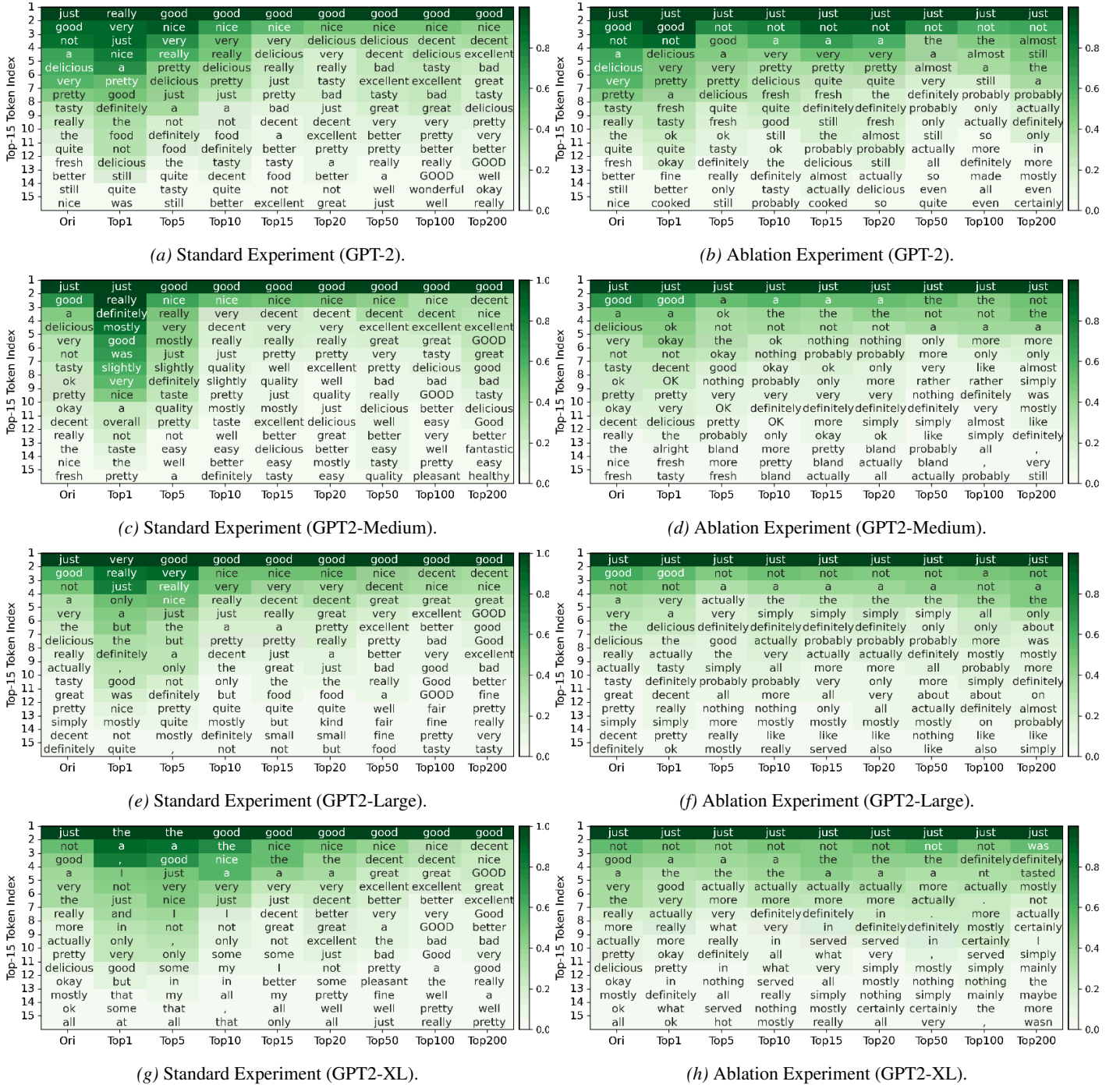
Case 12: Prompt “The food was not bad, it was” with Target Token “good”.

Figure 29. Reconstruction of Middle-to-Late MLP Layers via Pathways within Top- k High-Contribution Subspaces: Standard vs. Ablation Studies. This figure illustrates a case study using the prompt “The food was not bad, it was” with the target token “good”. We visualize the top-15 predicted tokens across GPT-2 variants under standard and ablation settings.

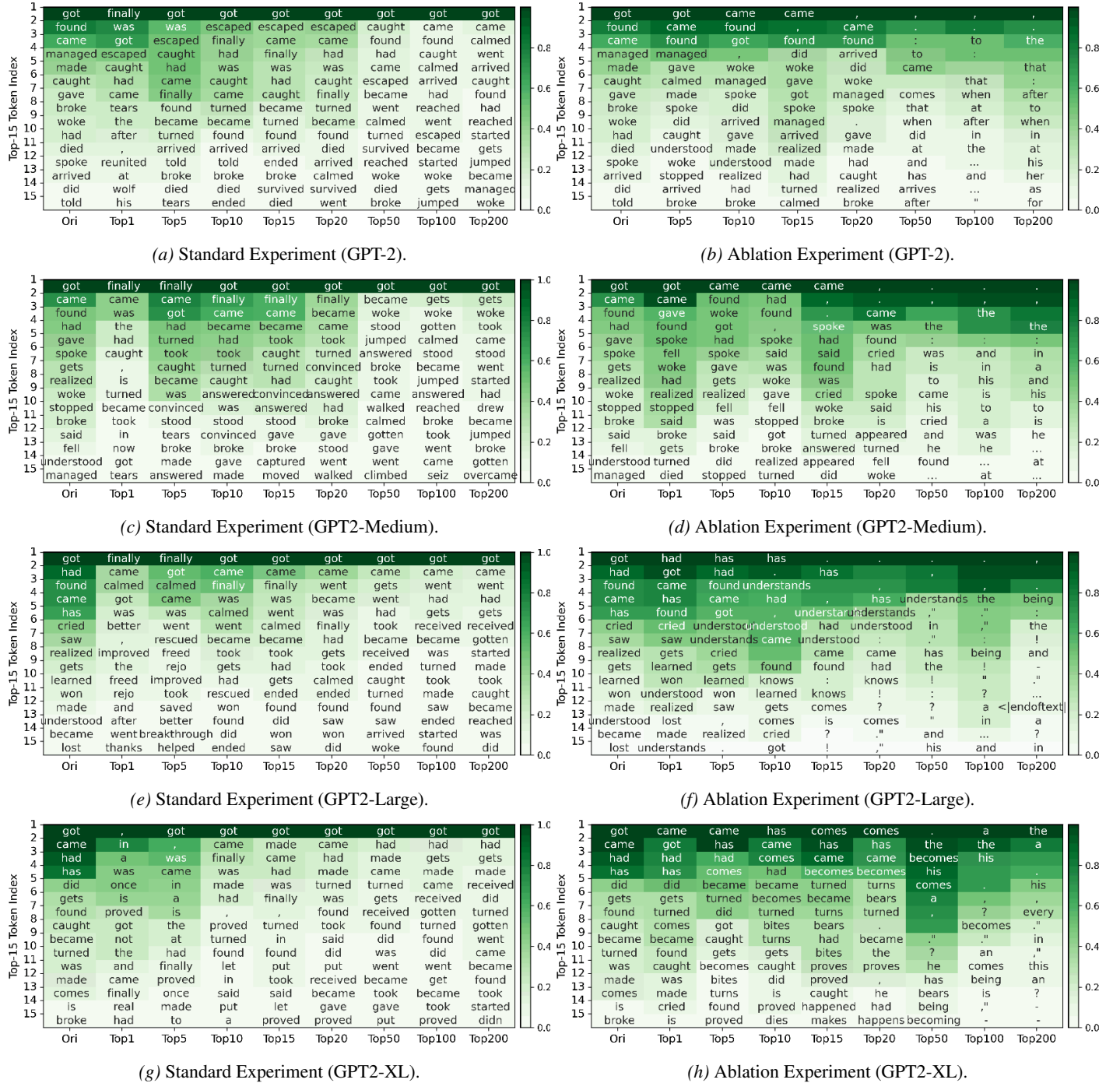
Case 13: Prompt “The boy who cried wolf finally” with Target Token “got”.

Figure 30. Reconstruction of Middle-to-Late MLP Layers via Pathways within Top- k High-Contribution Subspaces: Standard vs. Ablation Studies. This figure illustrates a case study using the prompt “The boy who cried wolf finally” with the target token “got”. We visualize the top-15 predicted tokens across GPT-2 variants under standard and ablation settings.

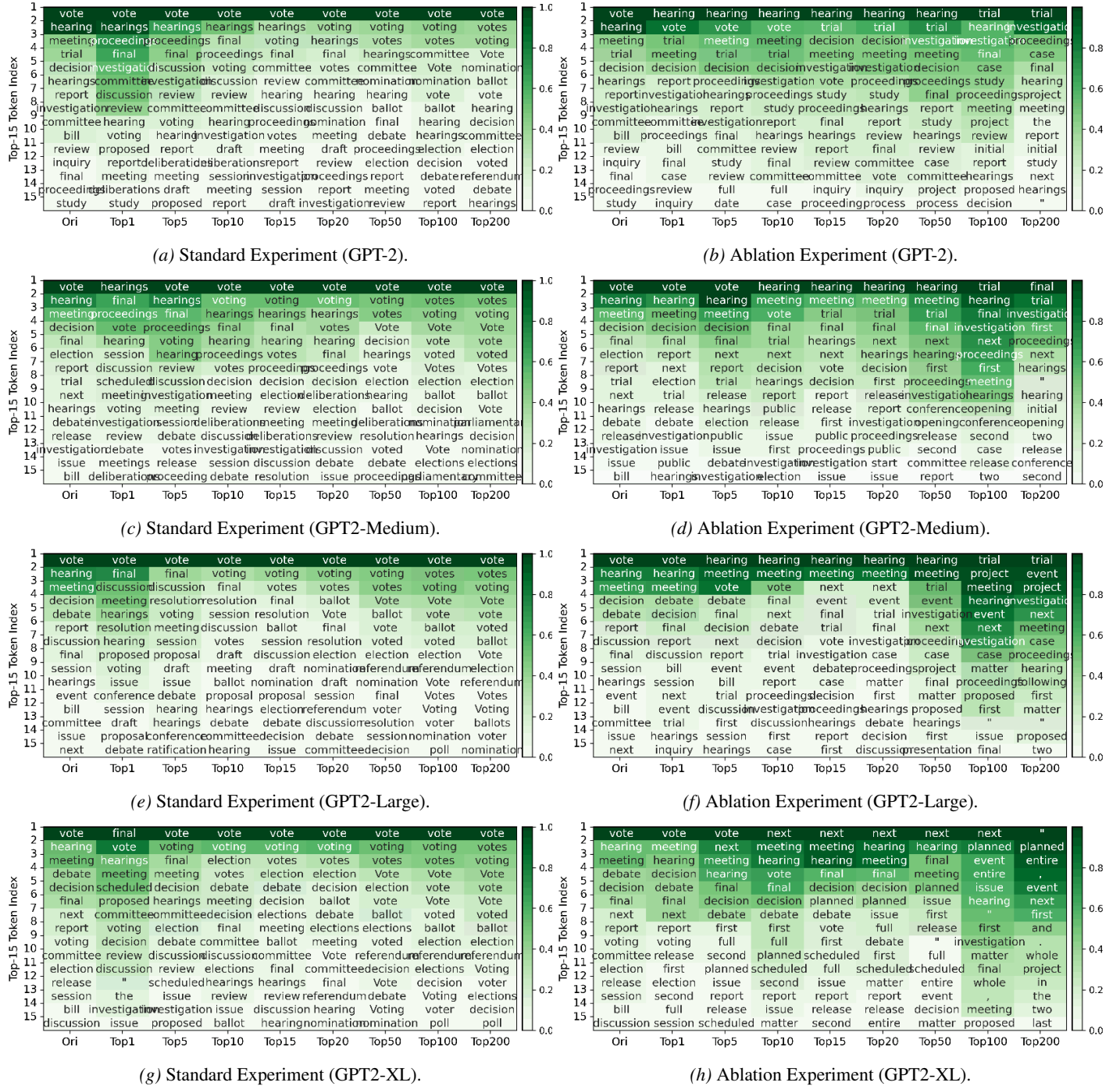
Case 14: Prompt “The committee decided to postpone the” with Target Token “vote”.

Figure 31. Reconstruction of Middle-to-Late MLP Layers via Pathways within Top- k High-Contribution Subspaces: Standard vs. Ablation Studies. This figure illustrates a case study using the prompt “The committee decided to postpone the” with the target token “vote”. We visualize the top-15 predicted tokens across GPT-2 variants under standard and ablation settings.

Case 15: Prompt “During the thunderstorm, the lights” with Target Token “of”.

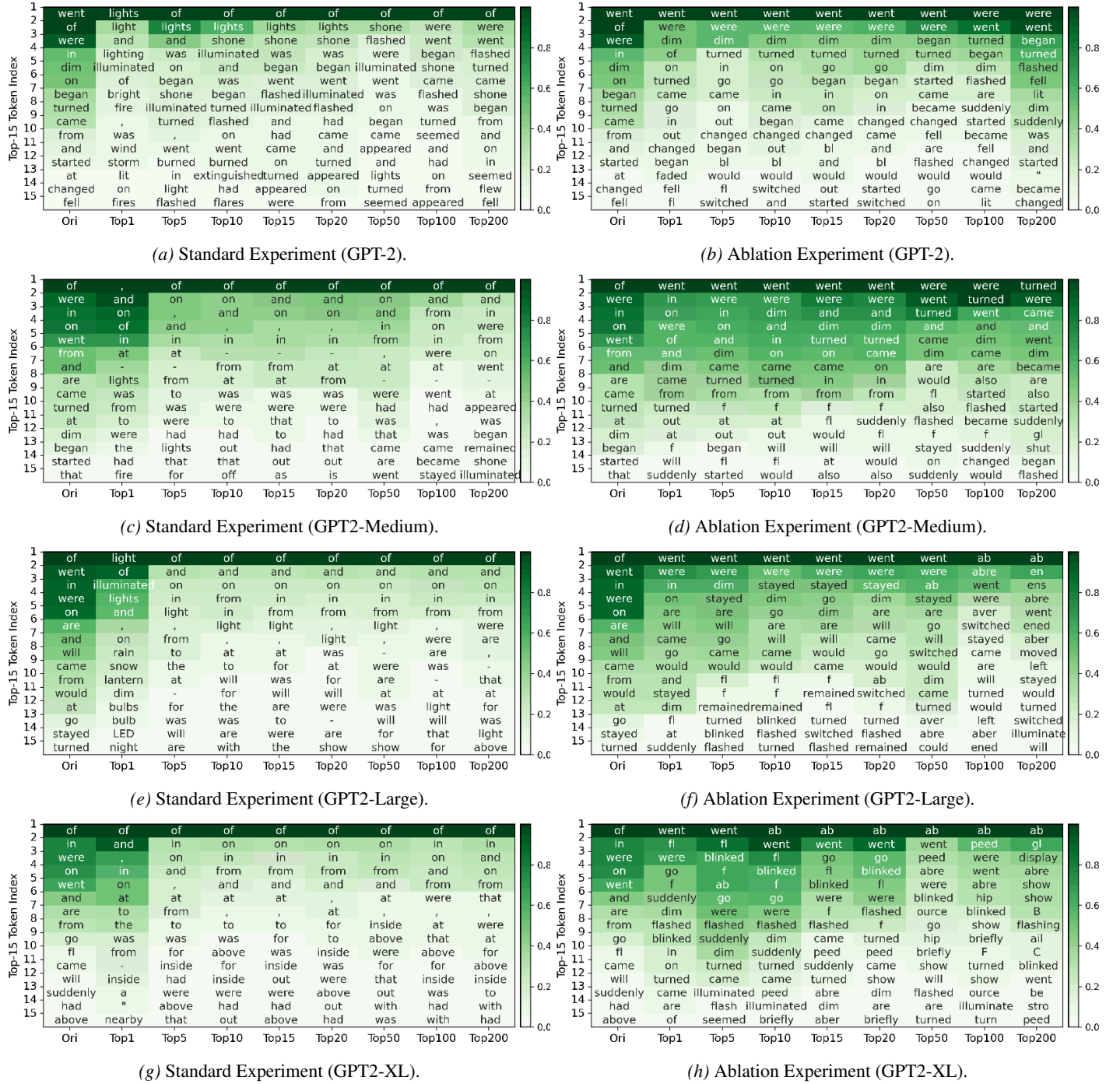


Figure 32. **Reconstruction of Middle-to-Late MLP Layers via Pathways within Top- k High-Contribution Subspaces: Standard vs. Ablation Studies.** This figure illustrates a case study using the prompt “During the thunderstorm, the lights” with the target token “of”. We visualize the top-15 predicted tokens across GPT-2 variants under standard and ablation settings.

Case 16: Prompt “The movie is thrilling but also” with Target Token “sad”.

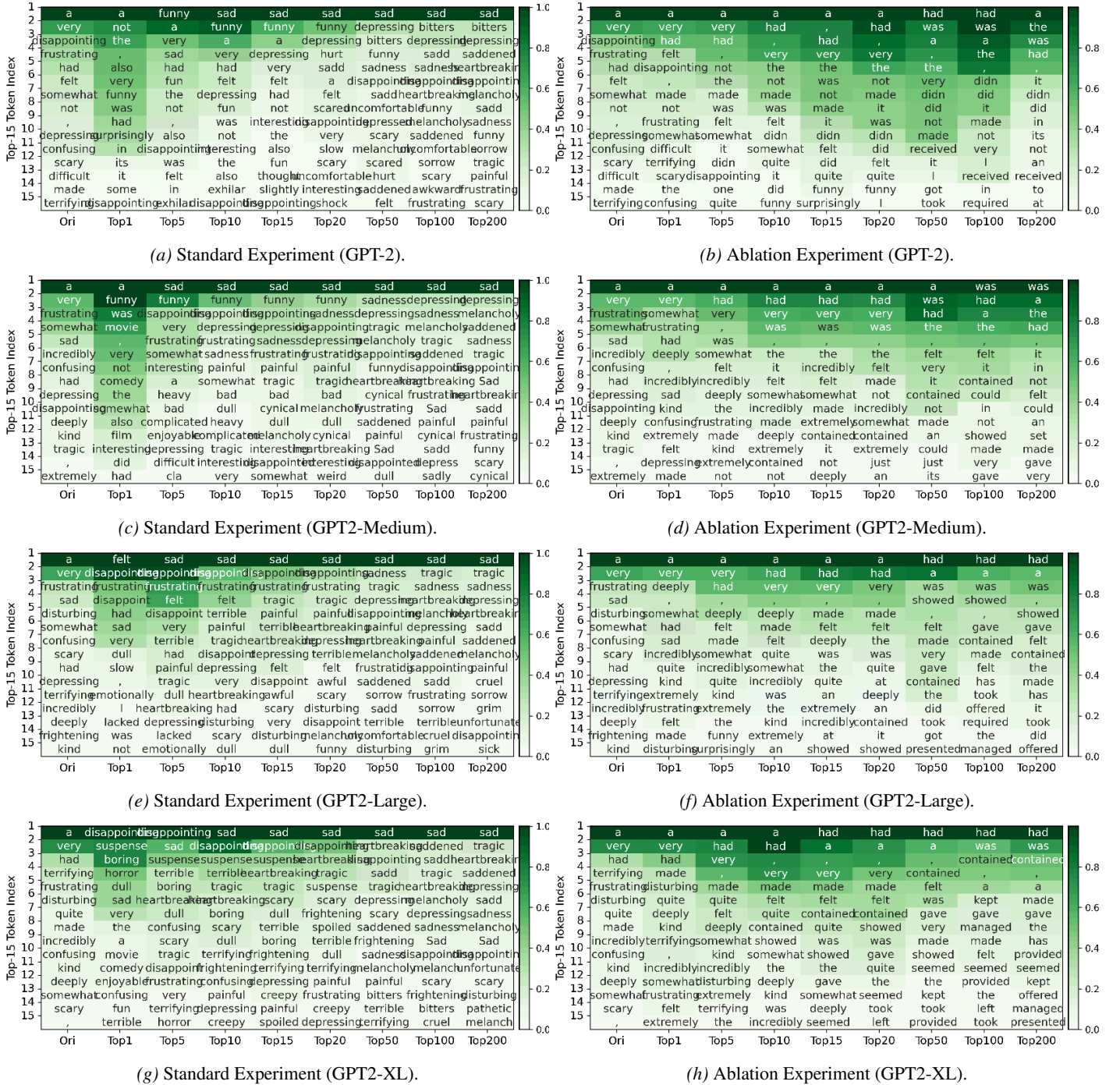


Figure 33. **Reconstruction of Middle-to-Late MLP Layers via Pathways within Top- k High-Contribution Subspaces: Standard vs. Ablation Studies.** This figure illustrates a case study using the prompt “The movie was thrilling but also” with the target token “sad”. We visualize the top-15 predicted tokens across GPT-2 variants under standard and ablation settings.

Case 17: Prompt “The recipe calls for two cups of” with Target Token “sugar”.

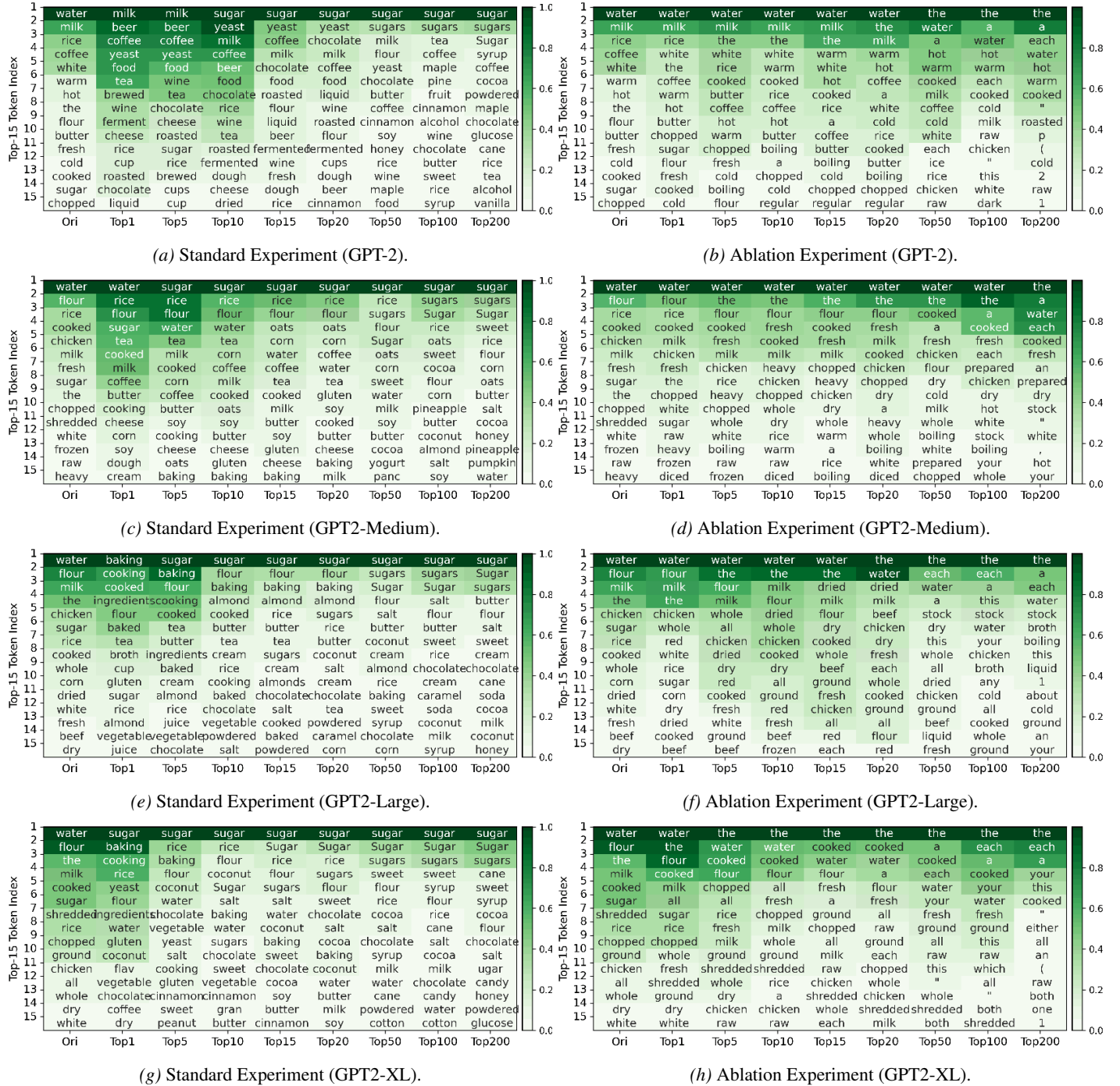


Figure 34. **Reconstruction of Middle-to-Late MLP Layers via Pathways within Top- k High-Contribution Subspaces: Standard vs. Ablation Studies.** This figure illustrates a case study using the prompt “The recipe calls for two cups of” with the target token “sugar”. We visualize the top-15 predicted tokens across GPT-2 variants under standard and ablation settings.

Case 18: Prompt “To solve the equation, one must first” with Target Token “simplify”.

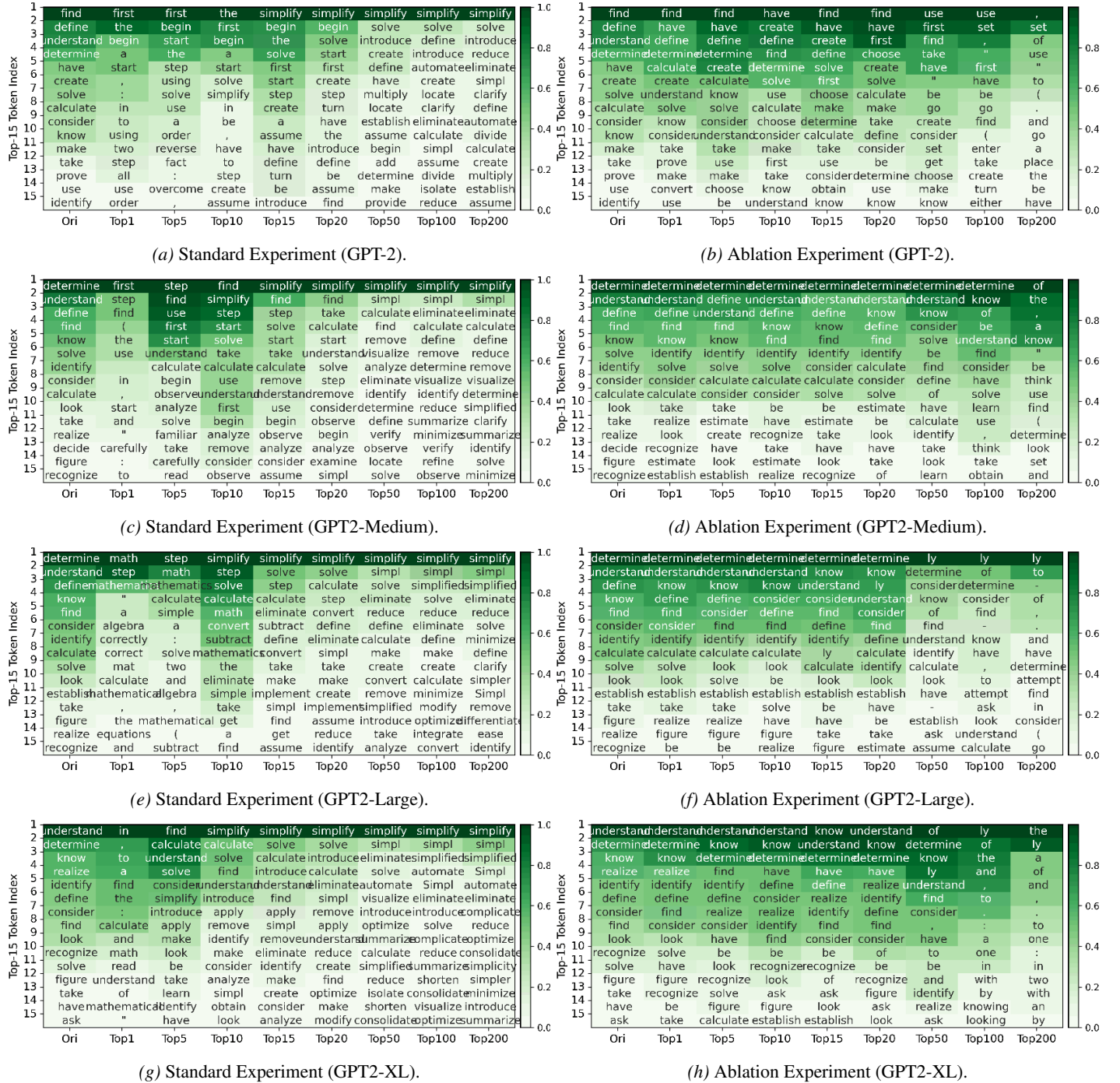


Figure 35. **Reconstruction of Middle-to-Late MLP Layers via Pathways within Top- k High-Contribution Subspaces: Standard vs. Ablation Studies.** This figure illustrates a case study using the prompt “To solve the equation, one must first” with the target token “simplify”. We visualize the top-15 predicted tokens across GPT-2 variants under standard and ablation settings.

Case 19: Prompt “Alice gave Bob the book because she” with Target Token “wanted”.

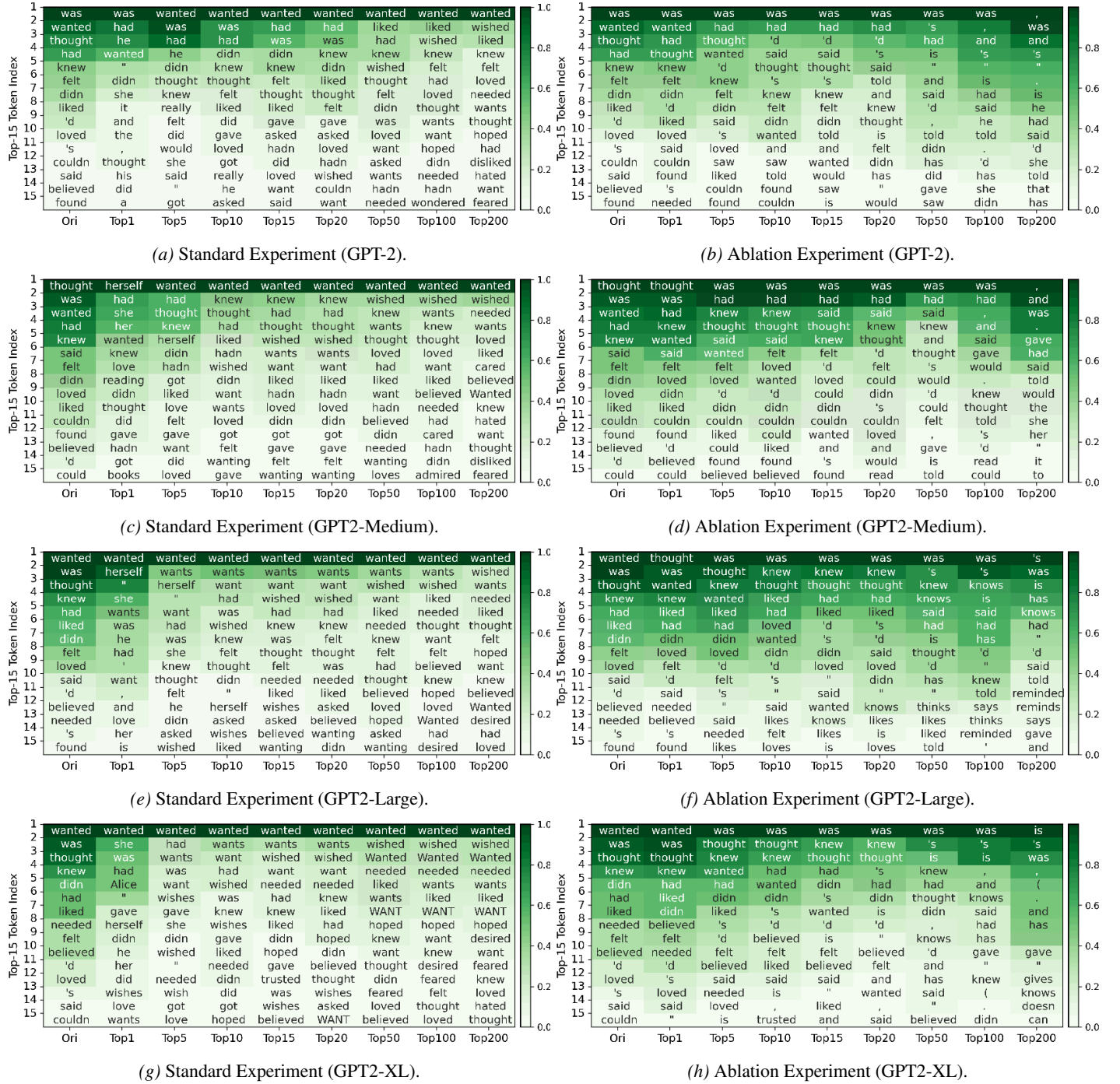


Figure 36. **Reconstruction of Middle-to-Late MLP Layers via Pathways within Top- k High-Contribution Subspaces: Standard vs. Ablation Studies.** This figure illustrates a case study using the prompt “Alice gave Bob the book because she” with the target token “wanted”. We visualize the top-15 predicted tokens across GPT-2 variants under standard and ablation settings.

Case 20: Prompt “If I had studied harder, I” with Target Token “ would”.

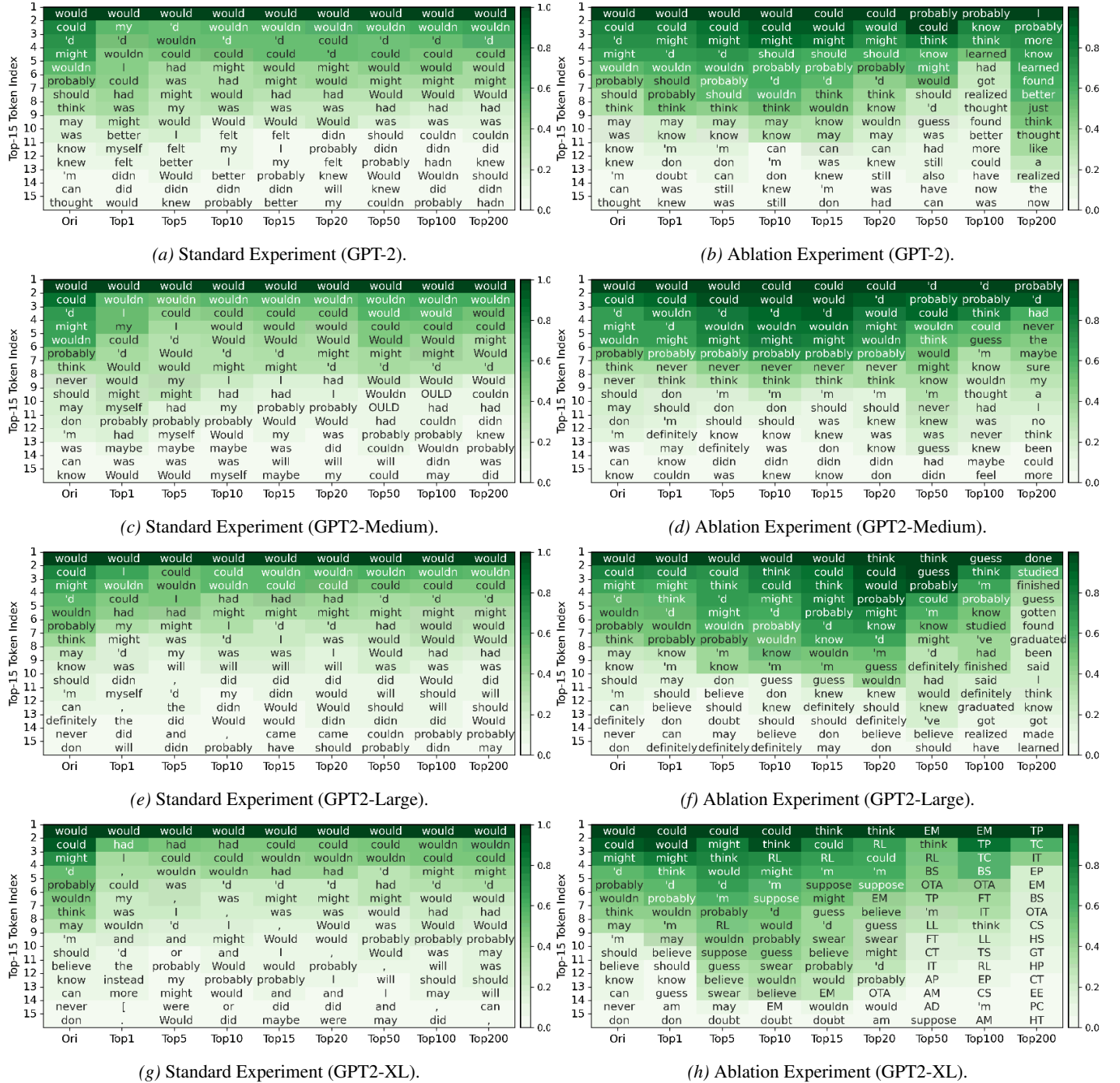


Figure 37. **Reconstruction of Middle-to-Late MLP Layers via Pathways within Top- k High-Contribution Subspaces: Standard vs. Ablation Studies.** This figure illustrates a case study using the prompt “If I had studied harder, I” with the target token “ would”. We visualize the top-15 predicted tokens across GPT-2 variants under standard and ablation settings.

I. Subspace-Level Analysis of Model Behavior and Lexical Preferences

I.1. Case Study: Progressive Effect of Top- k Subspaces

To elucidate the internal mechanisms of subspace pathways, we conduct a case study focused on positive semantics (e.g., “the cat looks very” $\rightarrow$ “happy”). We assess the fidelity and faithfulness of the identified pathways through reconstruction and ablation experiments. In Figure 38, we intervene on the MLP down-projection matrices across the middle-to-late layers of GPT-2 models of varying scales, and monitor the resulting changes in the target token’s *rank*, *logit*, and *probability*. The result demonstrates that ablating the top- k subspaces identified by SCA substantially degrades target prediction performance. Conversely, exclusively retaining these subspaces leads to a swift recovery of the target token, thereby verifying their pivotal role in the generation process.

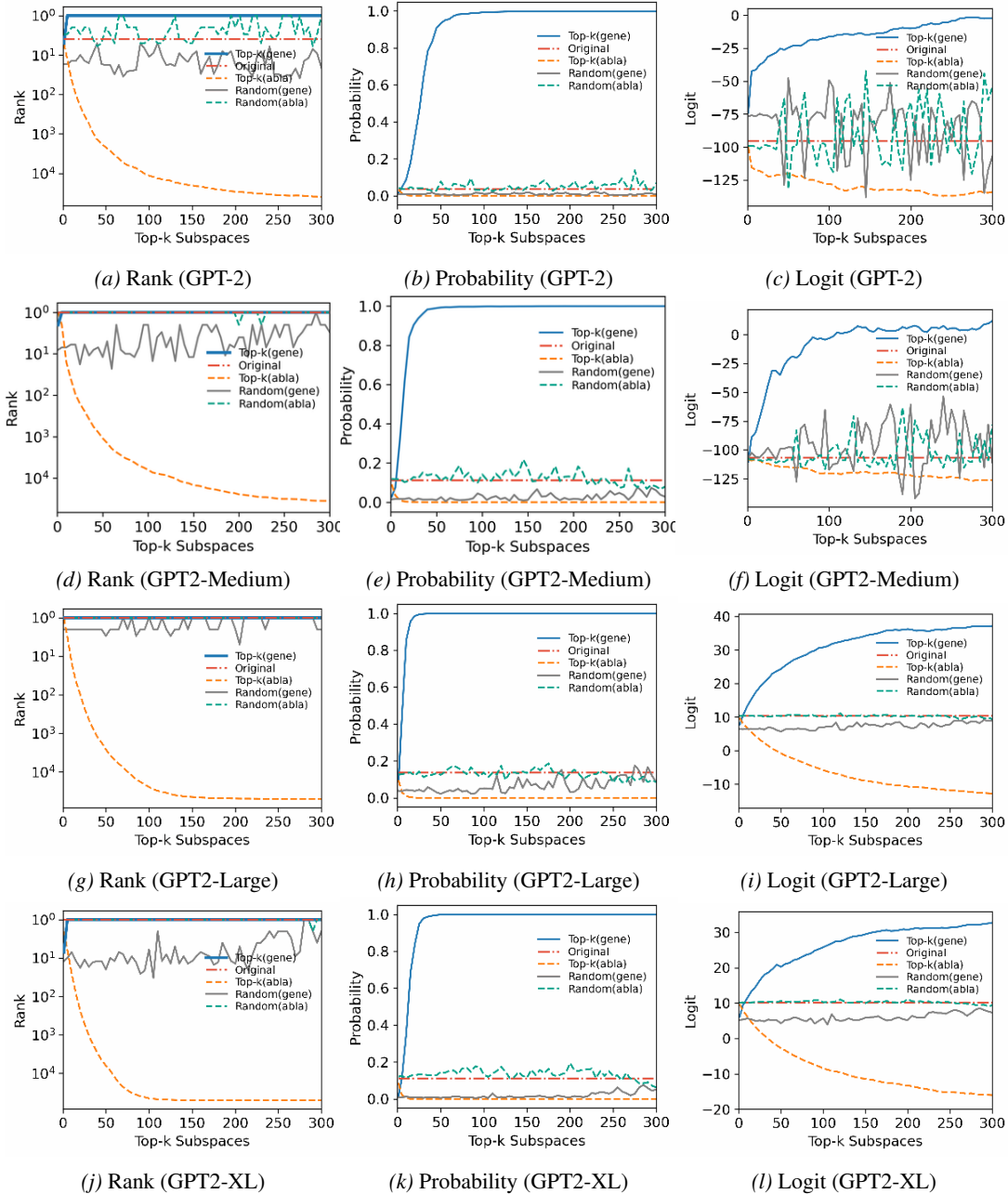


Figure 38. **Pathway-Level Analysis via Top- k Subspace Interventions.** We examine how the incremental inclusion of high-contribution subspaces influences the target token’s *probability*, *rank*, and *logit* for the case (“the cat looks very” $\rightarrow$ “happy”) across GPT-2 models of varying scales. **Original** denotes the unmodified model. **Top- k (gene/abla)**: Reconstructs or ablates the MLP down-projection matrix (W_{down}) using the top- k high-contribution subspaces. **Random (gene/abla)**: Reconstructs or ablates W_{down} using k randomly selected subspaces. Here, interventions are applied to layers 4–11 (GPT-2), 7–23 (GPT2-Medium), 10–35 (GPT2-Large), and 12–47 (GPT2-XL).

I.2. Layer-wise Distribution of Interpretable Subspaces

In this section, we analyze the layer-wise distribution of the top-20 high-contribution subspaces in GPT2-Medium, Large, and XL models for the case “*The cat looks very happy*” → “*happy*”. As shown in Figure 45, dominant subspaces are concentrated at lower indices, reflecting the strong influence of principal components on model predictions. Meanwhile, higher-index subspaces provide complementary contributions that refine the leading components.

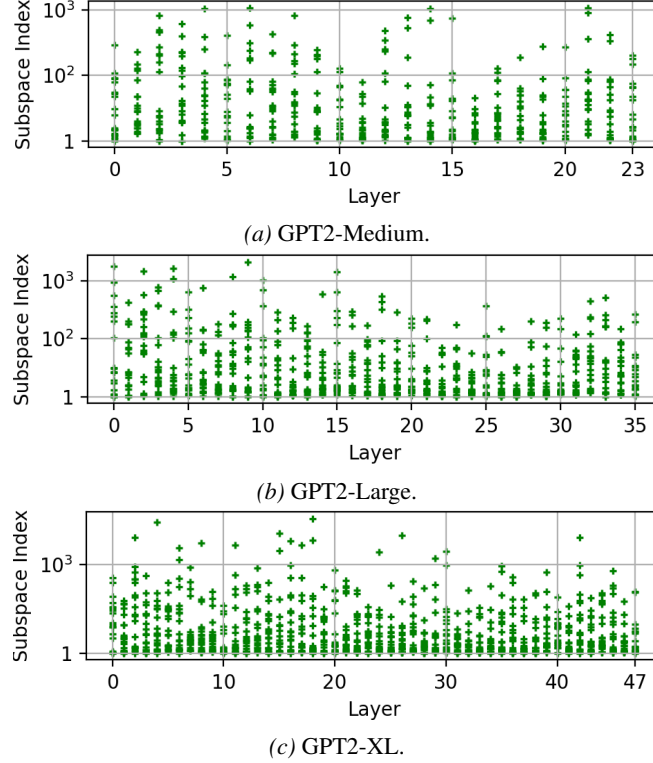


Figure 39. **Layer-wise Distribution of Top-20 High-Contribution Subspaces in GPT-2 Models (Medium, Large, XL).** Analysis focuses on the case “*The cat looks very happy*” → “*happy*”.

I.3. Investigating Subspace Contributions to Non-Rank-1 Token Generation

Transformer MLP layers exhibit structured subspaces that play a key role in token generation. In this work, we focus on a case study to examine how these subspaces shape a specific prediction. Using the prompt “*The capital of Germany is*”, where the model predicts “*Frankfurt*” instead of the correct answer “*Berlin*”, we analyze how subspace-based pathways contribute to both the emergence and ranking of the predicted token.

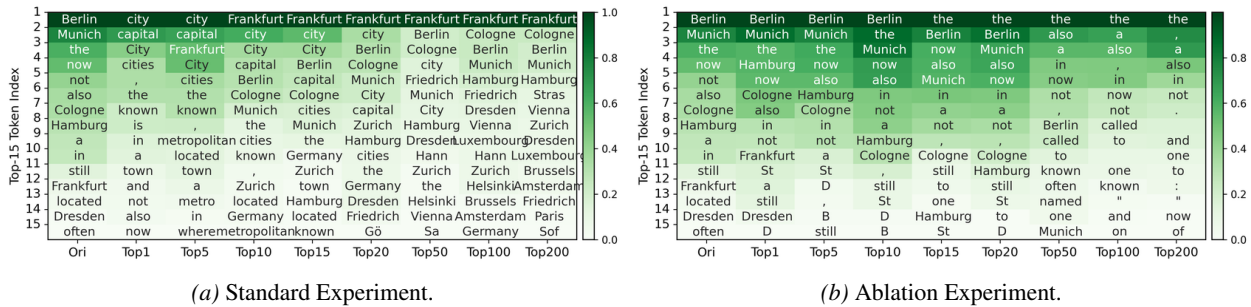


Figure 40. **Reconstruction of Middle-to-Late MLP Layers via Pathways within Top-k High-Contribution Subspaces: Standard vs. Ablation.** Here, we consider the case “*The capital of Germany is*” → “*Frankfurt*”.

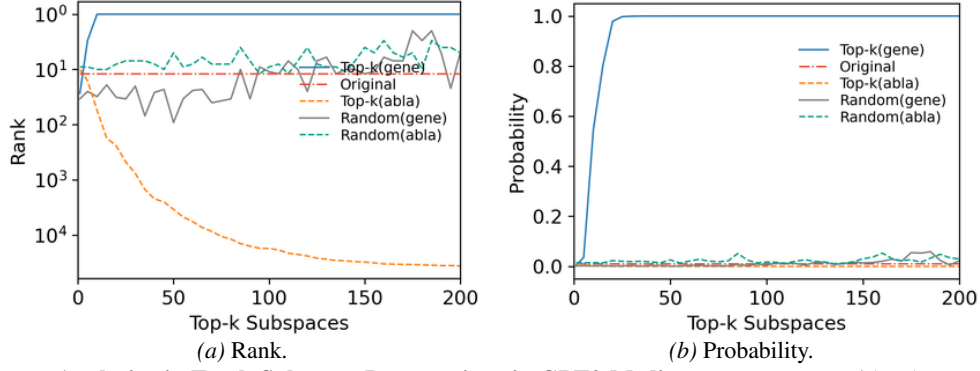


Figure 41. **Pathway Analysis via Top-k Subspace Interventions in GPT2-Medium.** Here, we consider the case “The capital of Germany is” $\rightarrow$ “Frankfurt”.

I.4. Additional Experiments on Model Lexical Preferences under Subspace Intervention

Following the setup in Section 4.2 (Section 4.2 (Experiment 3)), we investigate the effect of intervening on Subspaces 5 and 9 in Layer 16 of GPT-2 Medium using the prompt “The cat looks very”. We visualize the model outputs under amplification and suppression of these subspaces. The results in Figure 42 show that modulating each subspace leads to semantically consistent changes aligned with its learned direction. Here, Subspace 5 has a negative direction associated with “*insufficiency concepts*” and a positive direction corresponding to “*inspection actions*”; Subspace 9 exhibits a negative direction aligned with “*criticism and feedback*”, and a positive direction associated with “*historical decades*”. Since Subspaces 5 and 9 are not high-contribution subspaces, their contribution scores are only 0.073 and 0.0623, respectively. Consequently, their relative perturbation coefficients are higher than those used for the high-contribution Subspace 7 in Figure 4.

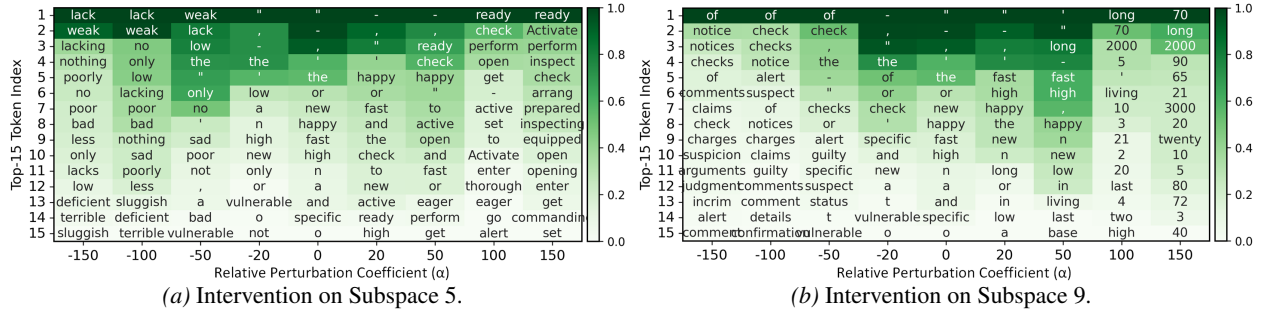


Figure 42. **Model Lexical Preferences under Intervention on Subspaces 5 and 9** Here, we consider the prompt “The cat looks very”.

J. Additional Evaluation on Benchmark Datasets and Advanced SAE Variants

J.1. Evaluation on the Gender Pronoun Task

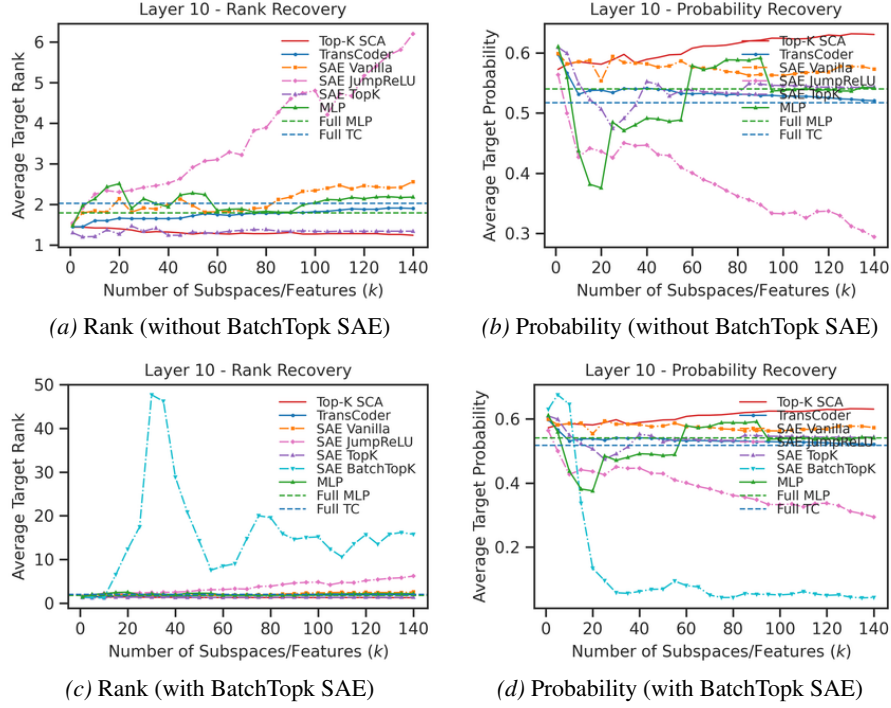


Figure 43. **Evaluation on the Gender Pronoun Task.** Due to the large fluctuations of the BatchTopk SAE curve, Figures 43a and 43b do not include BatchTopk SAE. Figures 43c and 43d include BatchTopk SAE.

J.2. Evaluation on the Subject-Verb Agreement Task (Relative Clause)

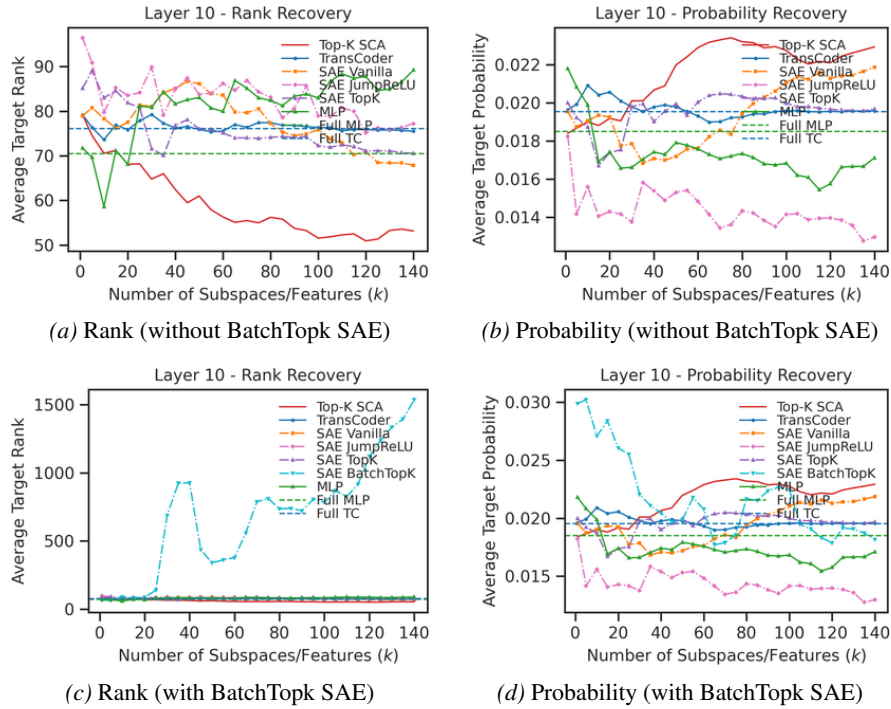


Figure 44. **Evaluation on the Subject-Verb Agreement Task (Relative Clause).** Due to the large fluctuations of the BatchTopk SAE curve, Figures 44a and 44b do not include BatchTopk SAE. Figures 44c and 44d include BatchTopk SAE.

J.3. Evaluation on the Subject–Verb Agreement Task (Simple Structure)

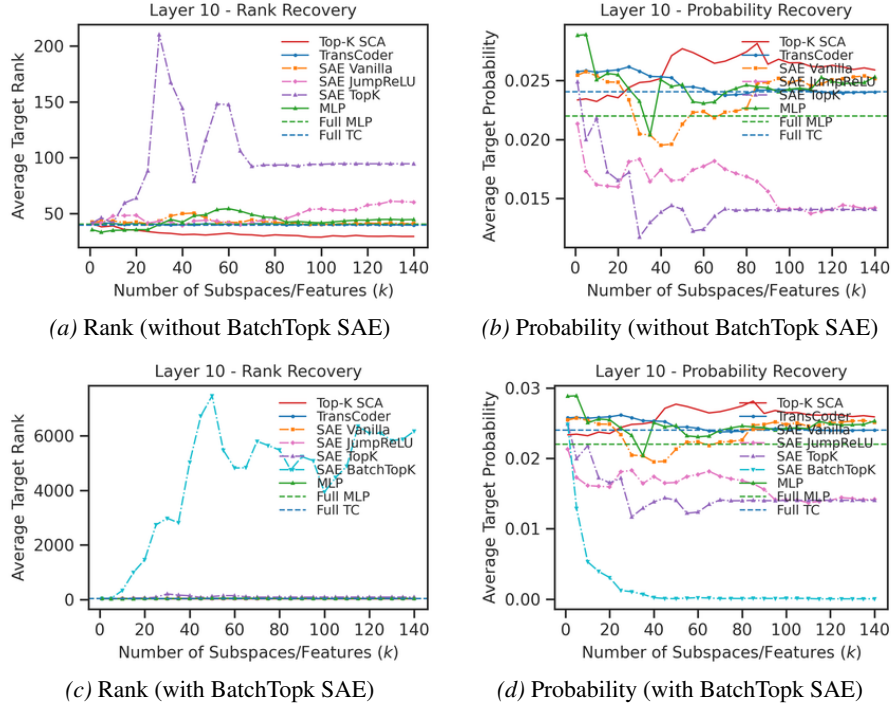


Figure 45. **Evaluation on the Subject–Verb Agreement Task (Simple Structure).** Due to the large fluctuations of the BatchTopk SAE curve, Figures 45a and 45b do not include BatchTopk SAE. Figures 45c and 45d include BatchTopk SAE.

K. Properties of Highly Interpretable MLP Subspaces

K.1. Layer-wise Count of Highly Interpretable Subspaces

In this section, we perform a layer-wise statistical analysis of interpretable subspaces derived from the Singular Value Decomposition (SVD) of MLP weight matrices. We analyze all DEU subspaces within the projection matrices across models of varying scales. Formally, a subspace is considered interpretable if it satisfies at least one of the following criteria: (i) the detector component in the up-projection matrix W_{up} exhibits high interpretability; or (ii) the effector component in the down-projection matrix W_{down} demonstrates high interpretability. Here, we focus exclusively on highly interpretable subspaces.

Relationship Between Model Size and Interpretability. In Figure 46, our empirical analysis reveals a robust positive correlation between model size and the absolute quantity of interpretable subspaces. This observation supports the hypothesis that increased parameter capacity alleviates feature superposition (Elhage et al., 2022; Arora et al., 2018).

Asymmetric Layer-wise Specialization. Figure 47 presents a detailed layer-wise analysis of the proportion of interpretable subspaces in both the up-projection (W_{up}) and down-projection (W_{down}) matrices.

Layer-wise Distribution of Interpretable Subspaces in the Up-Projection Matrix: In the up-projection matrices, interpretable subspaces are predominantly concentrated in the network’s middle layers. This pattern aligns with the Transformer’s hierarchical feature extraction paradigm: the initial layers primarily encode low-level lexical information, whereas the middle layers serve as the core reasoning engine, with W_{up} detectors responsible for extracting abstract, high-level contextual features from the aggregated residual stream.

Layer-wise Distribution of Interpretable Subspaces in the Down-Projection Matrix: In contrast, interpretable subspaces within the down-projection matrices exhibit a pronounced shift toward the middle-to-late layers. This pattern indicates a mechanistic progression from feature detection to output generation. In deeper layers, the model moves beyond merely detecting context (feature construction) to actively steering the residual state toward the target output. These subspaces project the extracted features onto the output manifold, thereby enforcing specific token predictions.

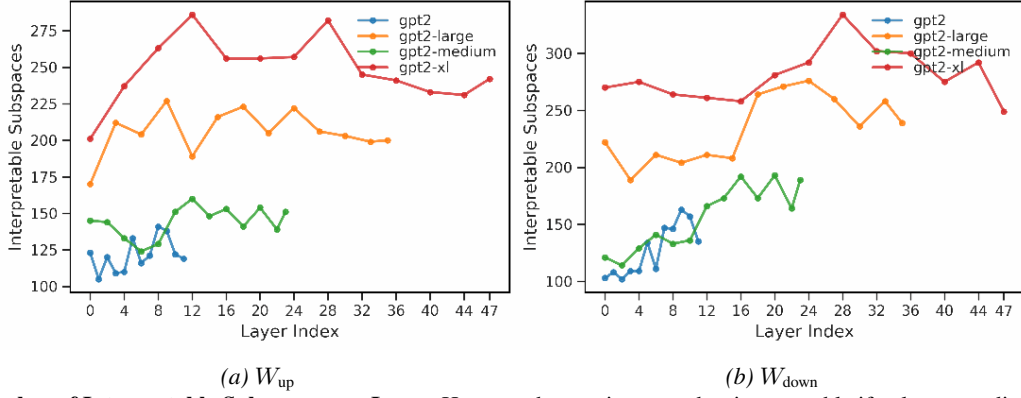


Figure 46. **Number of Interpretable Subspaces per Layer.** Here, a subspace is counted as interpretable if at least one direction (positive or negative) is interpretable.

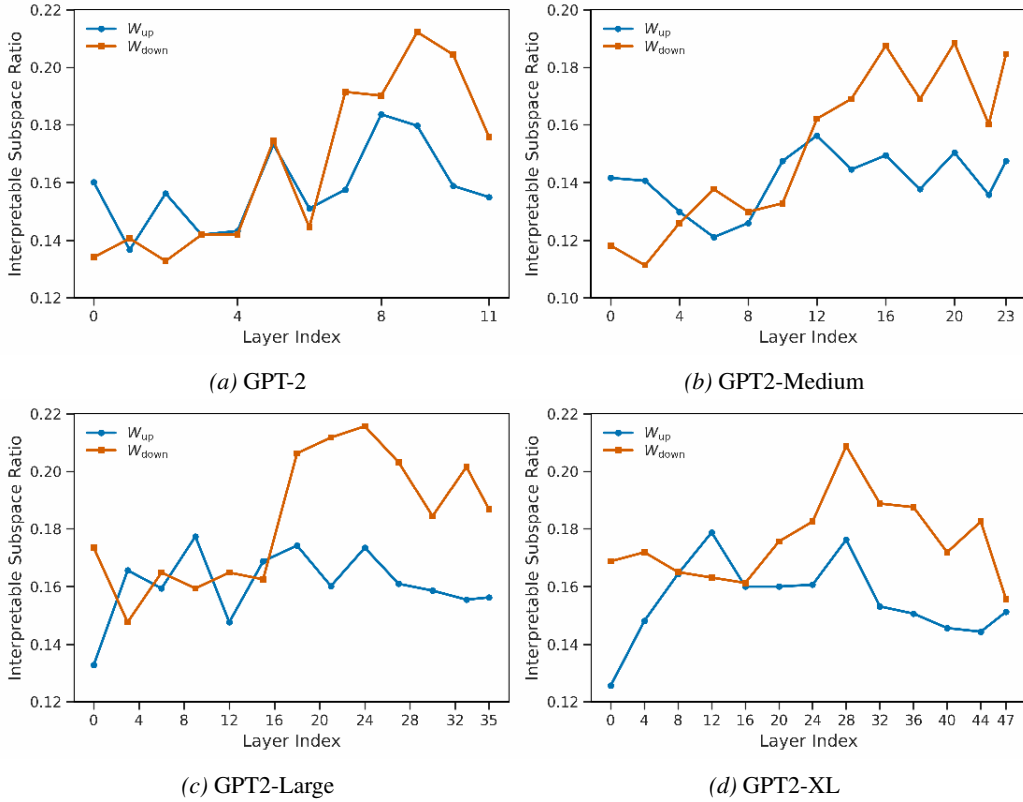


Figure 47. **Layer-wise Proportion of Interpretable Subspaces Across GPT-2 Family.** Each figure shows the proportion of interpretable subspaces within the W_{up} and W_{down} projection matrices.

K.2. Layer-wise Distribution of Interpretable Subspaces

As shown in Figure 48, we examine the spectral localization of highly interpretable subspaces in both the up-projection and down-projection matrices, analyzing how their alignment varies with network depth and model size.

Layer-wise Spectral Concentration of Interpretable Subspaces. Across both down-projection and up-projection matrices, we observe a consistent layer-wise trend: as network depth increases, highly interpretable subspaces progressively concentrate within the principal component regions, i.e., the low-index subspaces. This concentration effect is more pronounced in larger-scale models. It suggests that deeper and larger Transformer architectures progressively compress and align semantically meaningful representation into a low-dimensional but highly expressive spectral core. Notably, in up-projection matrices, interpretable subspaces exhibit strong concentration in small-index regions even in the earliest layers, indicating that these layers perform early semantic structuring, prioritizing core features to facilitate efficient downstream processing.

Non-trivial Contribution of Tail Subspaces. The dominant trend indicates that semantic information is primarily concentrated in low-index subspaces, which encode the core representations driving model predictions. At the same time, high-index (tail) subspaces retain non-trivial semantic content, providing complementary, fine-grained information rather than mere stochastic noise. This suggests that while the principal components capture the bulk of the model’s semantics, the tail subspaces contribute to its expressive capacity.

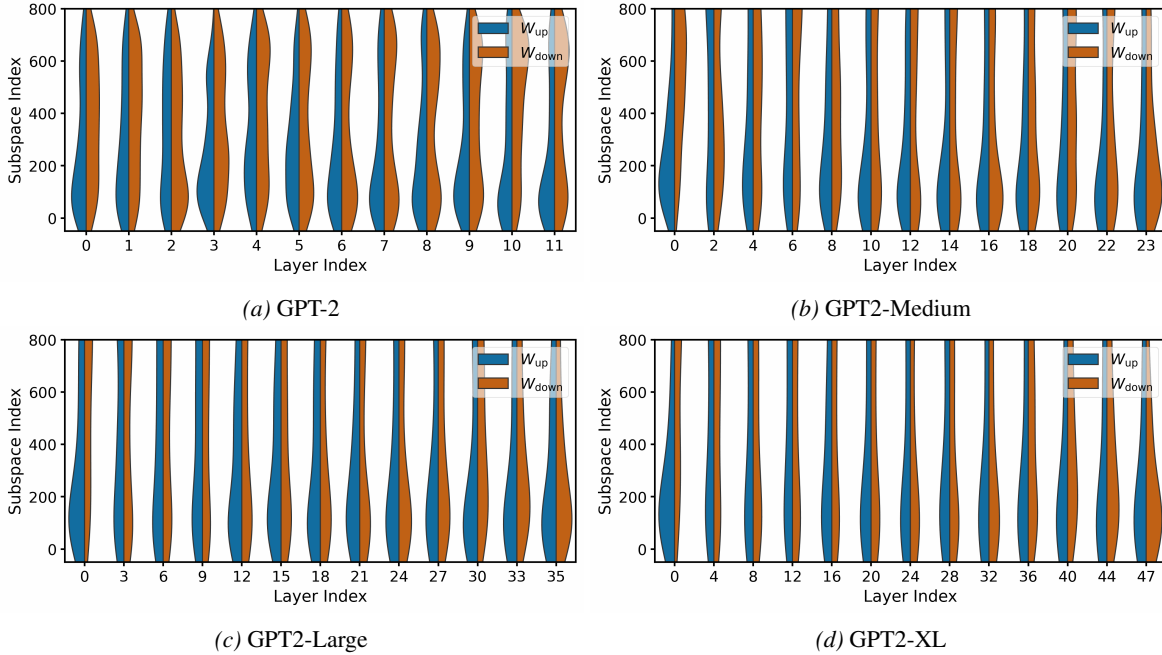


Figure 48. **Distribution of Interpretable Subspaces in MLP Modules Across GPT-2 Family.** Blue bars correspond to the up-projection matrix W_{up} , and red bars correspond to the down-projection matrix W_{down} .

L. Effect of Starting Layer on Pathway Intervention Efficacy

In this section, we investigate how the starting depth of pathway interventions affects performance recovery. Experiments are conducted on GPT2-Medium using the $\mathcal{D}_{gene}$ dataset, with interventions initiated at varying depths (Layers 5–8) and extending through Layer 23.

As illustrated in Figure 49, we evaluate performance using three metrics: mean probability, mean rank, and top-1 accuracy. The results indicate that the performance curves across different starting layers largely overlap, demonstrating that pathway interventions are robust to the choice of initial layer.

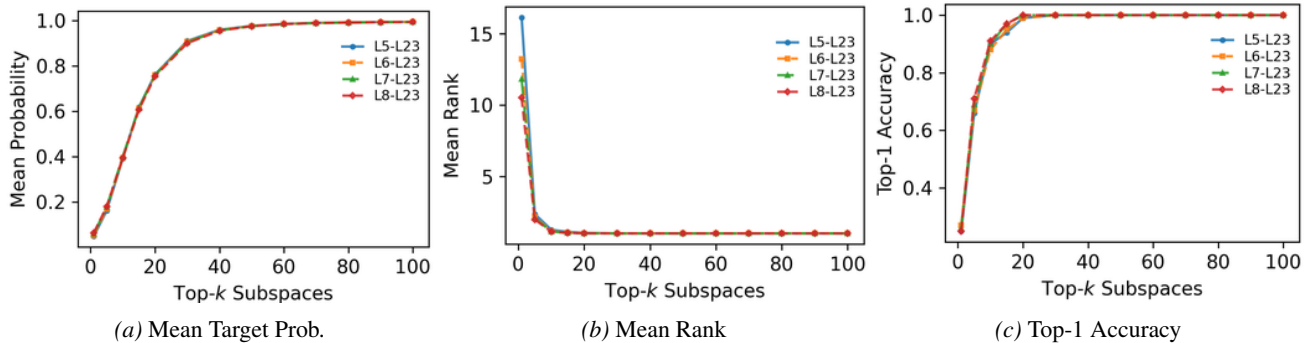


Figure 49. **Impact of Starting Layer Initialization on Pathway Efficacy (GPT2-Medium).** The plot illustrates performance recovery trends for interventions initiating at Layers 5-8, evaluated against increasing pathway widths (subspaces per layer).

M. Illustrative Examples of Interpretable Subspaces

In this section, we present illustrative examples of interpretable subspaces. Specifically, the interpretability of the up-projection matrix W_{up} is attributable to its detector component, whereas the interpretability of the down-projection matrix W_{down} derives from its effector component. Furthermore, we consider both the “Positive” and “Negative” directions, corresponding to the semantic concepts aligned with the opposing directions ($\pm \mathbf{u}_i$ or $\pm \mathbf{v}_i$) of the singular vector, thereby reflecting the inherent sign ambiguity in SVD (Kolda & Bader, 2009).

M.1. Interpretation of DEU Subspaces Across Models

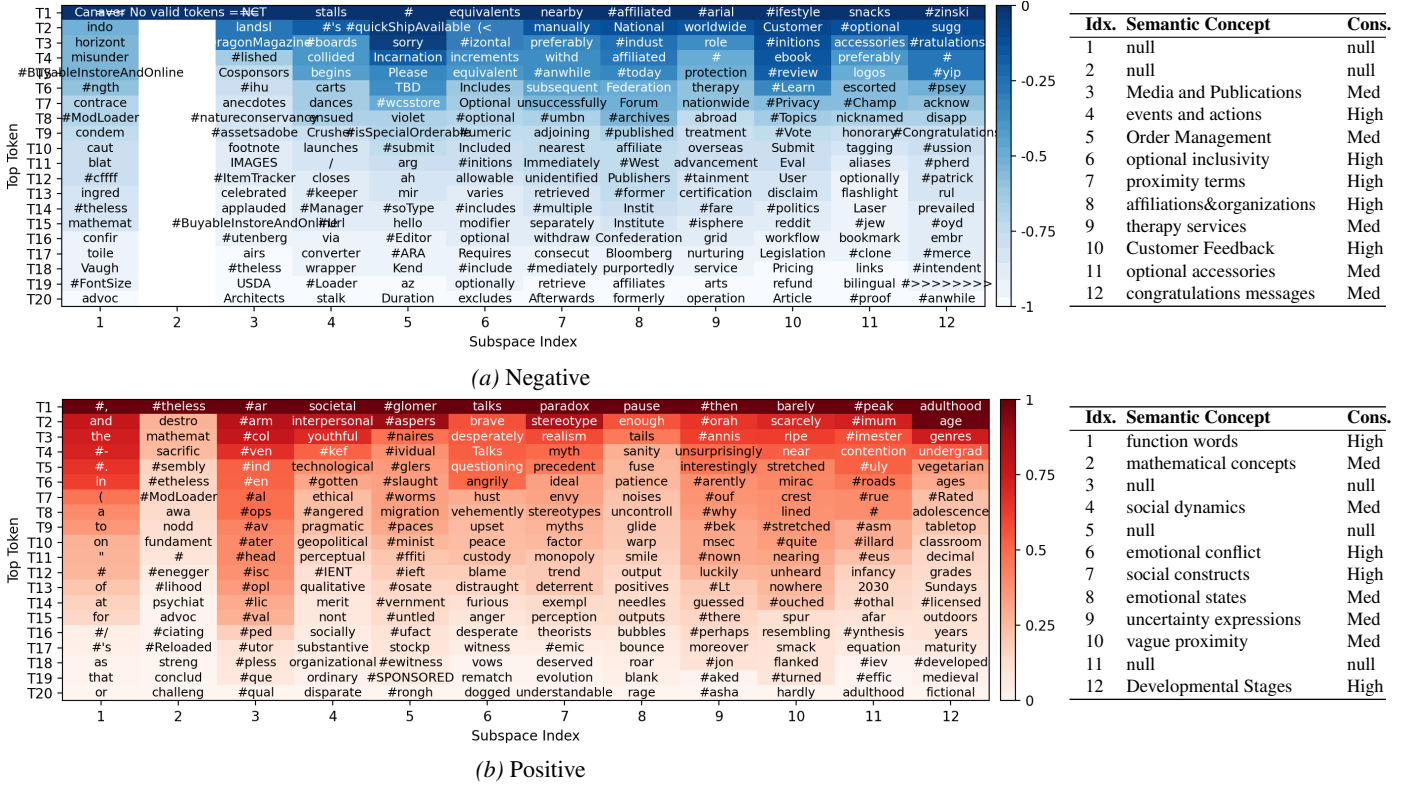


Figure 50. Example of DEU Subspace Interpretability via Semantic Consistency in the Up-Projection Matrix (GPT-2, Layer 7). Here, the color intensity represents the consistency score, while columns with blank entries correspond to uncaptured or unreadable tokens and have been omitted. “Idx.” refers to the subspace index, and “Cons.” indicates the subspace interpretability degree.

SVD as a Fast Interpretability Method for Transformers

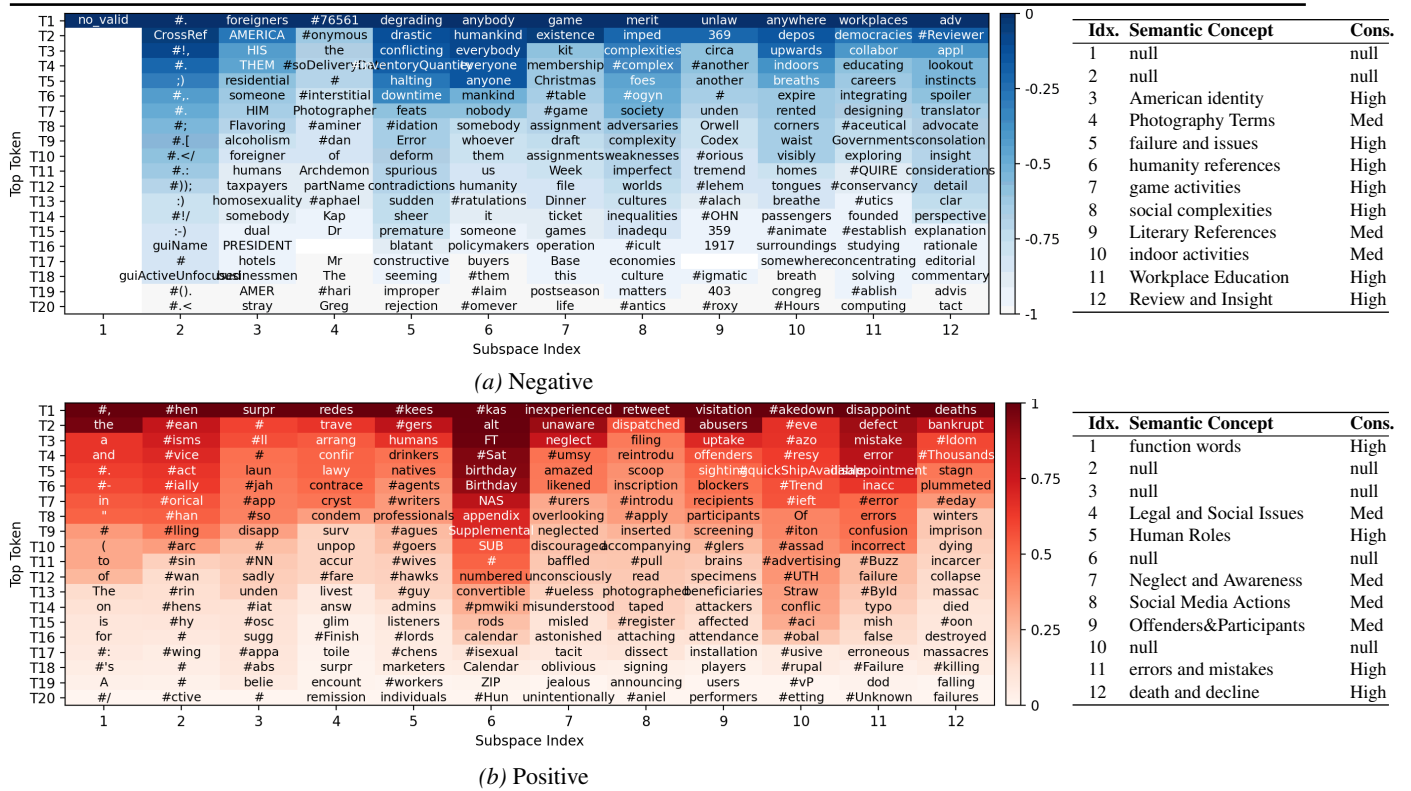


Figure 51. Example of DEU Subspace Interpretability via Semantic Consistency in the Down-Projection Matrix (GPT-2, Layer 7). Here, the color intensity represents the consistency score, while columns with blank entries correspond to uncaptured or unreadable tokens and have been omitted. “Idx.” refers to the subspace index, and “Cons.” indicates the subspace interpretability degree.

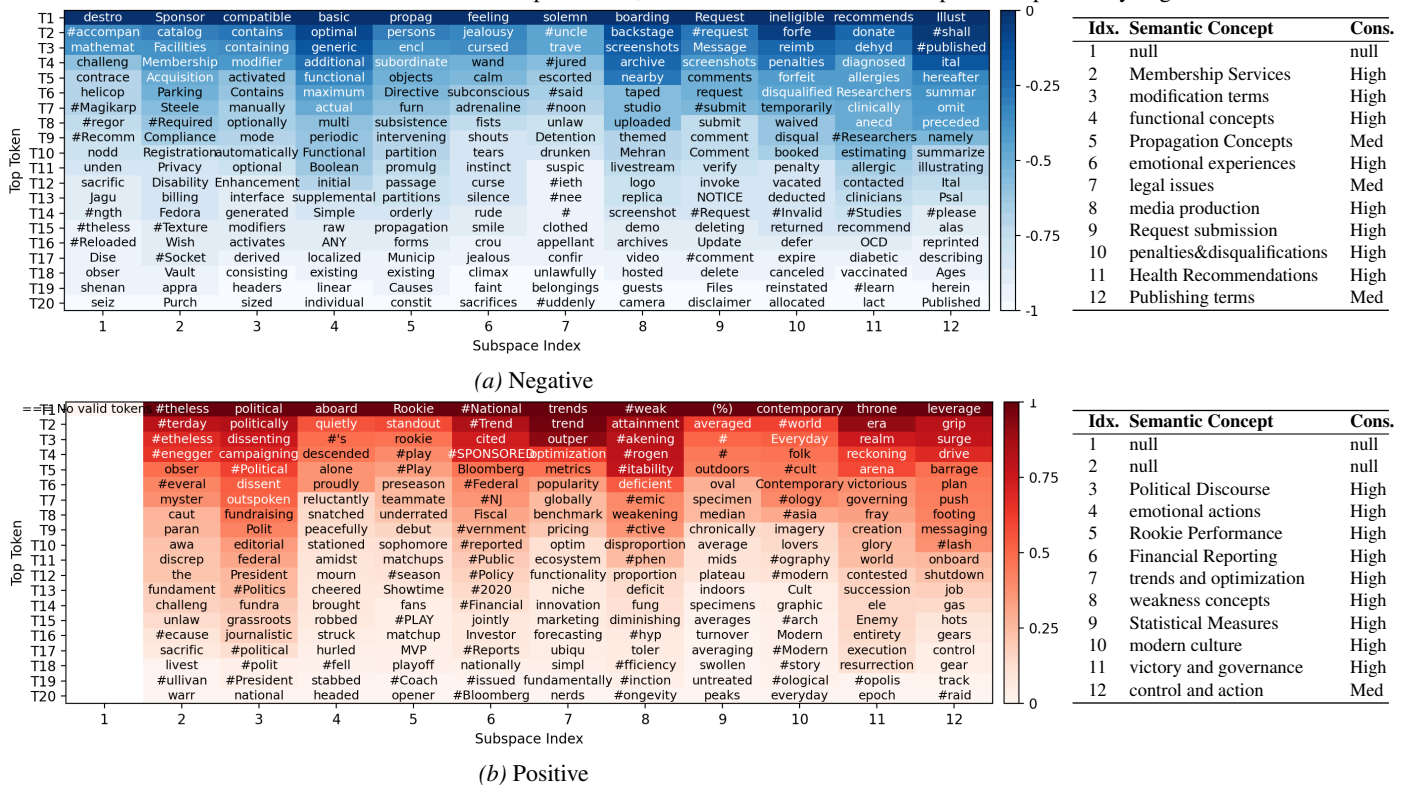


Figure 52. Example of DEU Subspace Interpretability via Semantic Consistency in the Up-Projection Matrix (GPT2-Medium, Layer 16). Here, the color intensity represents the consistency score, while columns with blank entries correspond to uncaptured or unreadable tokens and have been omitted. “Idx.” refers to the subspace index, and “Cons.” indicates the subspace interpretability degree.

SVD as a Fast Interpretability Method for Transformers

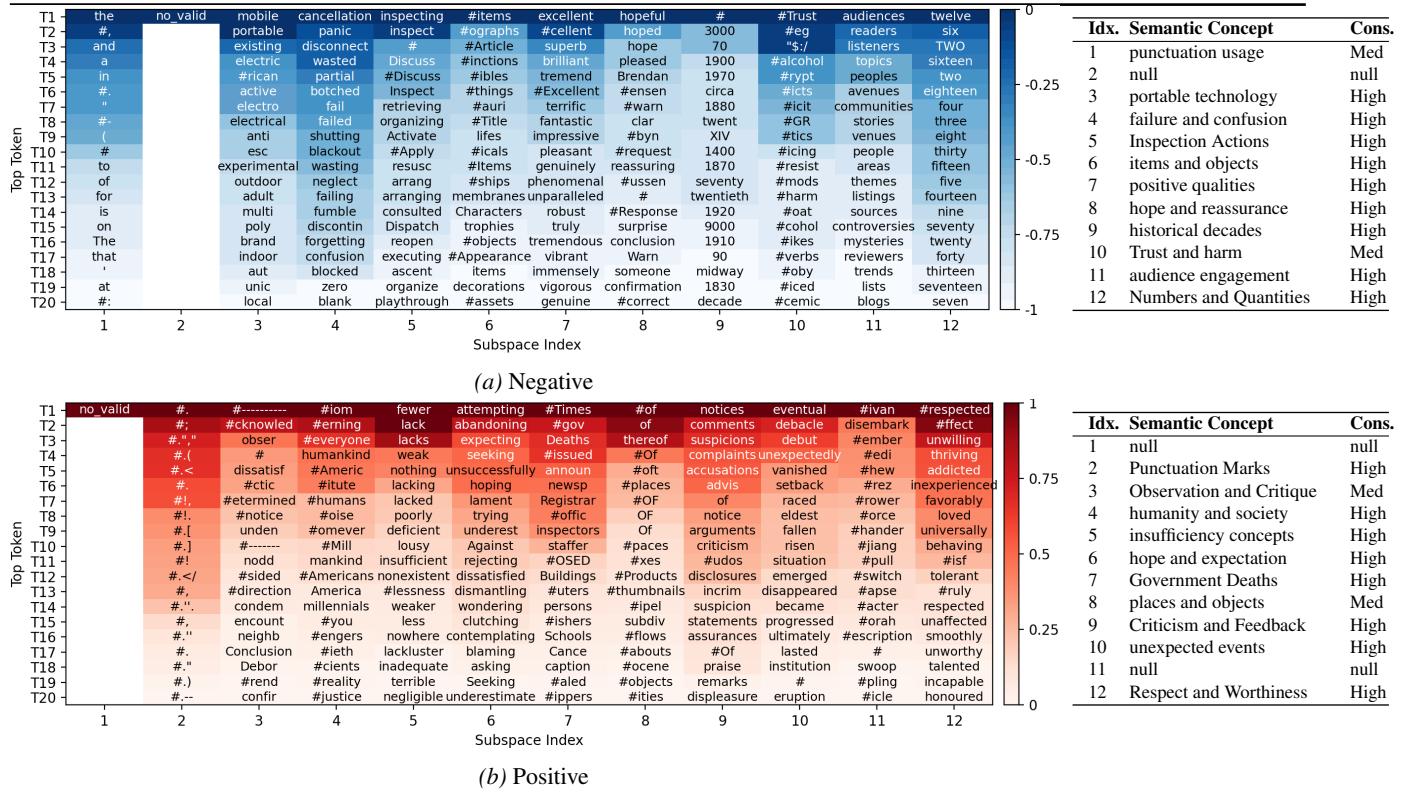


Figure 53. Example of DEU Subspace Interpretability via Semantic Consistency in the Down-Projection Matrix (GPT2-Medium, Layer 16). Here, the color intensity represents the consistency score, while columns with blank entries correspond to uncaptured or unreadable tokens and have been omitted. “Idx.” refers to the subspace index, and “Cons.” indicates the subspace interpretability degree.

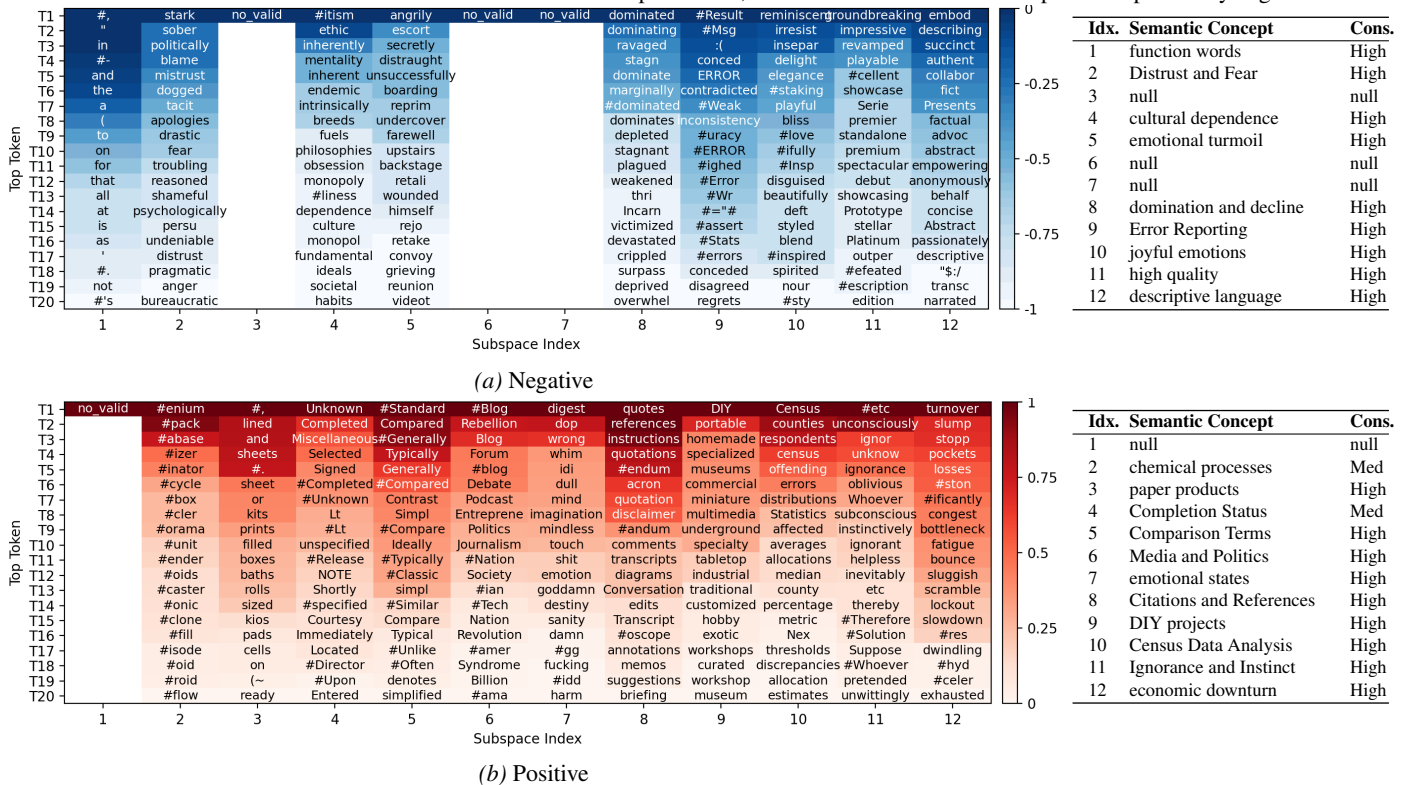
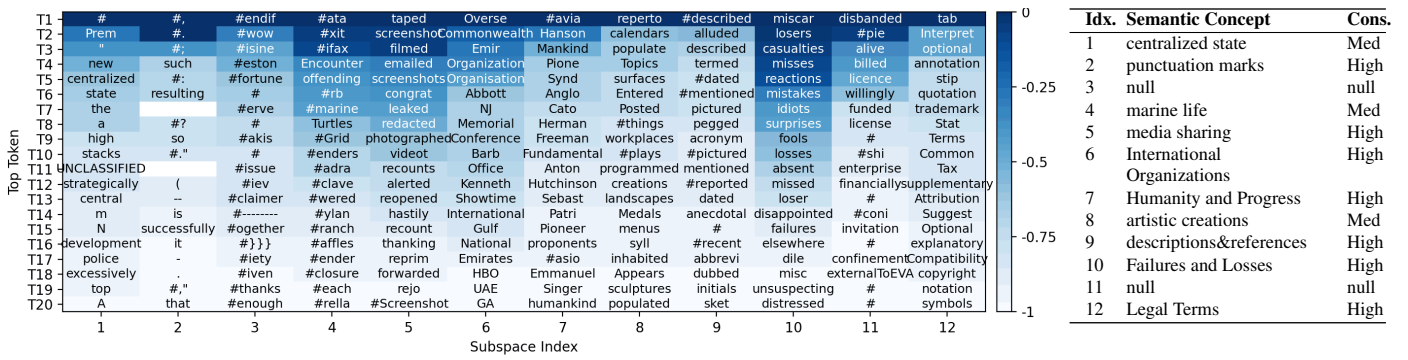
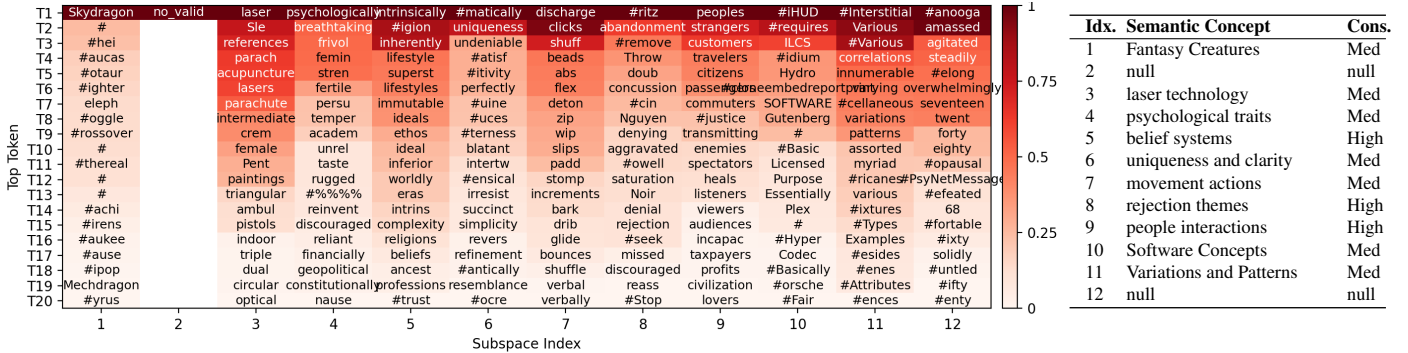


Figure 54. Example of DEU Subspace Interpretability via Semantic Consistency in the Up-Projection Matrix (GPT2-Large, Layer 18). Here, the color intensity represents the consistency score, while columns with blank entries correspond to uncaptured or unreadable tokens and have been omitted. “Idx.” refers to the subspace index, and “Cons.” indicates the subspace interpretability degree.

SVD as a Fast Interpretability Method for Transformers

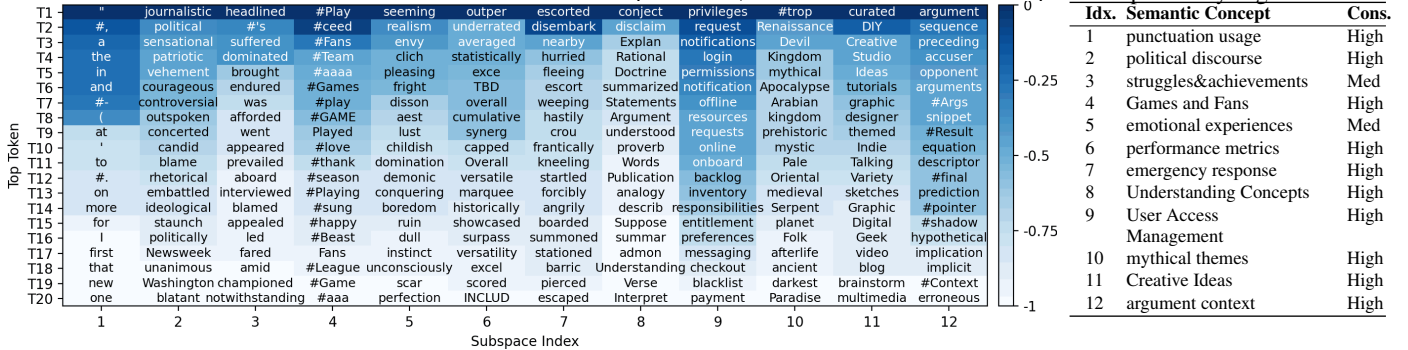


(a) Negative

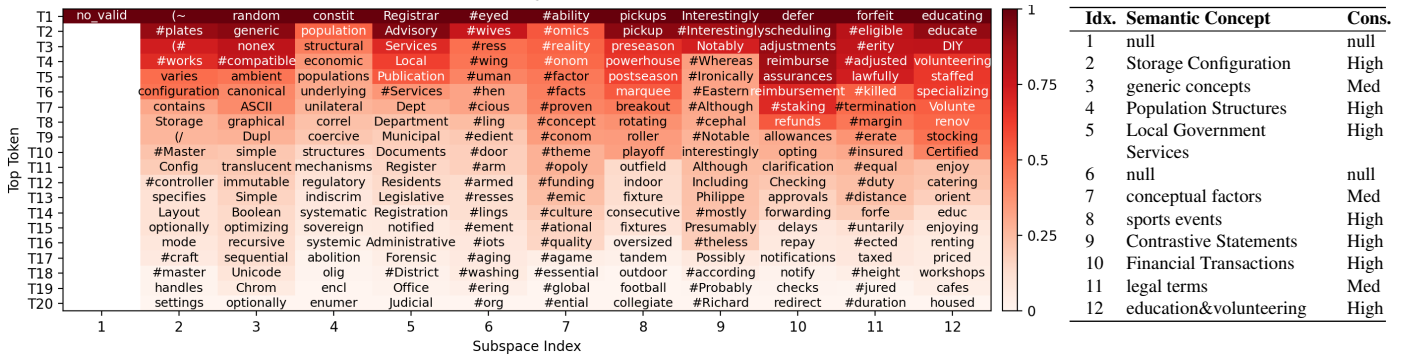


(b) Positive

Figure 55. Example of DEU Subspace Interpretability via Semantic Consistency in the Down-Projection Matrix (GPT2-Large, Layer 18). Here, the color intensity represents the consistency score, while columns with blank entries correspond to uncaptured or unreadable tokens and have been omitted. “Idx.” refers to the subspace index, and “Cons.” indicates the subspace interpretability degree.



(a) Negative



(b) Positive

Figure 56. Example of DEU Subspace Interpretability via Semantic Consistency in the Up-Projection Matrix (GPT2-XL, Layer 26). Here, the color intensity represents the consistency score, while columns with blank entries correspond to uncaptured or unreadable tokens and have been omitted. “Idx.” refers to the subspace index, and “Cons.” indicates the subspace interpretability degree.

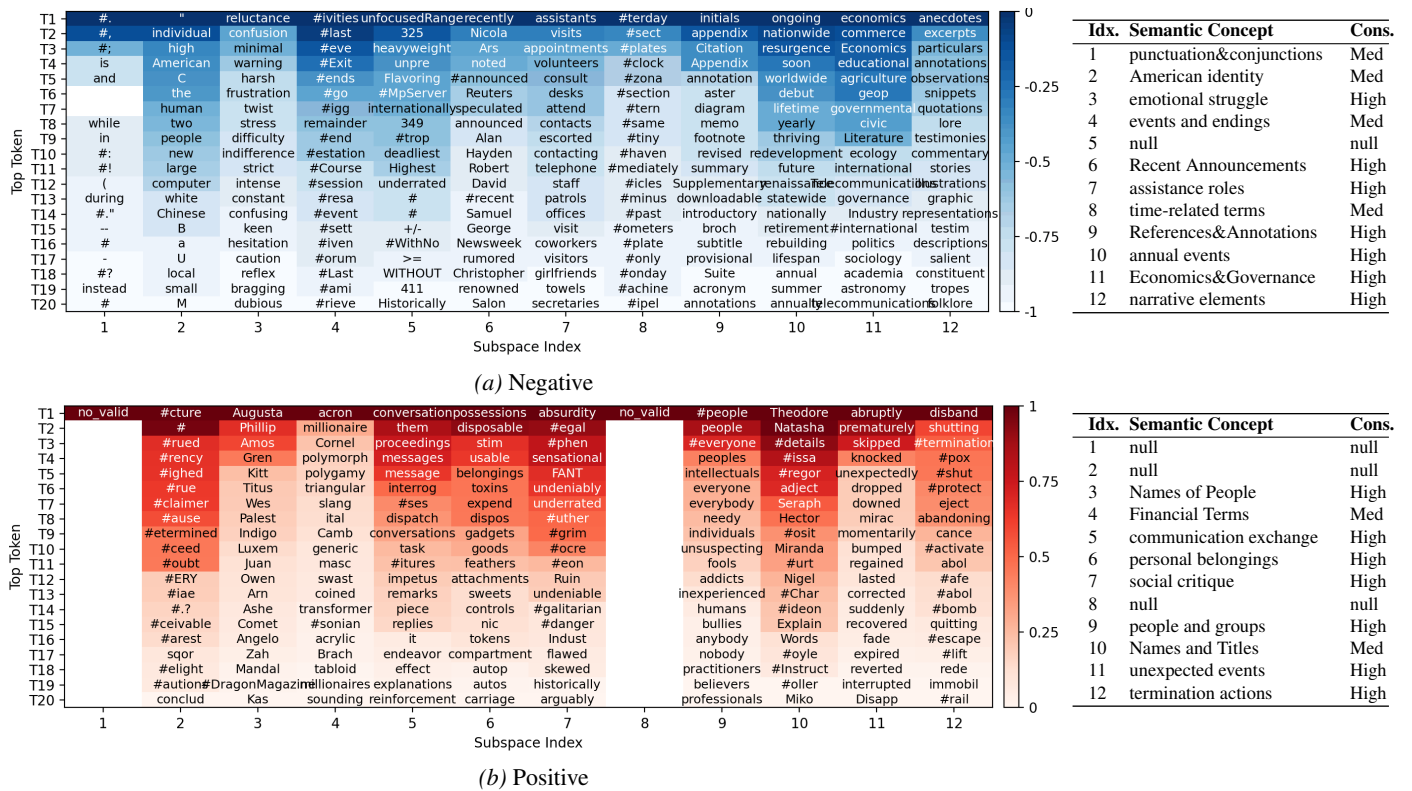


Figure 57. Example of DEU Subspace Interpretability via Semantic Consistency in the Down-Projection Matrix (GPT2-XL, Layer 26). Here, the color intensity represents the consistency score, while columns with blank entries correspond to uncaptured or unreadable tokens and have been omitted. “Idx.” refers to the subspace index, and “Cons.” indicates the subspace interpretability degree.

M.2. Semantic Consistency of Top-100 Tokens Captured by DEU Subspaces

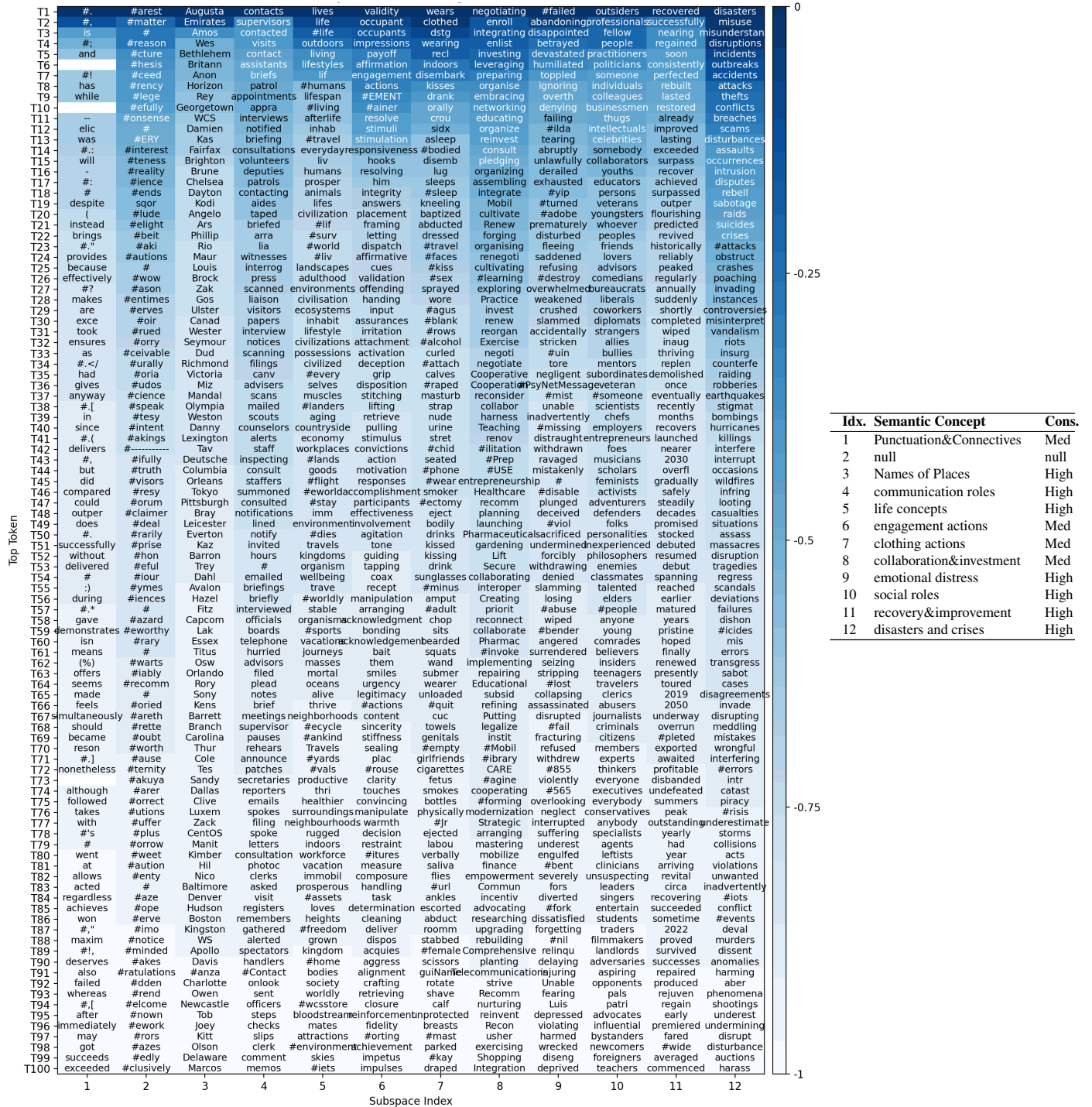


Figure 58. Semantic Consistency of Subspaces in the Down-Projection Matrix via Top-100 Tokens (GPT2-XL, Layer 25). The table lists the top-100 tokens associated with the top-12 subspaces (negative), demonstrating high semantic coherence. Here, the color intensity represents the consistency score, while columns with blank entries correspond to uncaptured or unreadable tokens and have been omitted. “Idx.” refers to the subspace index, and “Cons.” indicates the subspace interpretability degree.